

Moments Of Clarity in Machine Learning for Jet Physics

Rikab Gambhir

Based on [RG, Mastandrea, Nachman, Thaler; [2502.14036](#)]

[Ba, Dogra, RG, Tasissa, Thaler; [2302.12266](#)]

[RG, Thaler, Wu; WIP]

[RG, Larkoski, Thaler; [2410.05379](#)]

[RG, Osathapan, Thaler; [2403.08854](#)]

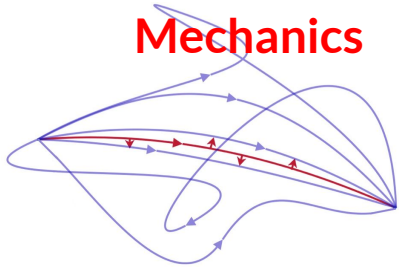
[RG, Mastandrea; WIP]

Our hammer: Machine Learning

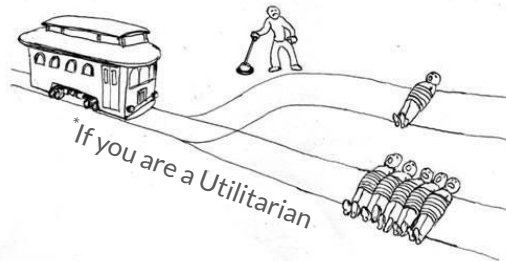
Machine learning is the art of *numerical functional optimization*.

Almost every interesting problem in life involves finding the “best” function for something!

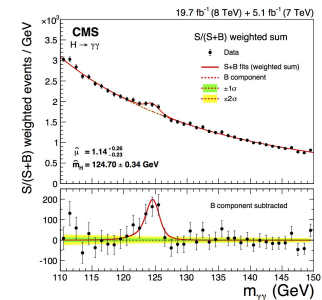
Classical Mechanics



Ethics and Morality*



Statistics and Fitting

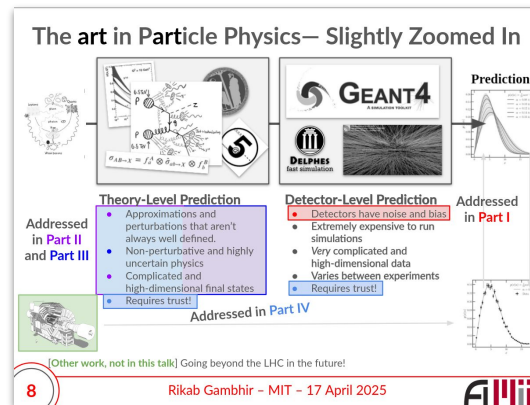


Human Connection

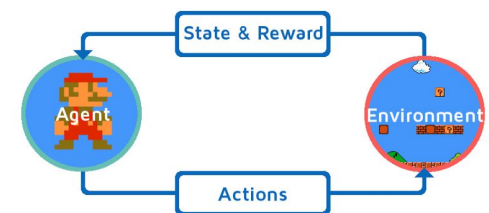
I'm sorry, but as an AI language model, I am unable to pretend to be



Particle Physics



Games

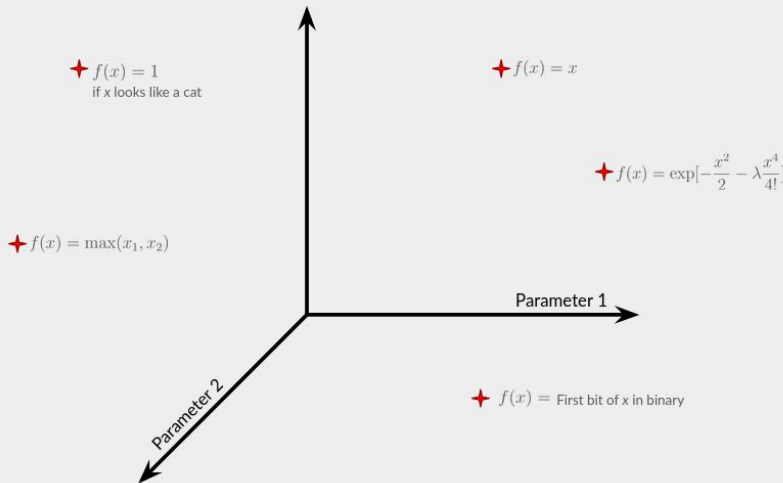


Not an exhaustive list of interesting problems in life.

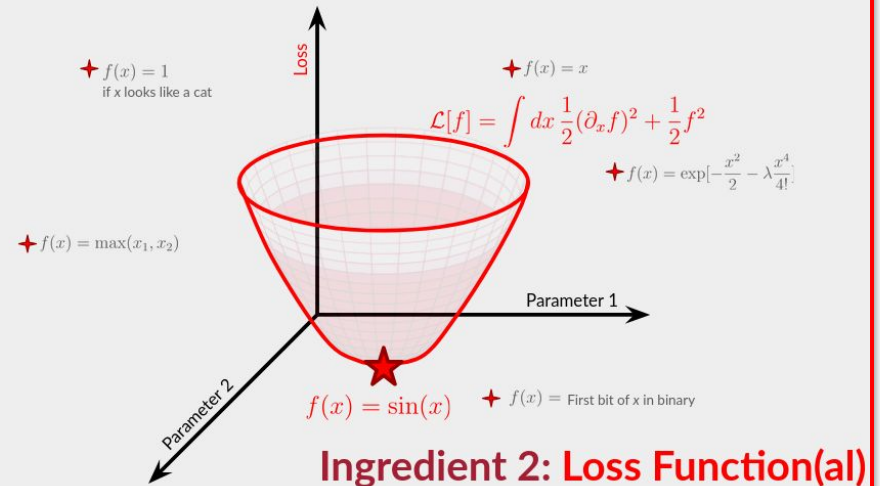
What even is Machine Learning?

Our approach: Rephrase our problems as numerically finding the best function in a large space of parameterized functions! The trick is (1) defining where we look and (2) what “best” means.

Ingredient 1: Function Space



Ingredient 1: Function Space

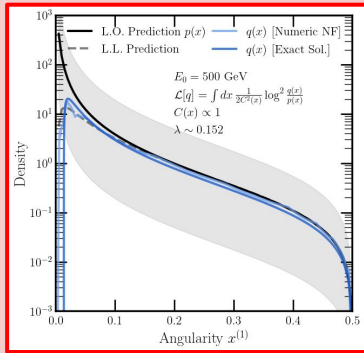


We can approximate large classes of functions using **universal function approximation theorems** and **neural networks**, and build in useful properties!

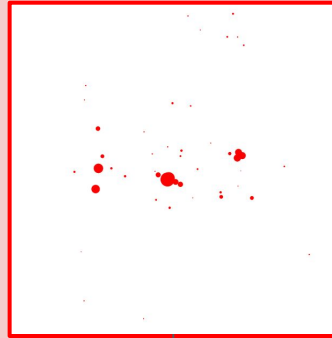
We can search through classes of functions and perform fits through **numerical minimization**, and analyze what we learn using functional analysis!

Things a **machine** can understand

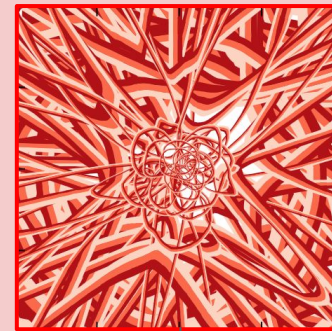
Nontrivially constrained function spaces



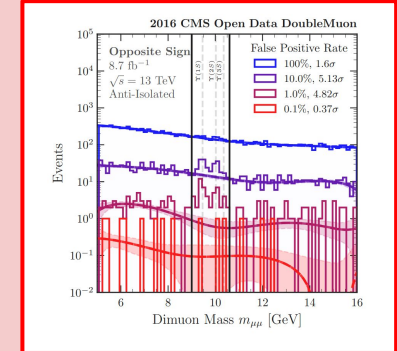
Complicated point clouds



Huge Latent Spaces



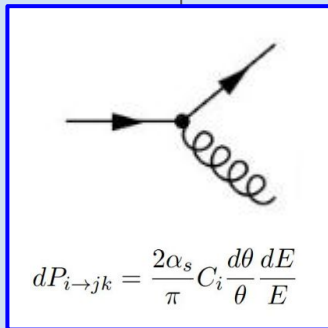
Optimal Statistical Analyses



Interface: Targeted and goal-motivated ML function spaces and loss functionals!

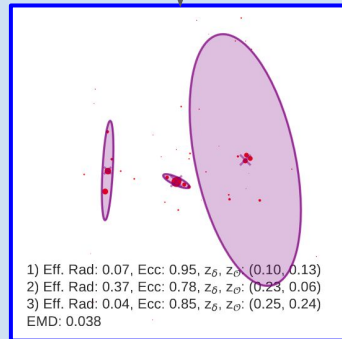
Using Normalizing
Flows

And a perturbative
matching Loss



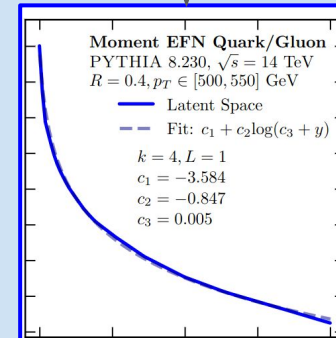
Tree-level Feynman
Diagrams

Using faithful
optimal transport



Basic Kindergarten Shapes

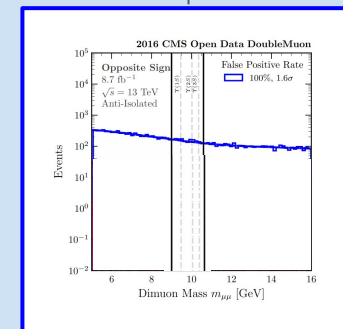
Using Moment
Pooling



Addition, multiplication, and a few
elementary functions

Using Normalizing
Flows

And a
Classifier



The fact that Upsilon's
sometimes decay to muons

Things my **mortal physicist brain** can understand

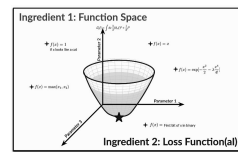
Outline

This talk is going to be **several mini-vignettes** about applying ML to jet physics.

Key takeaway: By enforcing physical and statistical principles and keeping your end goal in mind, we can design loss functionals and models to help us encode precisely the physics we want to learn! We have *full control* over our physics.

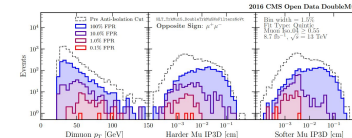
Part 0
(You are here)

Introduction



Part I

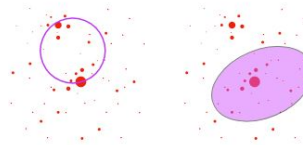
Isolating Unisolated Upsilon's: Finding Particles



[RG, Mastandrea, Nachman, Thaler;
[2502.14036](#)]

Part II

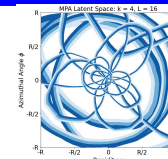
Can you Hear the Shape of a Jet?



[Ba, Dogra, RG, Tasissa, Thaler,
[2302.12266](#)]
[RG, Larkoski, Thaler,
[2410.05379](#)]

Part III

Moment of Clarity: Streamlining Latent Spaces



[RG, Osathapan, Thaler,
[2403.08854](#)]

Part IV

Renormalizing Flows: Taming Perturbation Theory

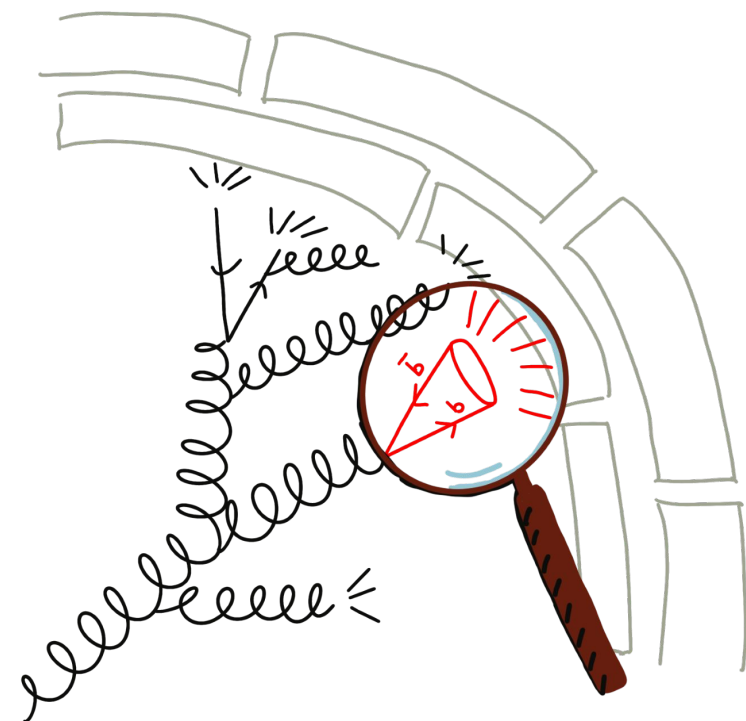
[RG, Mastandrea, WIP]

$$\exp\left[\alpha \frac{d}{d\alpha}\right] p(x) = p^0(x) + \alpha^1 q^1(x) + \dots + \alpha^N p^N(x) + \mathcal{O}(\alpha^{N+1})$$

$$\exp\left[\alpha \frac{d}{d\alpha}\right] q(x) = q^0(x) + \alpha^1 q^1(x) + \dots + \alpha^N q^N(x) + \mathcal{O}(\alpha^{N+1})$$

equal!

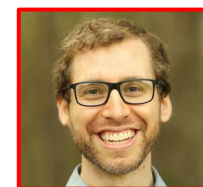
Part I *Isolating* **Unisolated** **Upsilons**



Anomaly Detection in CMS Open Data

In Collaboration with: Radha Mastandrea, Ben Nachman, and Jesse Thaler

Download and try it!

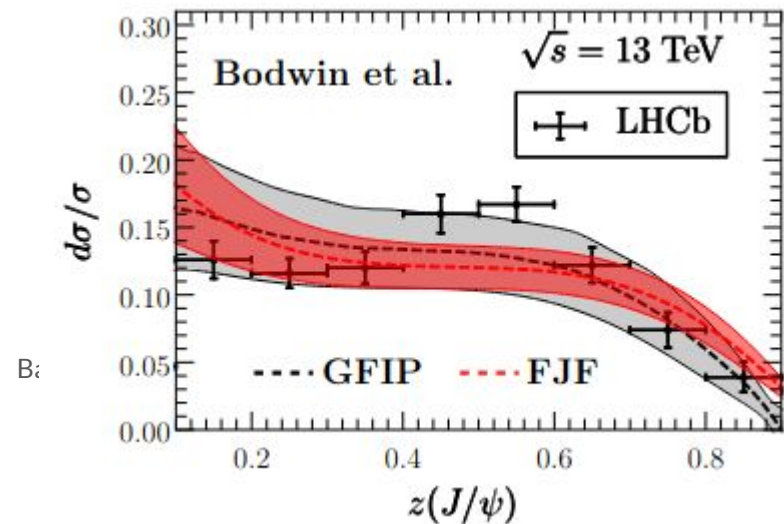
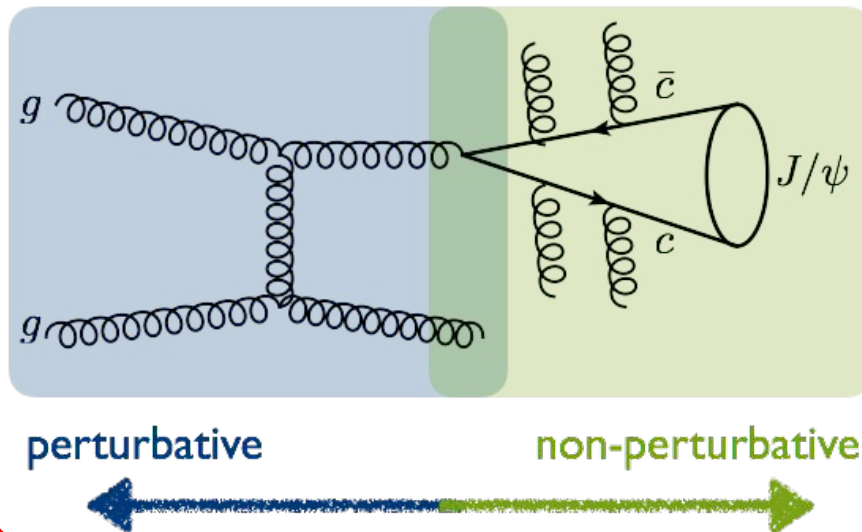


Upsilons Inside Jets

Production of heavy **quarkonia** (quark-antiquark) states in **jets** is an important probe of QCD.

Studies of J/ψ (charm-anticharm) production in jets has improved our understanding of (NR)QCD fragmentation functions.

Upsilons (bottom-antibottom) inside of jets (**anti-isolated**) have yet to be probed. They are rarer, and anti-isolated is an unusual channel!



Upsilons Inside Jets

Production of heavy **quarkonia** (quark-antiquark) states in **jets** is an important probe of QCD.

Studies
of QCD

Upsilon
They are

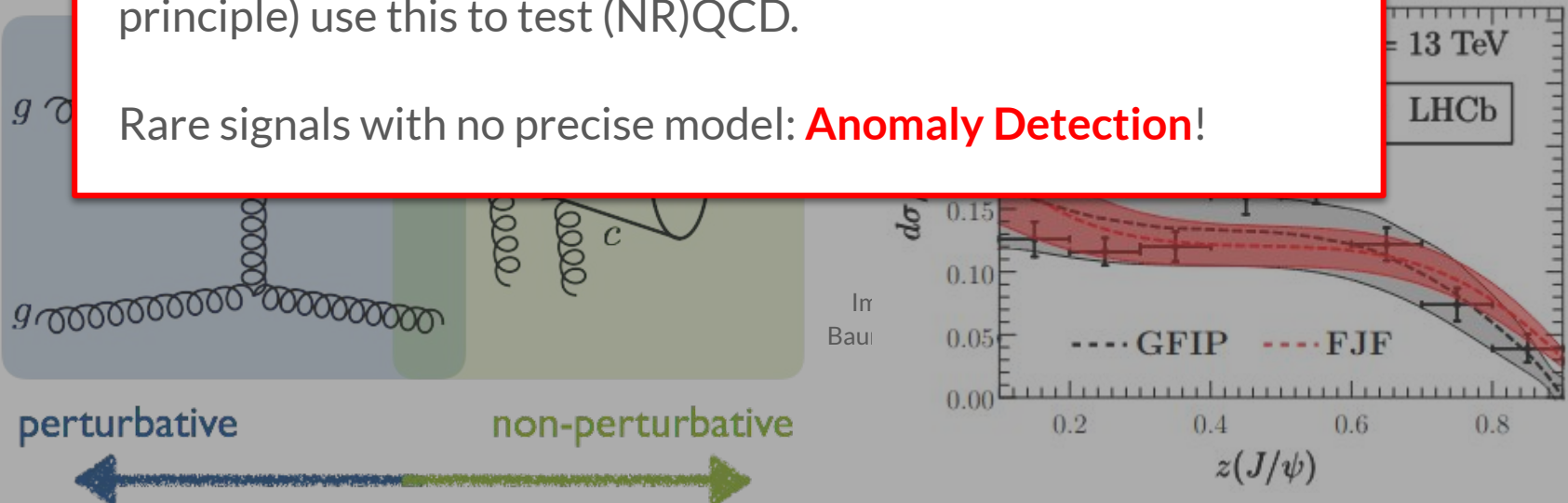
Understanding

probed.

Our Goal: Find anti-isolated upsilons in real data.

Moreover, we want to find them in a simulation-free or prior-independent way, since (a) heavy quarkonia fragmentation is difficult to simulate and (b) we can later (in principle) use this to test (NR)QCD.

Rare signals with no precise model: **Anomaly Detection!**



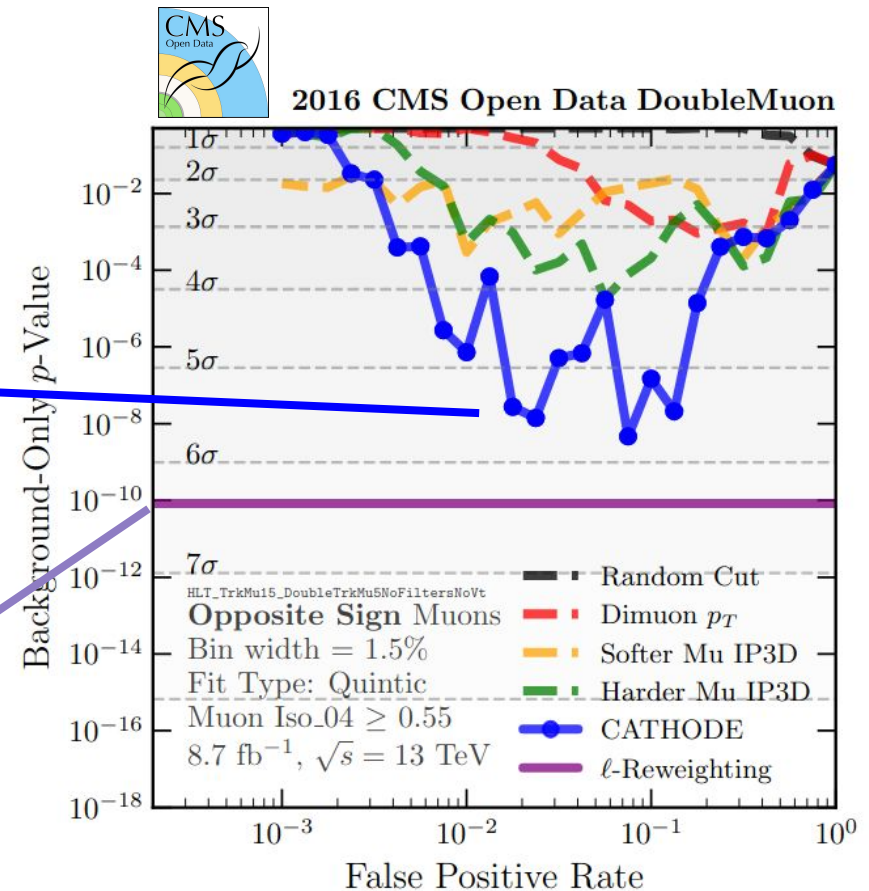
We found them.

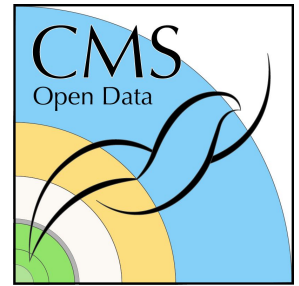
We find $>5\sigma$ evidence of anti-isolated upsilons decaying to dimuons in **CMS Open Data**.

The trick: We use **ML-based Anomaly Detection** (CATHODE¹) to look for subtle signals in multi-dimensional data.

We can use the relationship between ML classifiers and likelihood ratios to **increase the significance further!**

No simulation is used whatsoever. We can find nontrivial signals in **real data!**





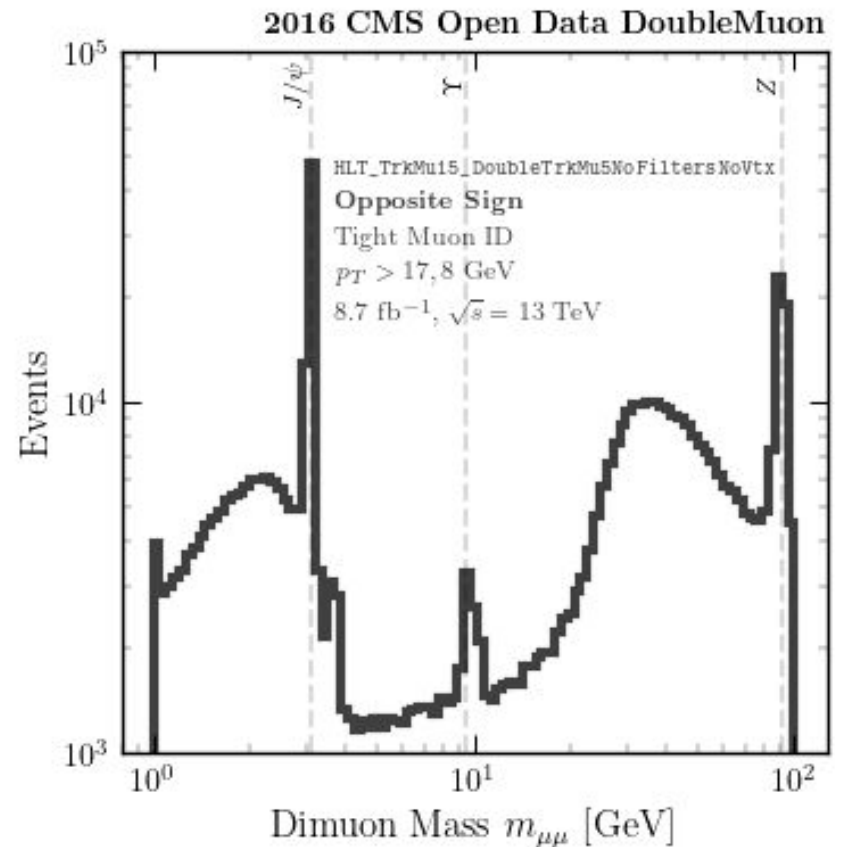
The Dataset: CMS Open Data

We look at 2016 CMS OpenData, corresponding to 13 TeV collisions with an integrated luminosity of 8.7fb^{-1} .

Trigger: HLT_TrkMu15_DoubleTrkMu5NoFiltersNoVtx, meaning 2 tracker muons with tracker $p_T > 15$ GeV and 5 GeV respectively.

Cuts: We require that the hardest two muons are oppositely charged**, meet tightID cuts, and have $p_T > 17$ GeV and 8 GeV respectively.

Result: A dataset of clean dimuon events to look for resonances in!



*The full 2016 run is $\sim 32\text{fb}^{-1}$, but only $\sim$ half is published as Open Data and another half is eaten by triggers.

**We also keep same-sign muons for extensive validation studies. Ask me about this later!

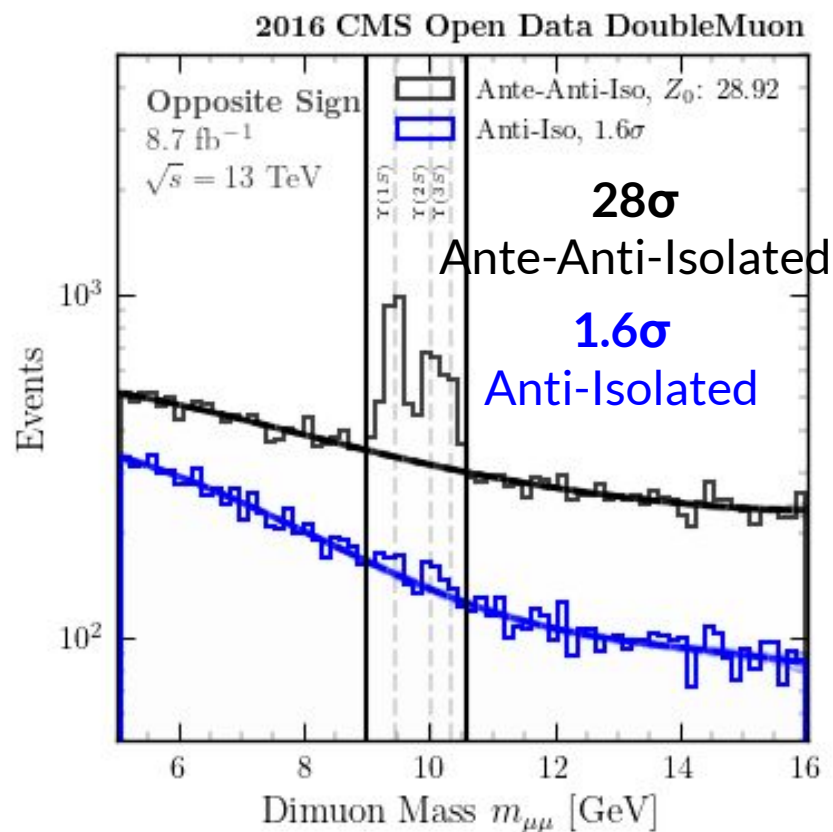
The Dataset: Anti-Isolation

We require that each muon is *anti-isolated*, meaning that each muon is surrounded by hadronic junk.

Precisely: The amount of hadronic p_T in a radius $\Delta R = 0.4$ should be at least 0.55 times the muon p_T

We emphasize that this is the *opposite* of what is usually done in dimuon BSM searches!

Result: The anti-isolated upsilon resonances are barely visible at 1.6σ . A mere statistical fluke!



*The full 2016 run is $\sim 32\text{fb}^{-1}$, but only $\sim$ half is published as Open Data and another half is eaten by triggers.

**We also keep same-sign muons for extensive validation studies. Ask me about this later!

The Dataset: Anti-Isolation

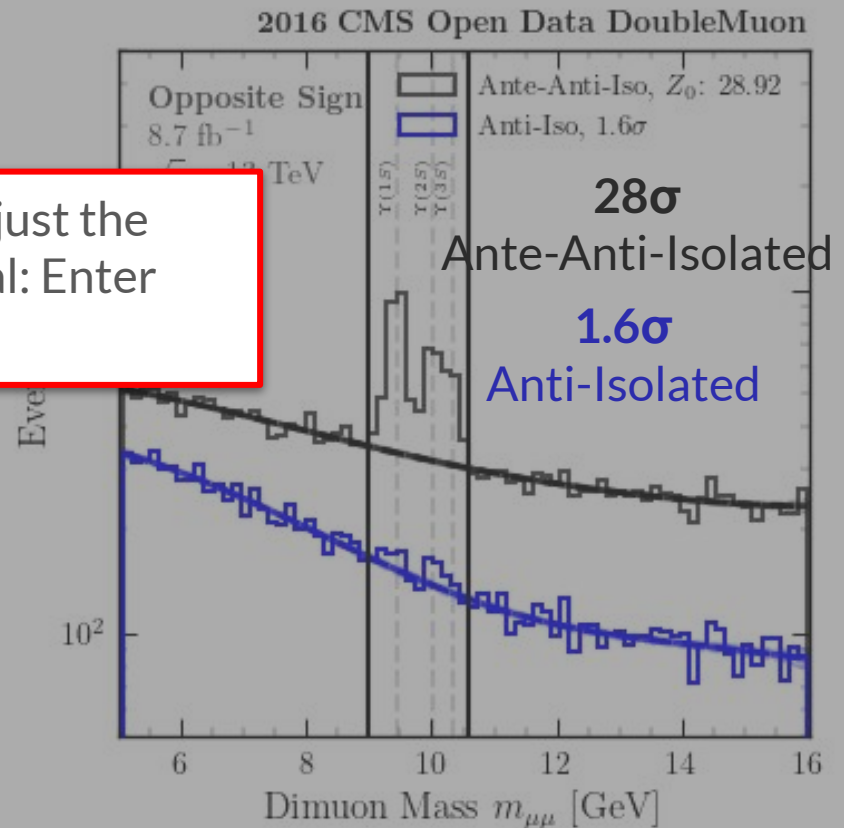
We require that each muon is *anti-isolated*, meaning that each muon is surrounded by hadronic junk.

Precisely: The radius $\Delta R = (\Delta\eta^2 + \Delta\phi^2)^{1/2}$ is less than 1.5 times the muon transverse momentum.

We emphasize that this is the *opposite* of what is usually done in dimuon BSM searches!

Result: The anti-isolated Υ resonances are barely visible at 1.6σ . A mere statistical fluke!

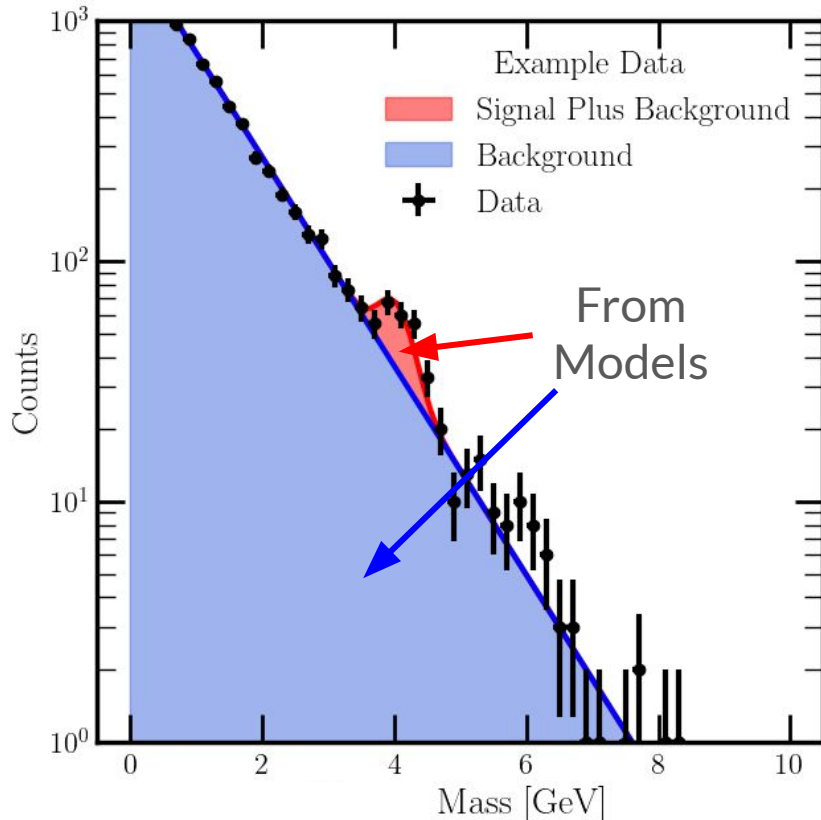
We need to use features beyond just the dimuon mass to elevate this signal: Enter **Anomaly Detection**.



*The full 2016 run is $\sim 32\text{fb}^{-1}$, but only $\sim$ half is published as Open Data and another half is eaten by triggers.

**We also keep same-sign muons for extensive validation studies. Ask me about this later!

Anomaly Detection Review



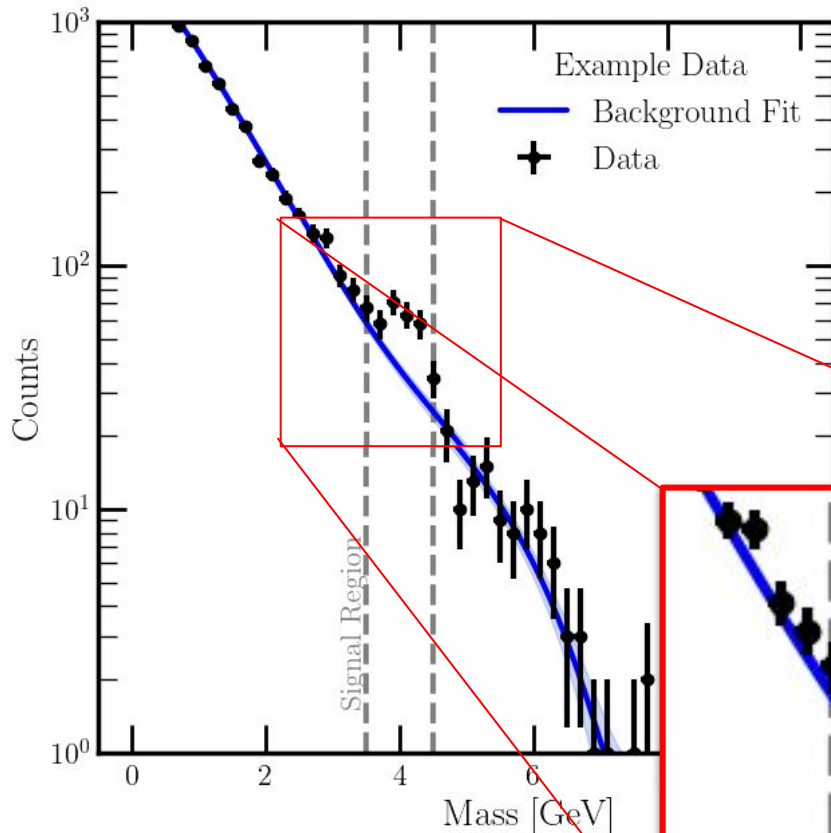
Typically, one wants to compare data to a **signal hypothesis*** versus a **background (null) hypothesis**.

Normally, one has a parametric form or simulations for both hypotheses.

In Anomaly Detection, we *don't* have a direct specification of these hypotheses. Instead, we implicitly and broadly define them via assumptions and fits.

*I will take "signal hypothesis" to really mean "signal plus background hypothesis".

Anomaly Detection Review: Bump Hunts



Assume the **signal** is resonant: It lives in a narrow **signal region**.

Assume it lives atop a **smooth background**, as defined via fit outside the signal region and interpolated into the signal region. This is our null hypothesis

We can then compare the

Observed Data

To the

Expected Background

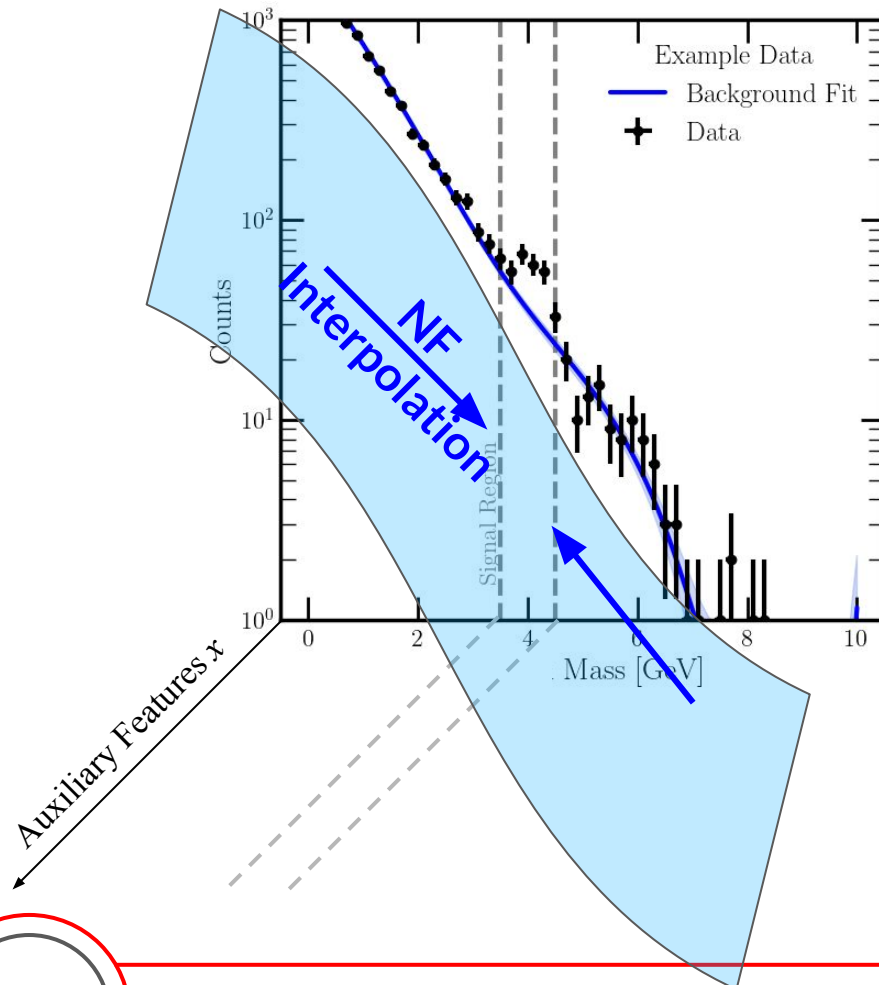
To see if they match or not!

Anomaly Detection Review: CATHODE

Let's do it again, but with more dimensions! We still assume we are resonant in 1 dimension, but other features might have useful info.

Step 1: Background fit in the sidebands and interpolation with a Normalizing Flow (a multidimensional distribution fitter)

Step 2: Compare data to interpolated background* using a BDT classifier (a multidimensional comparison method). Cutting on the classifier score reduces the problem back to an ordinary bump hunt.



*Technical Detail: Interpolated background = Generated pseudodata from the NF

The Classifier

We train a BDT to distinguish between real data and interpolated background.

Fact: Distinguishing (data vs. background) is equivalent to distinguishing (signal vs. background).

$$\begin{aligned}\text{Classifier Score} &= \frac{p_{\text{Data}}(x)}{p_{\text{Bkg}}(x)} \\ &= \mu \frac{p_{\text{Sig}}(x)}{p_{\text{Bkg}}(x)} + (1 - \mu)\end{aligned}$$

Higher score means more signal-like.

If we *can't* distinguish, then the two classes are the same → no signal!

view: CATHODE

Let's do it again, but with more dimensions! We still assume we are resonant in 1 dimension, but other features might have useful info.

Step 1: Background fit in the sidebands and interpolation with a Normalizing Flow (a multidimensional distribution fitter)

Step 2: Compare data to interpolated background* using a BDT classifier (a multidimensional comparison method). Cutting on the classifier score reduces the problem back to an ordinary bump hunt.

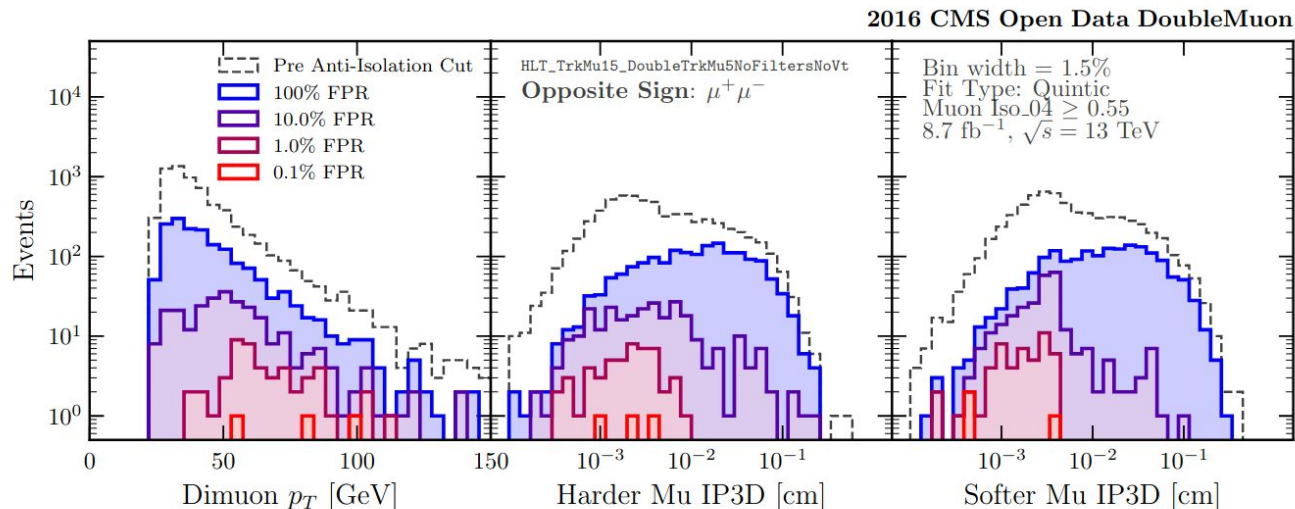
*Technical Detail: Interpolated background = Generated pseudodata from the NF

Features

There are lots of choices for auxiliary features sets! We choose the **dimuon p_T** and the **3D Impact Parameter of each muon (IP3D)**.

This set is motivated by:

- **No Sculpting**: We cannot choose a feature set that sculpts a peak
- **Numerics**: Too many features or non-informative features can ruin ML convergence.
- **Physics Priors**: An IP3D cut removes uncorrelated hadron decays, and a p_T cut removes Drell-Yan and QCD backgrounds



Features

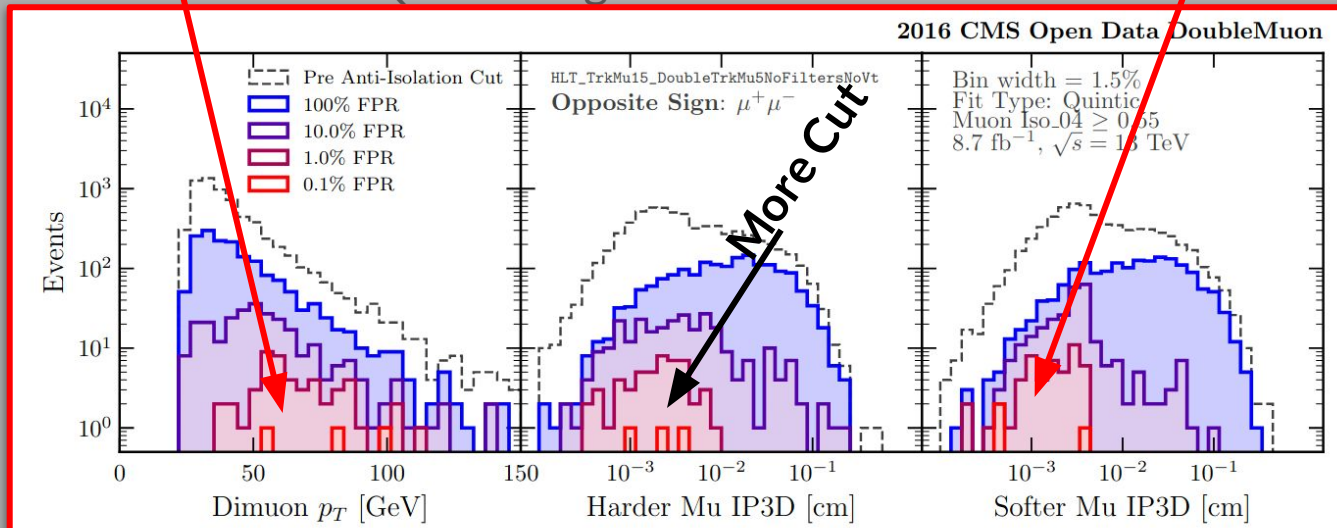
There are lots of choices for auxiliary features sets! We choose the **dimuon p_T** and the **3D Impact Parameter of each muon (IP3D)**.

This set is motivated by:

- **No Sculpting:** We cannot choose a feature set that sculpts the background distribution.
- **Numerics:** The features or non-informative features are not correlated with the signal.
- **Physics:** The IP3D cut removes uncorrelated hadron decays. The p_T cut removes Drell-Yan and QCD backgrounds.

Higher p_T rejects backgrounds

Quarkonia Formation is Prompt

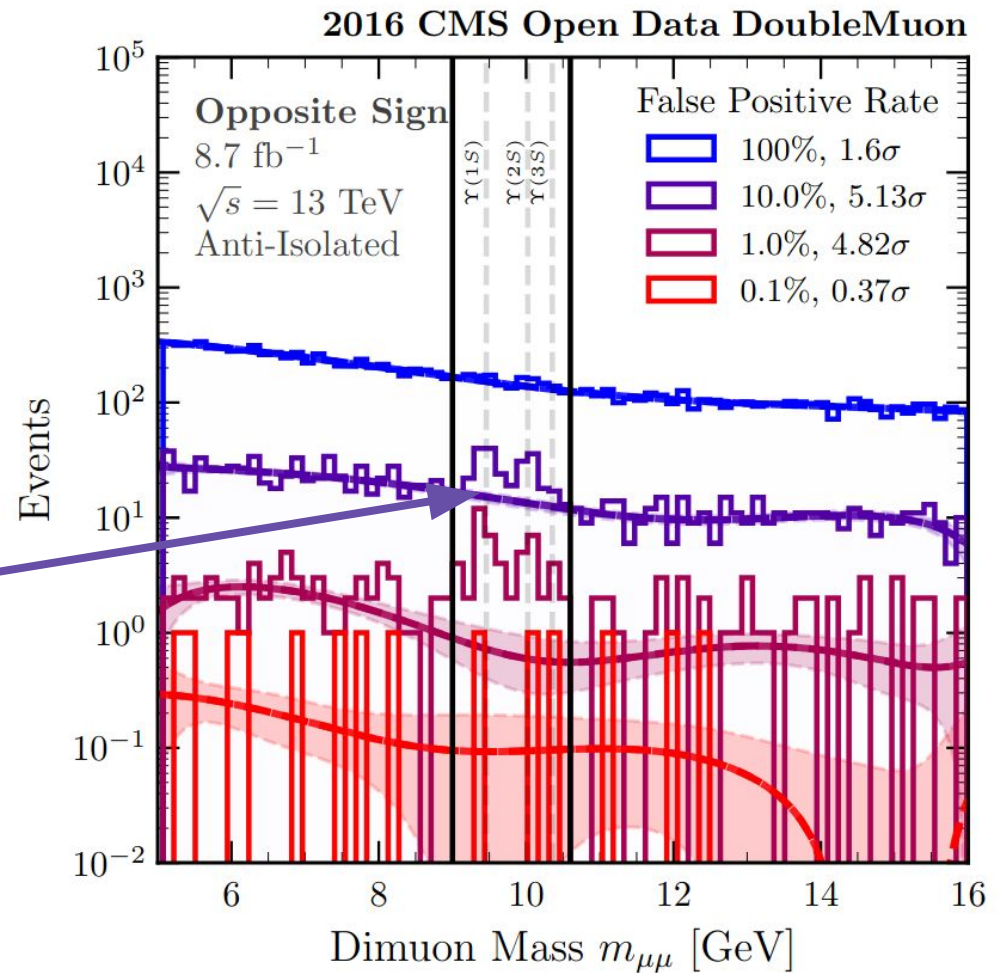


Finding Real Signals with CATHODE

Because **signal** (upsilons) is different from **background**, cutting on the classifier will elevate the signal!

A ~10% cut can significantly elevate the signal to over 5σ !

We can declare **victory**! We found the upsilons



... But we can do **better than victory!**

The classifier score isn't just an uninterpretable number to score events!

We are just doing fancy functional optimization with a known loss functional and function space with BDTs: **Neyman-Pearson** tells us we learn the likelihood ratio!

$$\ell(x) \equiv \frac{\overset{\text{Classifier score}}{z(x)} - (1 - \overset{\text{Naive signal estimate}}{\mu})(1 - z(x))}{\mu(1 - z(x))} = \frac{p_{\text{Sig}}(x)}{p_{\text{Bkg}}(x)}$$

We can show that by *weighing* events using the likelihood ratio *directly* rather than cutting on the classifier, we can achieve *much* more sensitivity!

Think of *cutting* as a special case of weighting: a weight of 0 is cut, and a weight of 1 is kept. By using weights, we allow more gradation.

... But we can do **better than victory!**

The classifier score isn't just an uninterpretable number to score events!

We are just doing fancy functional optimization with a known loss functional and function space with BDTs: **Neyman-Pearson** tells us we learn the likelihood ratio!

$$\ell(x) \equiv \frac{\overset{\text{Classifier score}}{z(x)} - (1 - \overset{\text{Naive signal estimate}}{\mu})(1 - z(x))}{\mu(1 - z(x))} = \frac{p_{\text{Sig}}(x)}{p_{\text{Bkg}}(x)}$$

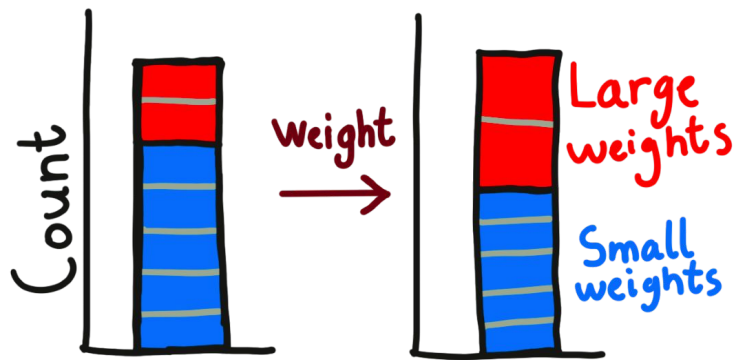
We can show that by *weighing* events using the likelihood ratio *directly* rather than cutting on the classifier, we can achieve *much* more sensitivity!

This is essentially a *free* increase in sensitivity just by using the fact that classifiers learn the likelihood ratio, plus we don't have to pick a cut working point!.

The worst thing to do with a classifier is classify!

Why should this work?

For simplicity, assume we are doing a single-bin counting experiment in the asymptotic limit.* Let's allow every event to have a weight w^{**} .



The total number of events is now:

$$S_{\text{eff}} = \int dx S(x)w(x), \quad B_{\text{eff}} = \int dx B(x)w(x)$$

The “significance” is:

$$Z = \frac{\int dx S(x)w(x)}{\sqrt{\int dx B(x)w^2(x)}} \quad (\text{Generalizes } Z = \frac{S}{\sqrt{B}})$$

The Likelihood Ratio!

We can then calculate what choice of weights optimizes the significance:

$$w(x) = \frac{S(x)}{B(x)} = \frac{N_{\text{Sig}}}{N_{\text{Bkg}}} \frac{p_{\text{Sig}}(x)}{p_{\text{Bkg}}(x)}$$

Unimportant Constants

*In the full analysis, we do a complete Poisson likelihood treatment with nuisance parameters due to fits.

When these are included, these weights may not be strictly optimal, but they are still better than nothing!

**Ordinary cutting can be seen as assigning a weight of 0.

The worst thing to do with a classifier is classify!

Final Results

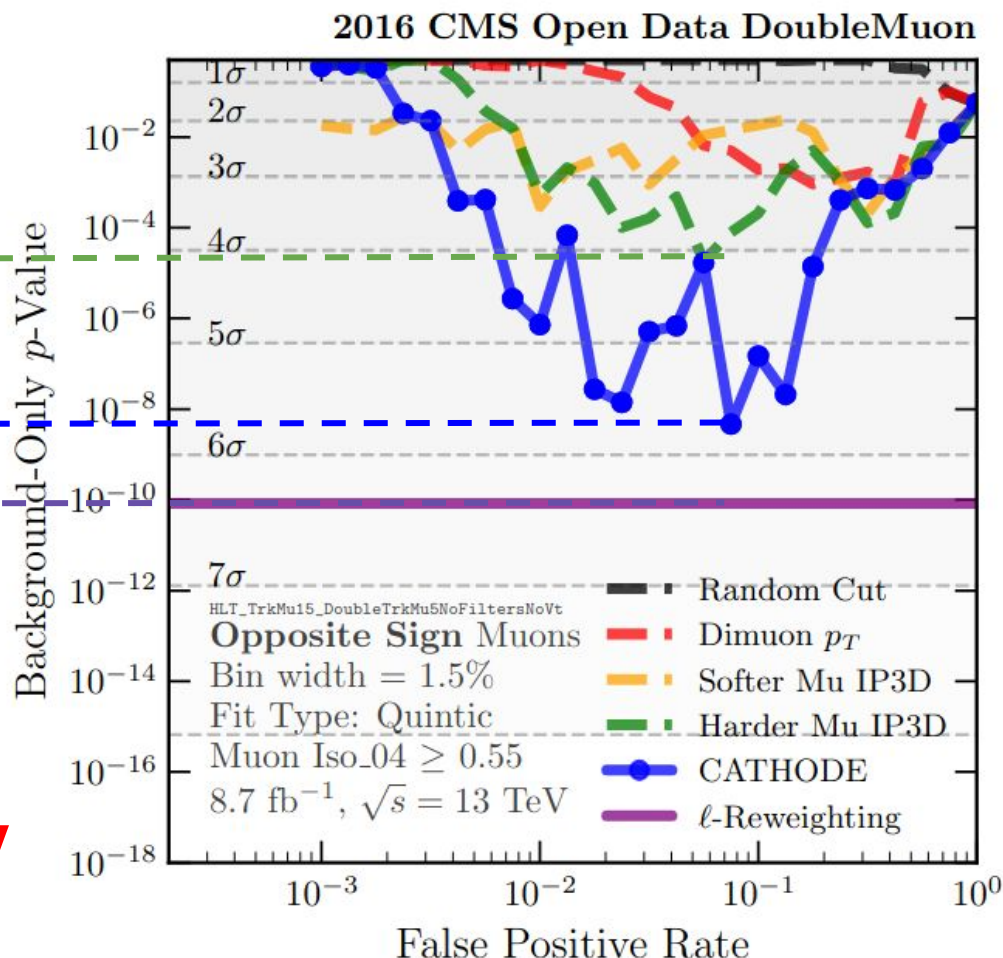
Best “naive” single-feature analysis: 4.1σ

Best cut CATHODE: 5.7σ

Neyman-Pearson Weighting: 6.4σ

We can find Upsilon
with CATHODE, and we
can find them *even better*
with weighting!

Down = Better!



*Also, weighting doesn't require a cut choice, so there is no Look-Elsewhere-Effect scanning over cut choices

Closing Thoughts

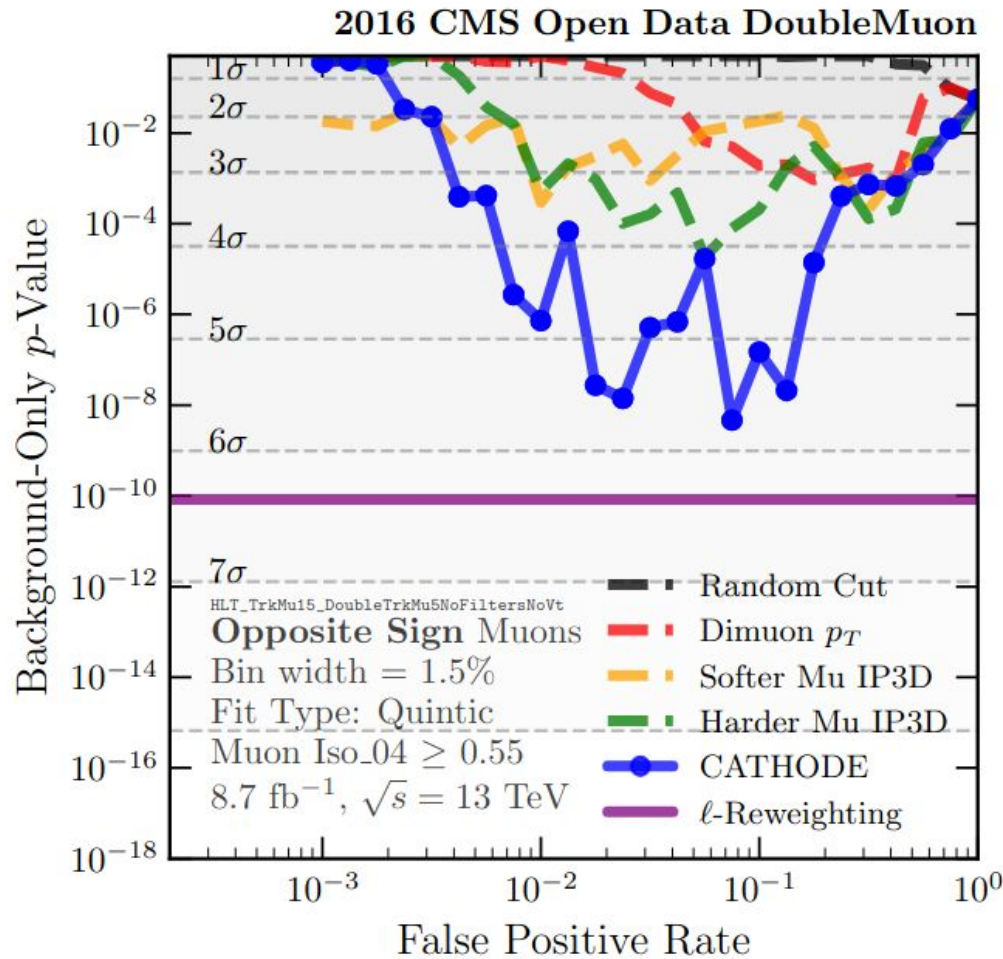
- **BSM Scan:** We showed AD can find real SM signals in real dimuon data. We are now trying to find real BSM signals in a follow up.
- **Energy Fraction Measurement:** An object of interest in QCD is the quarkonia energy fraction $z = (p_T) / (p_{T, Jet})$, which governs fragmentation and has been measured for J/ψ . In principle, now that we have *found* the Upsilon, this can be measured with a full unfolding.
- **Anomaly Detection Benchmarking:** AD Methods have primarily been tested on developed on toy datasets. We made the curated and formatted dimuon sample public so that AD can be developed on real data with the Upsilon as a “Standard Candle”.

<https://zenodo.org/records/14618719>

Download and try it!



Conclusion



- AD can be used to find *real, nontrivial signals* in *real data*, like hard-to-find upsilons.
- Likelihood reweighting can improve sensitivity even further

Conclusion

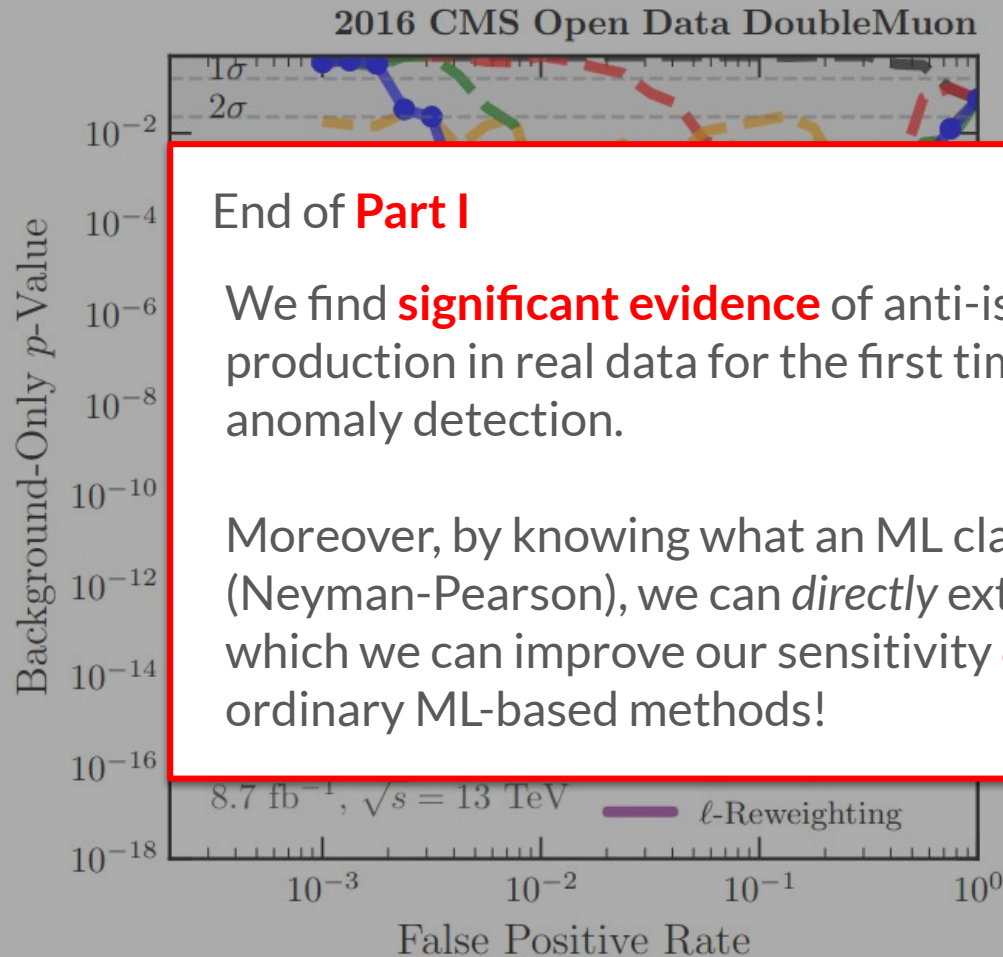
Isolating Unisolated
Upsilon: Finding
Particles

Can you Hear the
Shape of a Jet?

The Moments of
Clarity: Streamlining
Latent Spaces

Renormalizing Flows:
Taming Perturbation
Theory

Conclusion



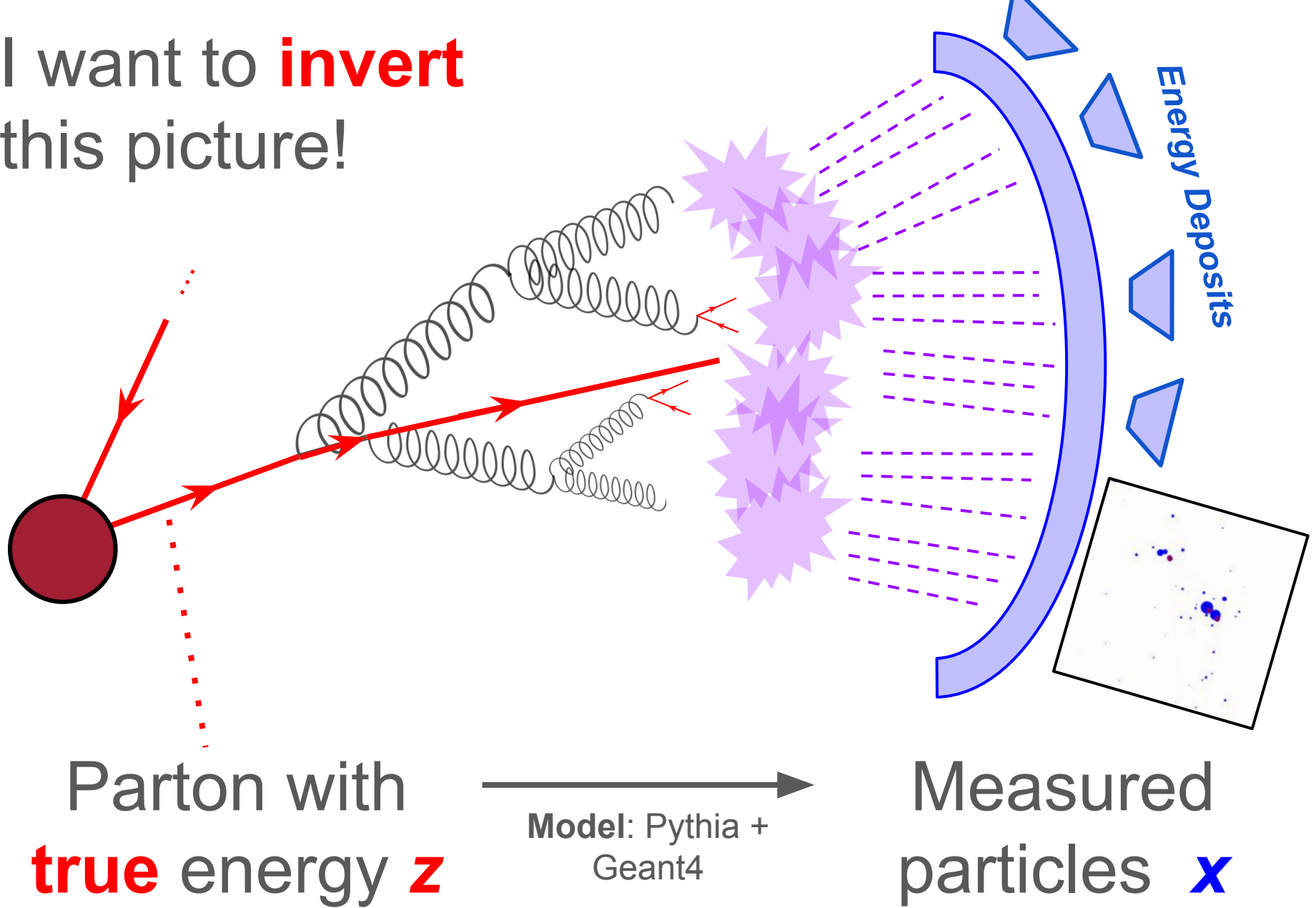
End of Part I

We find **significant evidence** of anti-isolated Upsilon production in real data for the first time, by using ML-based anomaly detection.

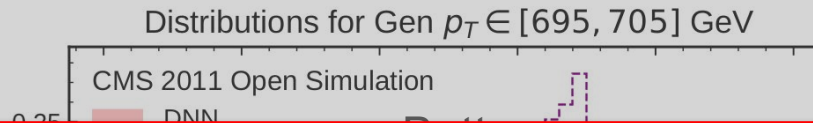
Moreover, by knowing what an ML classifier *actually learns* (Neyman-Pearson), we can *directly* extract the likelihood ratio, which we can improve our sensitivity **even further** beyond ordinary ML-based methods!

- Likelihood reweighting can improve sensitivity even further

I want to **invert**
this picture!



Example 2: Jet Energy Resolution



End of **Part I**

By choosing:

1. A Gaussian-like functional space
2. A loss that guarantees prior independence

We can perform inference and uncertainty estimation in a manifestly robust way, while taking advantage of the power of machines to crunch a lot of high-dimensional data!

Best of both worlds: Using ML to extract more information* than hand-crafted features, while still being able to extract resolutions in a prior-independent and unbiased way!

*I mean “information” literally. Ask me later about the cool information theory of the DV loss!

Part II

Can you Hear the **Shape** of a Jet?



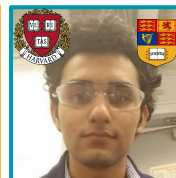
Download
SHAPER!

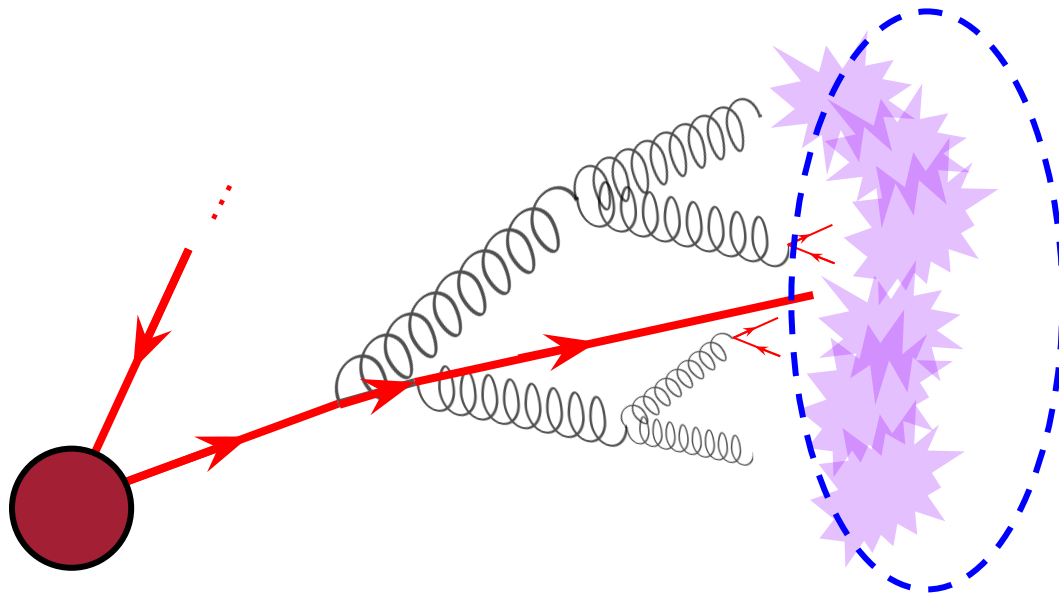


Download
SPECTER!

Try *pip install pyshaper*

(New!) *pip install specterpy*

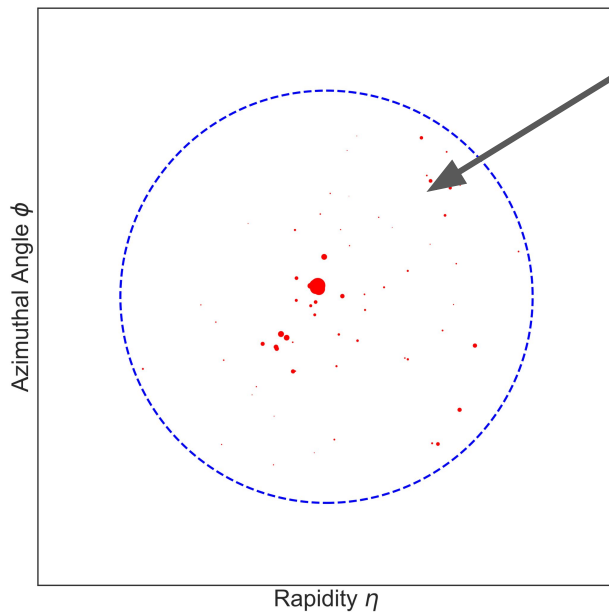




Jets are messy collections of 100's of particles — a lot of information to measure and predict!

Can we come up with “summary statistics” (**Observables**) that characterize jets and are easier to work with?

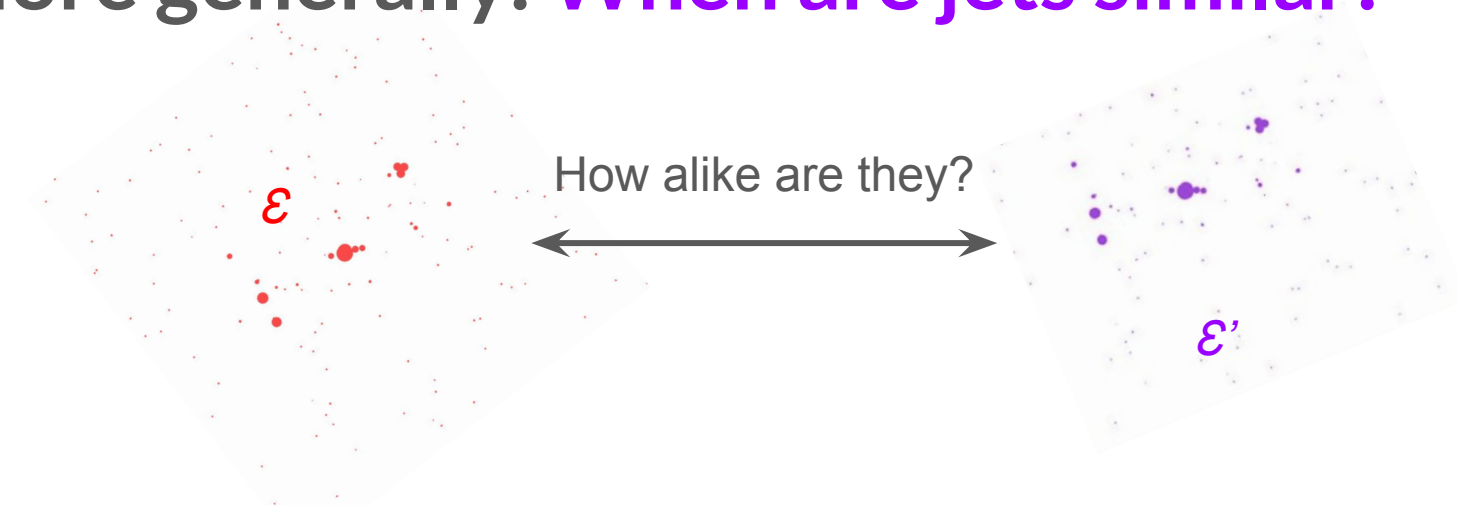
e.g. How “*big*” is this jet?
... What do we even mean by the “size” of a jet?



By following this line of thinking, we'll be able to come up with a new class of QCD observables!

I don't mean the fixed size of a jet algorithm, e.g. AK4, but rather a dynamical notion of a jet size!

More generally: When are jets similar?

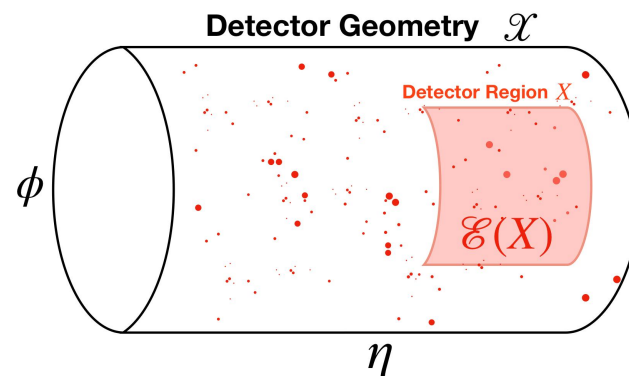


Jets can be represented as **point clouds** – let's scour through the ML and computer vision literature for a metric on point clouds we like!*

$$\mathcal{E}(y) = \sum_i z_i \delta(y - y_i)$$

Detector Coordinate (η, ϕ)

Energy Fraction E_i / E_{Tot}



*Or better yet, construct one!

Loss: The Wasserstein Metric

Let's demand the following reasonable properties of our metric on point clouds:

1. ... is a genuine metric: nonnegative, nondegenerate, finite
2. ... is **IRC-safe**^{*} (Calculable and robust)
3. ... is translationally invariant
4. ... is invariant to particle labeling
5. respects the detector metric **faithfully**^{**}

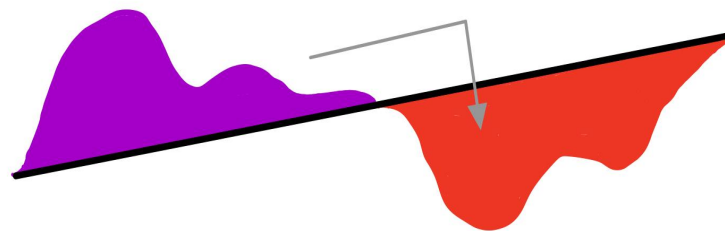
^{*}Ask me for more details on this offline!

^{**}Preserves distances between *extended* objects, not just points

Loss: The Wasserstein Metric

Let's demand the following reasonable properties of our metric on point clouds:

1. ... is a genuine metric: nonnegative, nondegenerate*, finite
2. ... is **IRC-safe*** (Calculable and robust)
3. ... is translationally invariant
4. ... is invariant to particle labeling
5. respects the detector metric **faithfully****



EMD = Work done to move “dirt” optimally

I can prove that the *only* metric that satisfies this is the **Wasserstein Metric / EMD**!

$$\text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}') = \min_{\pi \in \mathcal{M}(\mathcal{X} \times \mathcal{X})} \left[\frac{1}{\beta R^\beta} \left\langle \pi, d(x, y)^\beta \right\rangle \right] + |\Delta E_{\text{tot}}|$$

$$\pi(\mathcal{X}, Y) \leq \mathcal{E}'(Y), \quad \pi(X, \mathcal{X}) \leq \mathcal{E}(X), \quad \pi(\mathcal{X}, \mathcal{X}) = \min(E_{\text{tot}}, E'_{\text{tot}})$$

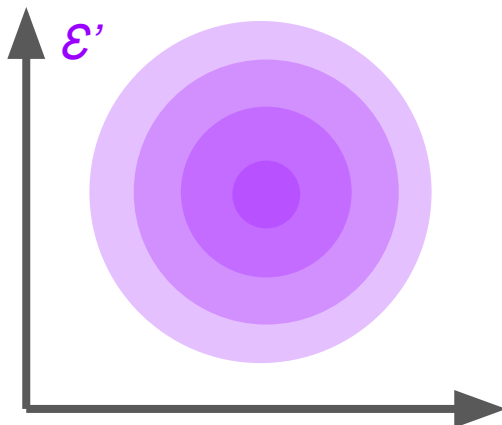
A staple of computer vision ML is also useful for jets!

Other common ML metrics, like Chamfer, MMDs, or Kernel metrics arent *faithful*.

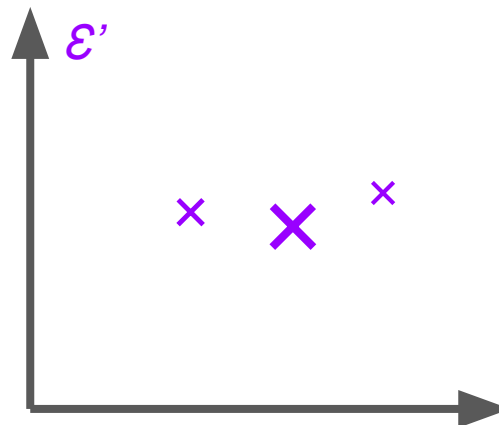
Shapes as Energy Flows

Energy flows don't have to be real events – they can be *any* energy distribution in detector space, or **shape**. In our framework, these shapes will later become our **Function Space**

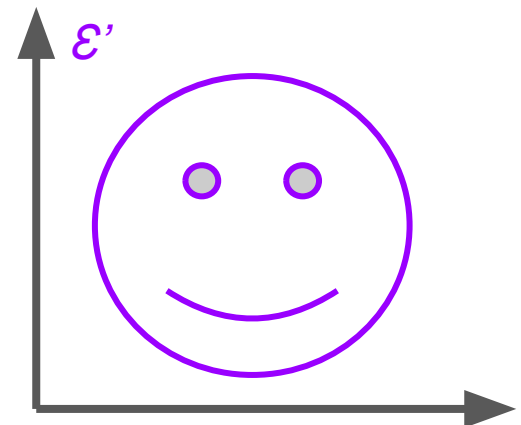
Can make anything you want! Even continuous or complicated shapes.
(Or, something you calculate in perturbative QCD)



Shape = 2D Gaussian



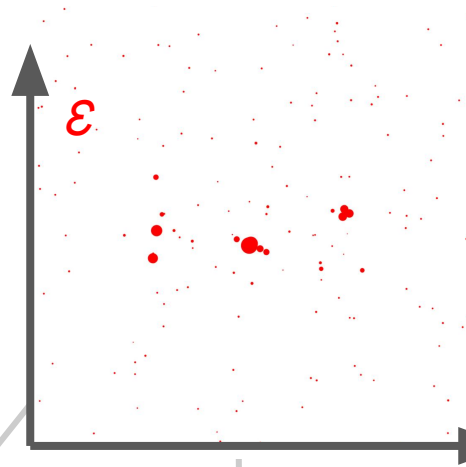
Shape = 3 Points



Shape = Smile

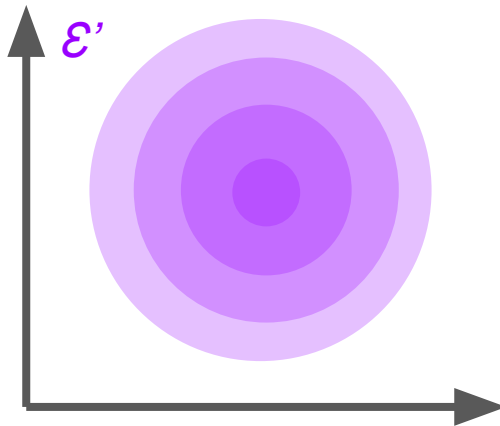
Shapiness

The EMD between a real event or jet $\mathcal{E}$ and idealized shape $\mathcal{E}'$ is the [shape]iness of $\mathcal{E}$ – a well defined IRC-safe observable!



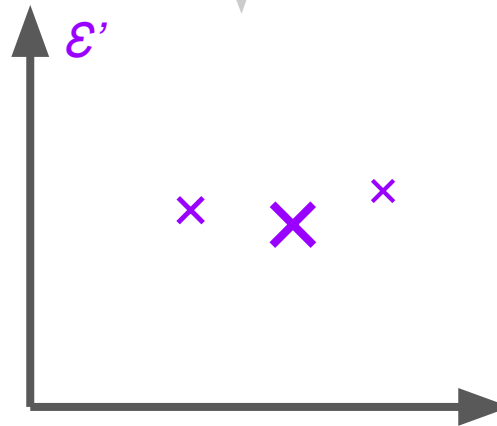
Answers the question:
“How much like the
shape $\mathcal{E}'$ is my event $\mathcal{E}$?”

Med EMD($\mathcal{E}, \mathcal{E}'$)
“Gaussiness”



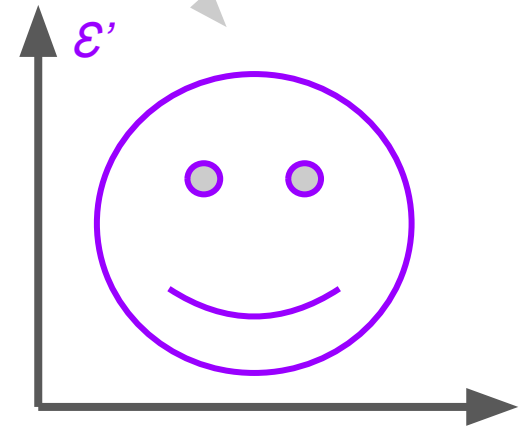
Shape = 2D Gaussian

Low EMD($\mathcal{E}, \mathcal{E}'$)
“3-Pointiness”
AKA “3-Subjettiness”



Shape = 3 Points

High EMD($\mathcal{E}, \mathcal{E}'$)
“Smileyness”



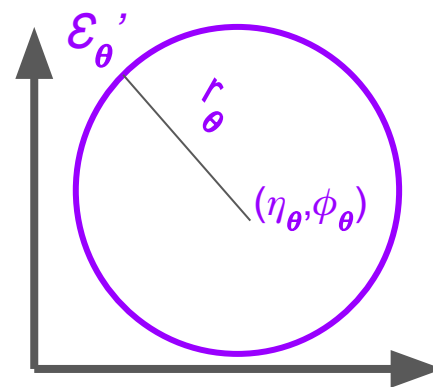
Shape = Smile

Mathematical Details - Shapiness

Rather than a single shape, consider a **parameterized manifold** $\mathcal{M}$ of **energy flows**.

e.g. The manifold of uniform circle energy flows:

$$\mathcal{E}_\theta'(y) = \begin{cases} \frac{1}{2\pi r_\theta} & |\vec{y} - \vec{y}_\theta| = r_\theta \\ 0 & |\vec{y} - \vec{y}_\theta| \neq r_\theta \end{cases}$$



Then, for an event $\mathcal{E}$, define the **shapiness** $\mathcal{O}_{\mathcal{M}}$ and **shape parameters** $\theta_{\mathcal{M}}$, given by:

$$\mathcal{O}_{\mathcal{M}}(\mathcal{E}) \equiv \min_{\mathcal{E}_\theta \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_\theta)$$

$$\theta_{\mathcal{M}}(\mathcal{E}) \equiv \operatorname{argmin}_{\mathcal{E}_\theta \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_\theta)$$

Unlike our previous function space in the last **Part**, we have only a few parameters

Observables $\Leftrightarrow$ Manifolds of Shapes

Observables can be specified solely by defining $\mathcal{O}_{\mathcal{M}}(\mathcal{E}) \equiv \min_{\mathcal{E}_{\theta} \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_{\theta})$,

a **manifold of shapes**:

$$\theta_{\mathcal{M}}(\mathcal{E}) \equiv \operatorname{argmin}_{\mathcal{E}_{\theta} \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_{\theta}),$$

Functional Minimization! Many well-known observables* already have this form!

Observable	Manifold of Shapes
N -Subjettiness	Manifold of N -point events
N -Jettiness	Manifold of N -point events with floating total energy
Thrust	Manifold of back-to-back point events
Event / Jet Isotropy	Manifold of the single uniform event

... and more!

All of the form “How much like **[shape]** does my **event** look like?”

Generalize to *any* shape.

*These observables are usually called event shapes or jet shapes in the literature – we are making this literal!

Hearing Jets Ring

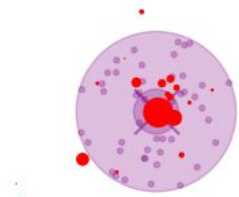
(and Disk, and Ellipse)

$\mathcal{O}_{\mathcal{M}}(\mathcal{E})$ answers: “How much like a **shape** in $\mathcal{M}$ does my **event** $\mathcal{E}$ look like?”

$\theta_{\mathcal{M}}(\mathcal{E})$ answers: “Which **shape** in $\mathcal{M}$ does my **event** $\mathcal{E}$ look like?”

Can define complex manifolds to probe increasingly subtle geometric structure!

Shape	Specification	Illustration
Ringiness $\mathcal{O}_R$	Manifold of Rings $\mathcal{E}_{x_0, R_0}(x) = \frac{1}{2\pi R_0}$ for $ x - x_0 = R_0$ $x_0 = \text{Center}, R_0 = \text{Radius}$	
Diskiness $\mathcal{O}_D$	Manifold of Disks $\mathcal{E}_{x_0, R_0}(x) = \frac{1}{\pi R_0^2}$ for $ x - x_0 \leq R_0$ $x_0 = \text{Center}, R_0 = \text{Radius}$	
Ellipsiness $\mathcal{O}_E$	Manifold of Ellipses $\mathcal{E}_{x_0, a, b, \varphi}(x) = \frac{1}{\pi ab}$ for $x \in \text{Ellipse}_{x_0, a, b, \varphi}$ $x_0 = \text{Center}, a, b = \text{Semi-axes}, \varphi = \text{Tilt}$	
(Ellipse Plus Point)iness	Composite Shape $\mathcal{O}_E \oplus \tau_1$ Fixed to same center x_0	
N-(Ellipse Plus Point)iness Plus Pileup	Composite Shape $N \times (\mathcal{O}_E \oplus \tau_1) \oplus \mathcal{I}$	



Disk + δ -function



Ring + δ -function

Probing **collinear** (δ -function) and **soft** (shape) structure!

Some examples of new shapes you can define!

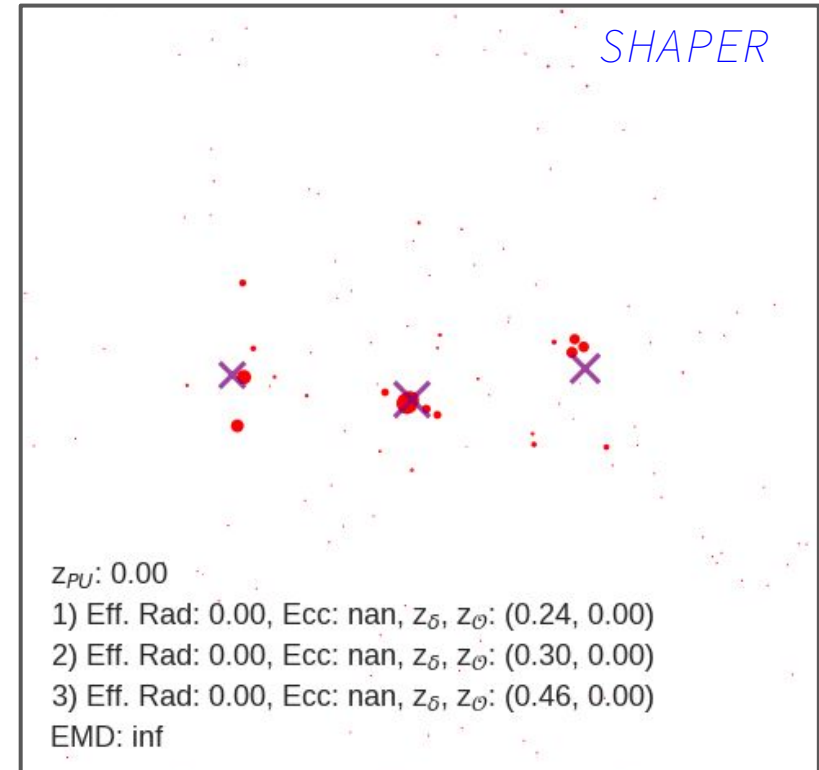
New IRC-Safe Observables

The *SHAPER* framework makes it easy to algorithmically invent new jet observables!

e.g. *N-(Ellipse+Point)iness+Pileup* as a jet algorithm:

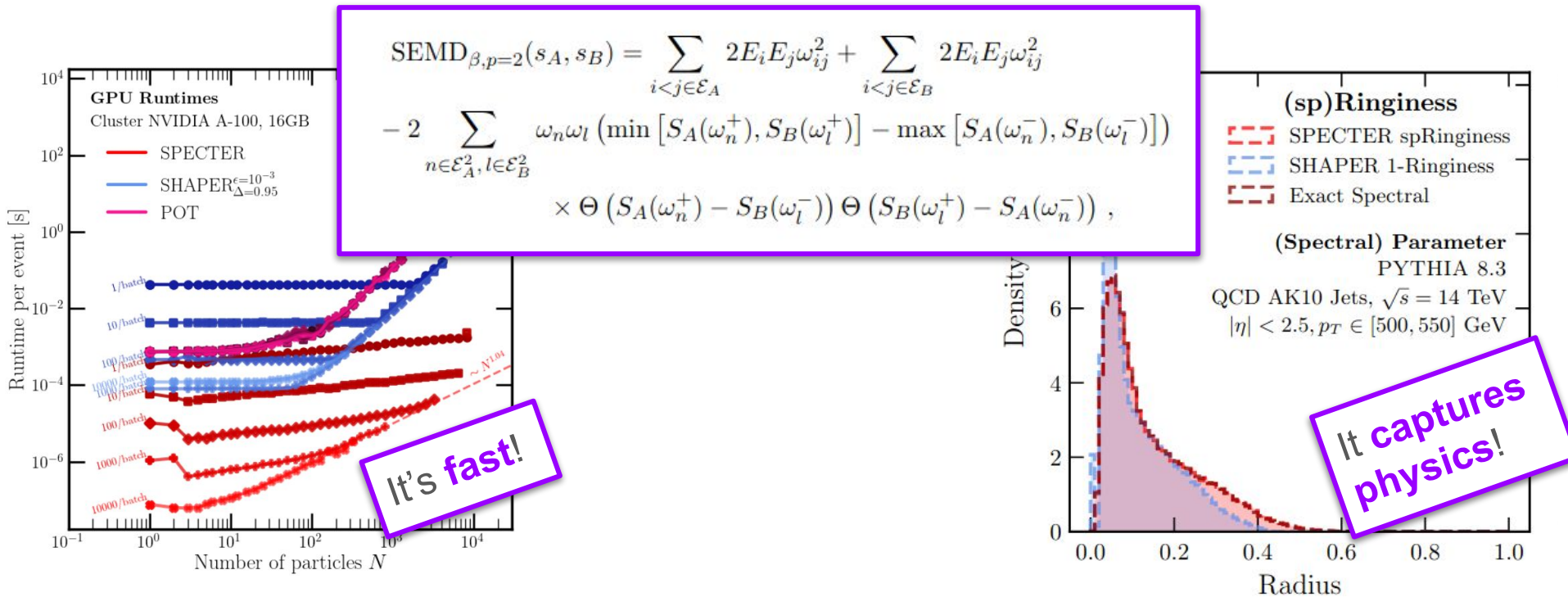
- Learn jet centers + collinear radiation
- Dynamic jet radii (no R parameter!)
- Dynamic eccentricities and angles
- Dynamic jet energies
- Dynamic pileup (no z_{cut} parameter!!!)
- Learned parameters for discrimination

Can design custom specialized jet algorithms to learn jet substructure!



SPECTER: We can do more with OT!

By demanding our loss is only *almost** always a metric on events rather than always a metric, we can build a new loss that is *completely and exactly solvable*!



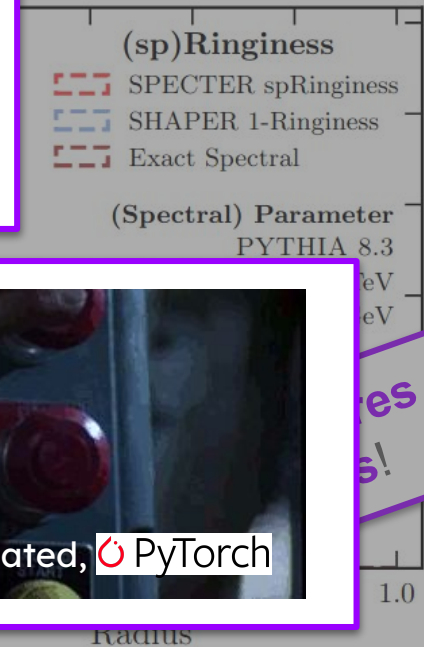
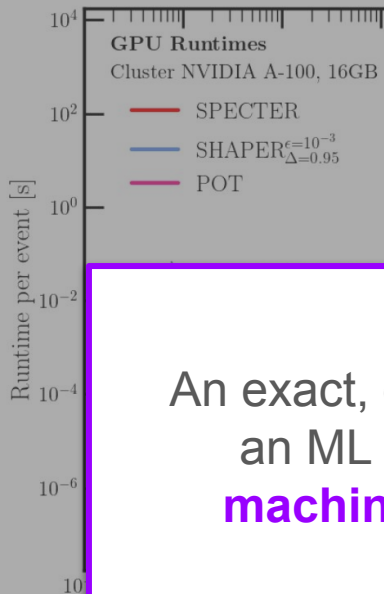
*By *almost*, I mean that there is a measure-0 set of events for which $d(x,y) = 0$ does not imply $x = y$
Ask me later about connections to EECs!



SPECTER: We can do more with OT!

By demanding our loss is only *almost** always a metric on events rather than always a metric, we can build a new loss that is *completely solvable*!

$$\begin{aligned} \text{SEMD}_{\beta,p=2}(s_A, s_B) = & \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 \\ & - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l \left(\min [S_A(\omega_n^+), S_B(\omega_l^+)] - \max [S_A(\omega_n^-), S_B(\omega_l^-)] \right) \\ & \times \Theta (S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta (S_B(\omega_l^+) - S_A(\omega_n^-)) , \end{aligned}$$



An exact, closed form for what started as an ML problem ... can we take the **machine** out of **machine learning**?



[Cameron et. al.; 1984]

*By *almost*, I mean that there is a measure-0 set of events for which $d(x,y) = 0$ does not imply $x = y$
Ask me later about connections to EECs!



[RG, Larkoski, T

Learning
Uncertainties the
Frequentist Way

Can you Hear the
Shape of a Jet?

The **Moments** of
Clarity: Streamlining
Latent Spaces

Isolating Unisolated
Upsilons: Finding
Particles

Conclusion

SPECTER: We can do more with OT!

By demanding our loss is only *almost** always a metric on events rather than always

End of **Part II**

By choosing:

1. A loss that guarantees IRC safety and respects geometry
2. Parametrized distribution spaces (shapes)

We can define new meaningful observables and jet algorithms that are (in-principle) calculable in QCD

*Ask me later about theory space and beta functions!

machine out of **machine learning**?

Your're terminated,  PyTorch

[Cameron et. al.; 1984]

Radius

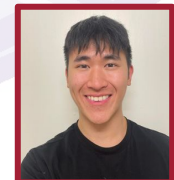
*By *almost*, I mean that there is a measure-0 set of events for which $d(x,y) = 0$ does not imply $x = y$
Ask me later about connections to EECs!

Part III

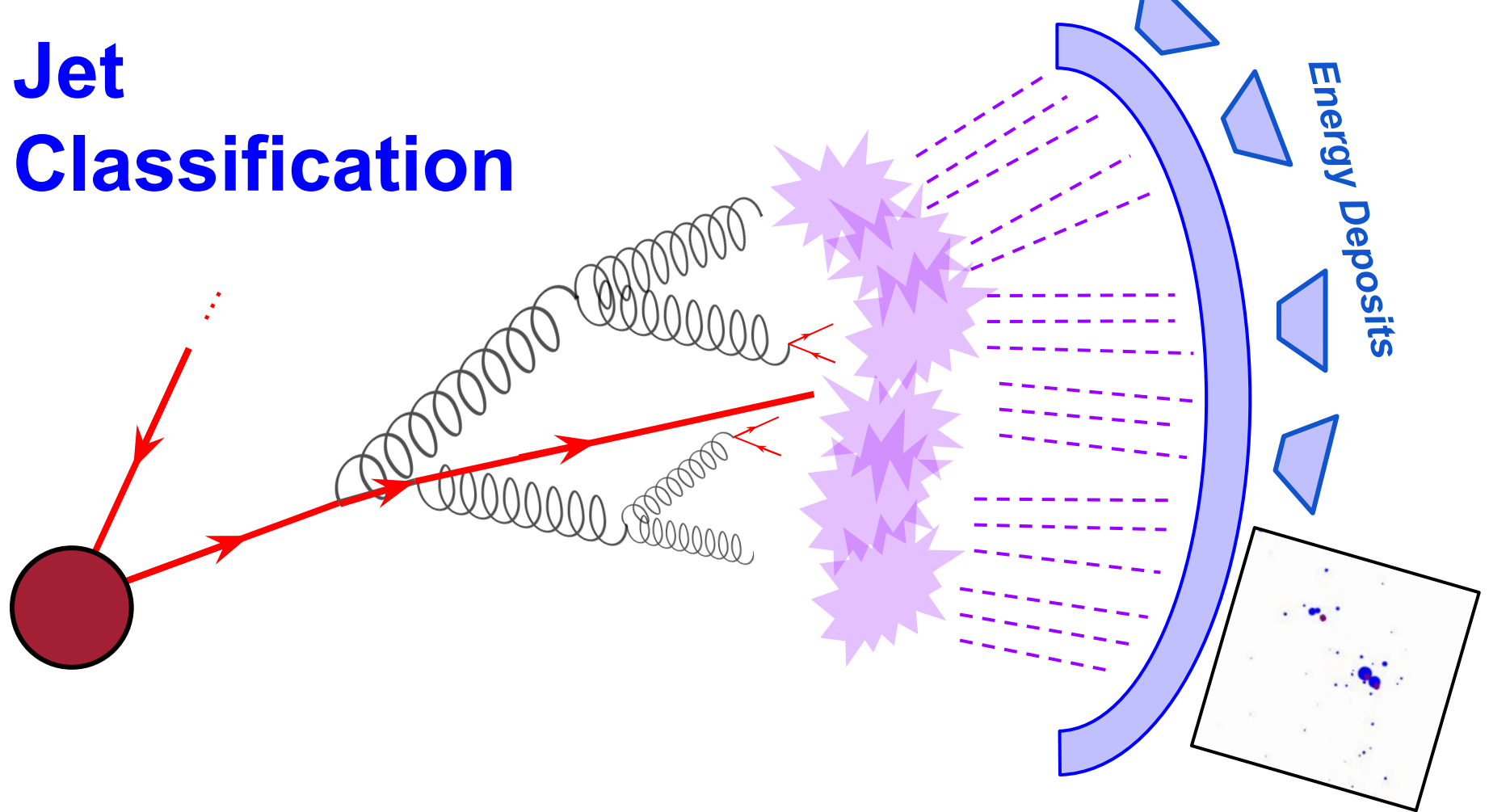
Moments of Clarity: Streamlining Latent Spaces



Download
Moment EFNs!



Jet Classification

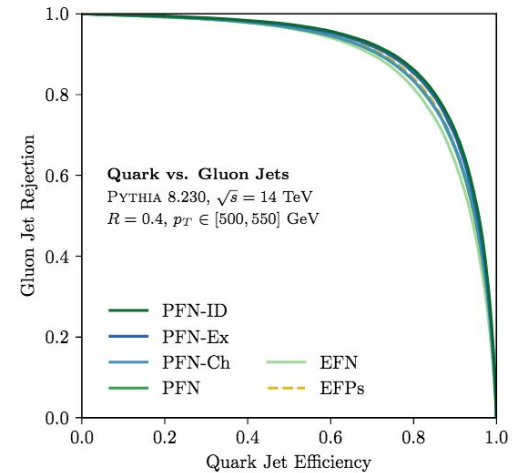
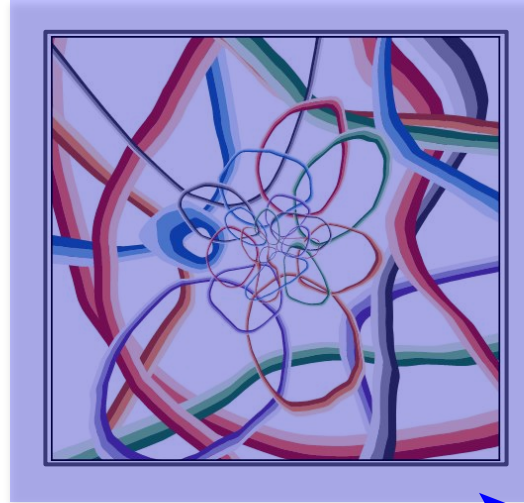
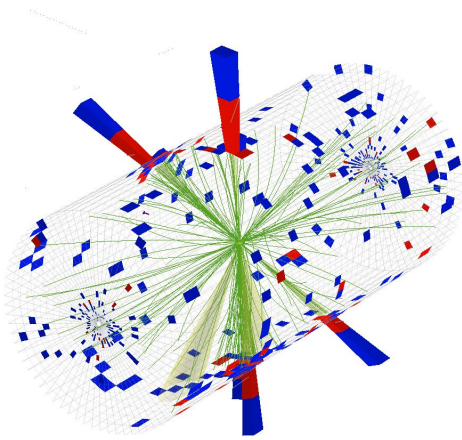


Given the final measured **point cloud**, was the **initiating parton** a **quark** or a **gluon**?

A staple machine learning task!

Rikab Gambhir – Harvard – 11 June 2025

Typical Machine Learning Setup



Collider Data
 ~1000 Dimensional

Latent Space
 ~10-100 Dimensional

Observables
 ~1-10 Dimensional

Pictured: An Energy Flow Network (**EFN**):

$$\mathcal{O}(\{p_1, \dots, p_M\}) = F \left(\sum_{i=1}^M z_i \Phi(\hat{p}_i) \right)$$

(How) can we understand and constrain this?

(How) can we be more efficient?

Deep Learning on Point Clouds

The **Deep Sets Theorem** tells us how to parameterize functions on point clouds*:

Set of momenta $\rightarrow$ $a = 1 \dots L$, the *Latent Dimension* index

$$\mathcal{O}(\mathcal{P}) = F(\langle \phi^a \rangle_{\mathcal{P}})$$

The Deep Sets Theorem guarantees that any function on sets $\mathcal{P}$ can be written this way, for “sufficiently complex” F and ϕ and large enough L .

$\langle \phi \rangle_{\mathcal{P}} \equiv \sum_i z_i \phi(\hat{p}_i)$

For this talk, I will focus primarily on Energy Flow Networks (EFNs) where the sums are energy-weighted.

*More sophisticated architectures still reduce to Deep Sets in some limit, e.g. graph networks

Deep Learning on Point Clouds

The **Deep Sets Theorem** tells us how to parameterize functions on point clouds*:

Set of momenta $\rightarrow$ $a = 1 \dots L$, the *Latent Dimension* index

$$\mathcal{O}(\mathcal{P}) = F(\langle \phi^a \rangle_{\mathcal{P}})$$

The Deep Sets Theorem guarantees that any function on sets $\mathcal{P}$ can be written this way, for “sufficiently complex” F and ϕ and large enough L .

$\langle \phi \rangle_{\mathcal{P}} \equiv \sum_i z_i \phi(\hat{p}_i)$

But how complex do F and ϕ need to be? Can I reduce the number of latent dimensions?

For this talk, I will focus primarily on Energy Flow Networks (EFNs) where the sums are energy-weighted.

*More sophisticated architectures still reduce to Deep Sets in some limit, e.g. graph networks

Moment Pooling

$$\mathcal{O}(\mathcal{P}) = F(\langle \phi^a \rangle_{\mathcal{P}})$$



Generalize to more moments! “**Moment Pooling**”

$$\mathcal{O}_k(\mathcal{P}) = F_k(\underbrace{\langle \phi^a \rangle_{\mathcal{P}}}_{1^{\text{st}} \text{ Moment}}, \underbrace{\langle \phi^{a_1} \phi^{a_2} \rangle_{\mathcal{P}}}_{2^{\text{nd}} \text{ Moment}}, \dots, \underbrace{\langle \phi^{a_1} \dots \phi^{a_k} \rangle_{\mathcal{P}}}_{k^{\text{th}} \text{ Moment}})$$

$k = \text{Highest order moment considered}$

$$L_{\text{eff}} = \binom{L+k}{k}$$

Moment Pooling – Why?

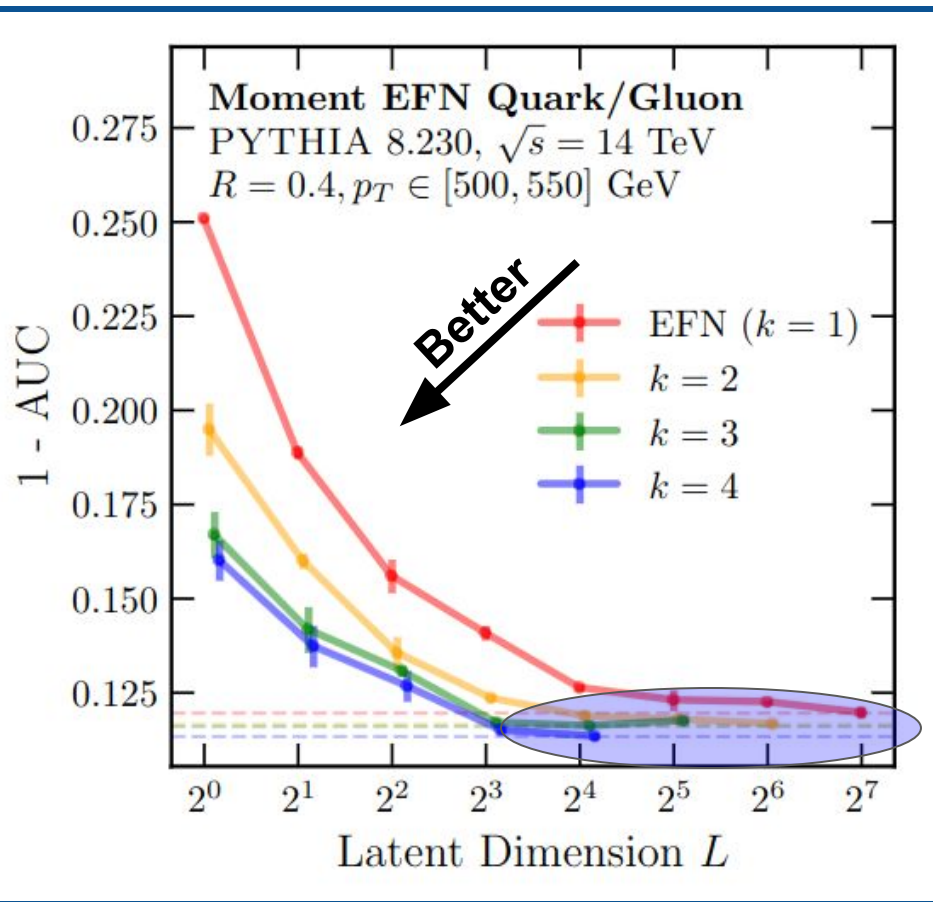
k = Highest order moment considered ↓

$$\mathcal{O}_k(\mathcal{P}) = F_k \left(\underbrace{\langle \phi^a \rangle_{\mathcal{P}}}_{1^{\text{st}} \text{ Moment}}, \underbrace{\langle \phi^{a_1} \phi^{a_2} \rangle_{\mathcal{P}}}_{2^{\text{nd}} \text{ Moment}}, \dots, \underbrace{\langle \phi^{a_1} \dots \phi^{a_k} \rangle_{\mathcal{P}}}_{k^{\text{th}} \text{ Moment}} \right)$$

$L_{\text{eff}} = \binom{L+k}{k}$

- **Explicit Multiplication:** Neural nets are mostly piecewise linear! But most functions we learn aren't. Moments are just multiplication, but for distributions! *Ask me about this later!*
- **Latent Space Compression:** For large L , there are effectively L^k latent dimensions due to combinatorics, but still made of only L functions! Fewer latent dimensions means easier analysis!

Moment Pooling Results

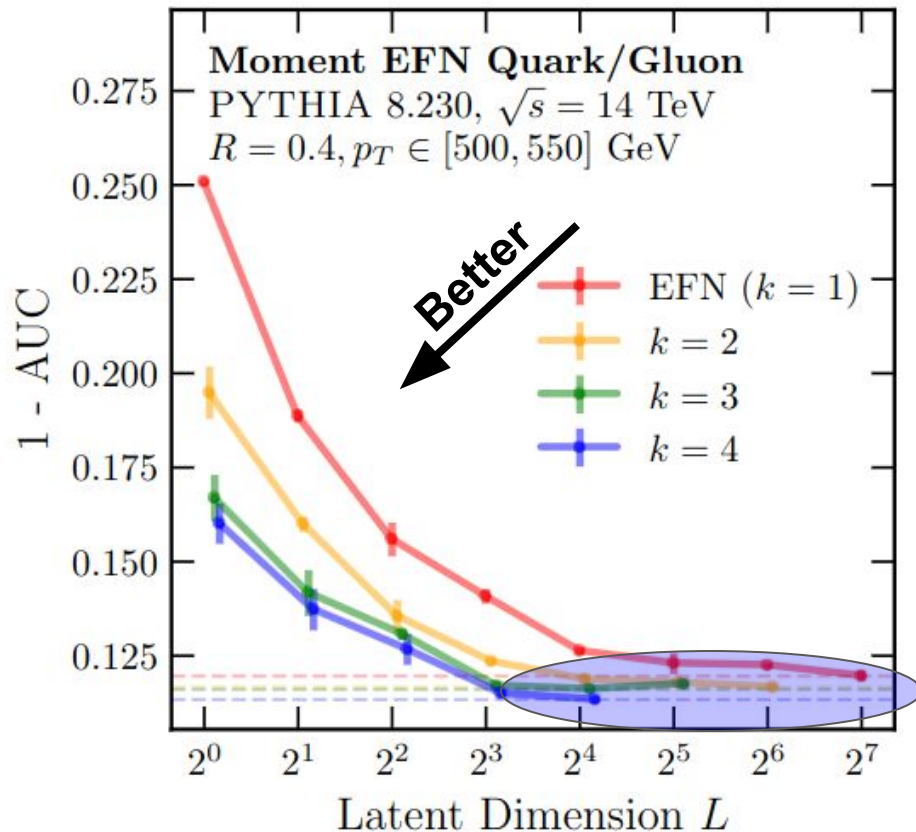


A high order **Moment EFN** can achieve the same top performance on quark/gluon classification as an **ordinary EFN**, but with far fewer latent dimensions!*

A Moment EFN is a more efficient encoding of the same data!

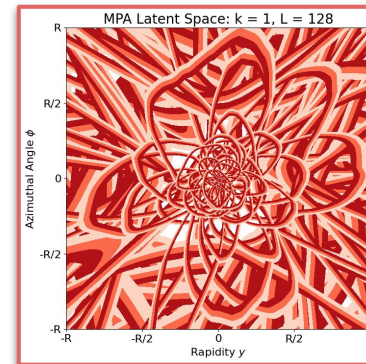
*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

The Black Box

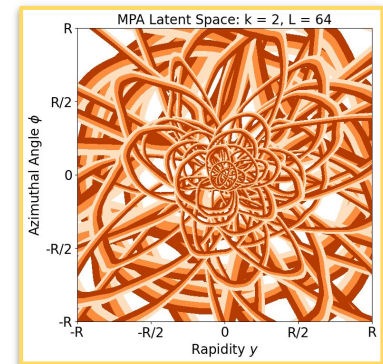


Latent Spaces

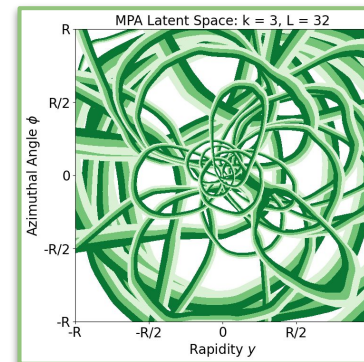
EFN ($k = 1$)
 $L = 128$



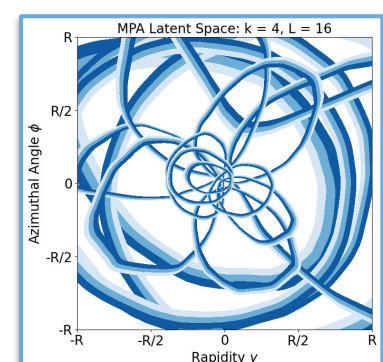
$k = 2$
 $L = 64$



$k = 3$
 $L = 32$



$k = 4$
 $L = 16$

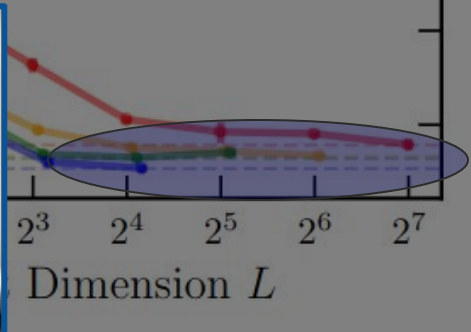


The 45% - 55% contours of each of the L different ϕ functions

*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

So what? The information is just being shuffled around. Is 16 latent dimensions any more interpretable than 128?

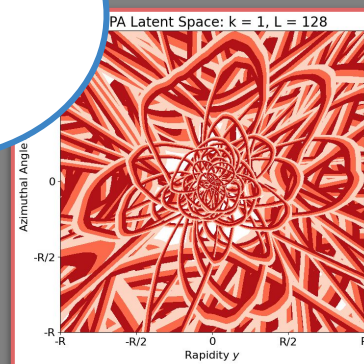
The machine probably learned something, but I didn't!



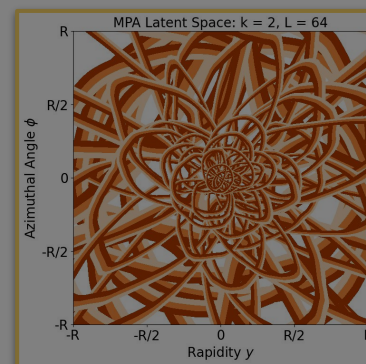
Me, just before the **moment** of clarity.

Latent Spaces

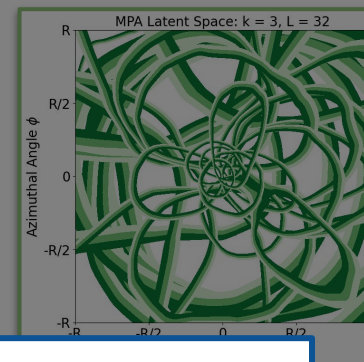
EFN ($k = 1$)
 $L = 128$



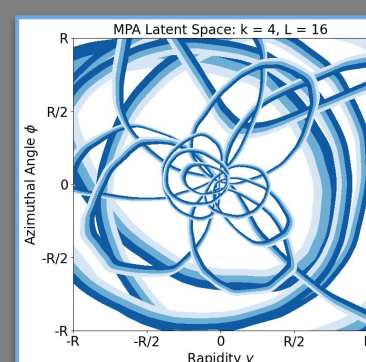
$k = 2$
 $L = 64$



$k = 3$
 $L = 32$



$k = 4$
 $L = 16$



*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

Latent Spaces

So what? The information is just being shuffled around. Is 16 latent dimensions any more interpretable than 128?

The machine probably learned something, but I didn't!

I'm allowed to control the ML function space!

Let me just choose one that's easy for me to understand: **One dimensional functions**

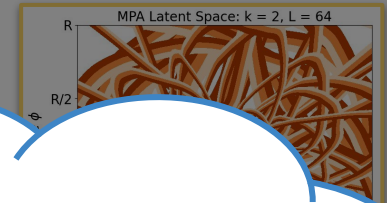
Because of latent space compression, even one dimension can contain a lot of info!

Me, just before the **moment** of clarity.

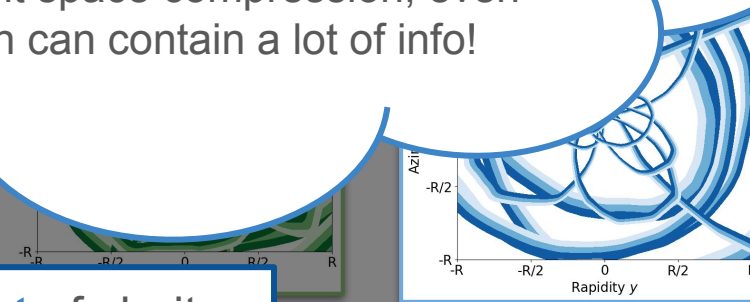
EFN ($k = 1$)
 $L = 128$



$k = 2$
 $L = 64$

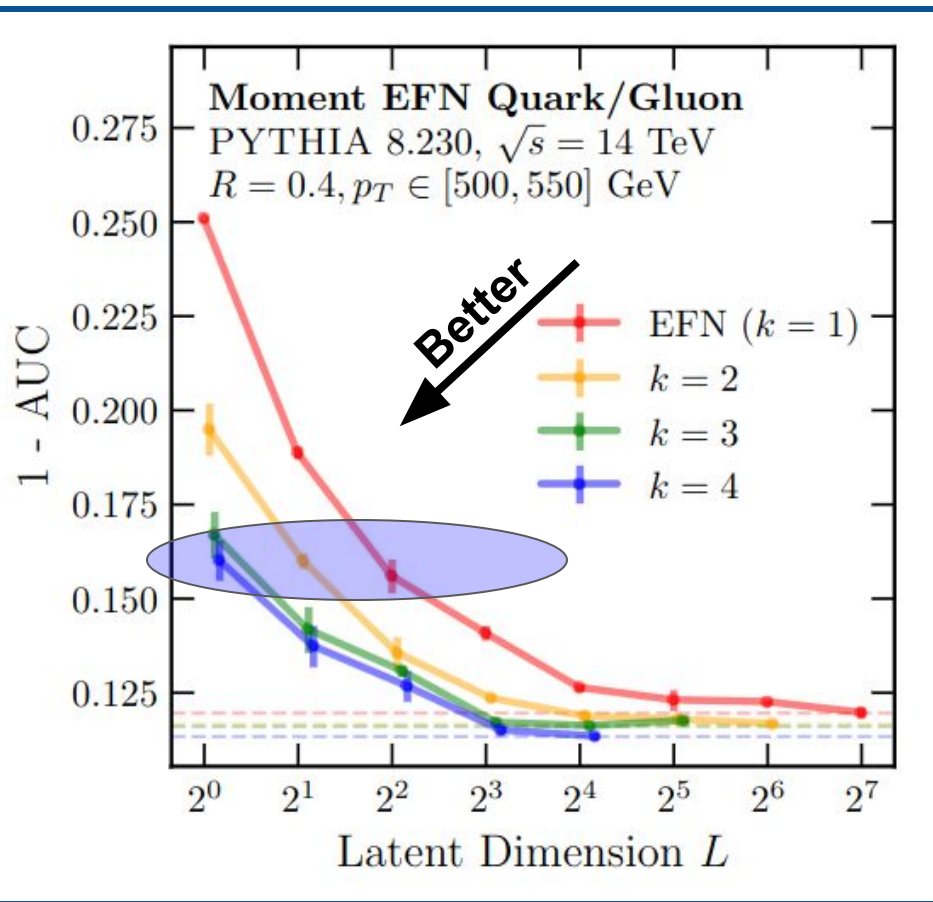


2^3 2^4 2^5
Dimension L



*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

Moment Pooling Results (Cont'd)



Let's instead look at networks with just a *single* latent dimension.

A **$k = 4$ Moment EFN** with *just one* latent dimension does just as good as an **ordinary EFN** with *four* latent dimensions!

Going down to $L = 1$

The more efficient encoding is extremely useful when we can get to $L = 1$!
Look how much simpler allowing neural networks to multiply makes things!

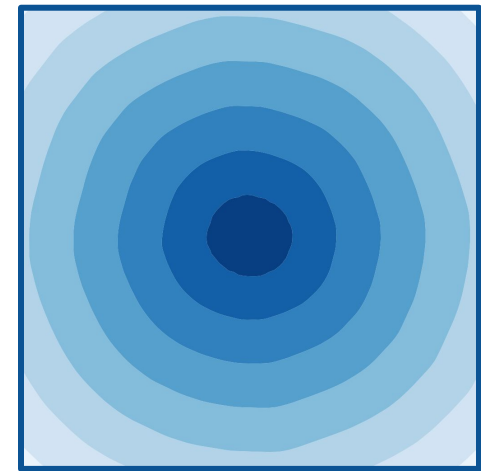
EFN ($k = 1$)
 $L = 4$



Same information!



Order $k = 4$
 $L = 1$



- 4 Different Functions
- Combined in a complicated way
- No symmetry
- ???

- Single function
- Expected to be, and is, radially symmetric
- We can **interpret** this!

The Moment of Truth

We can extract physics from this!

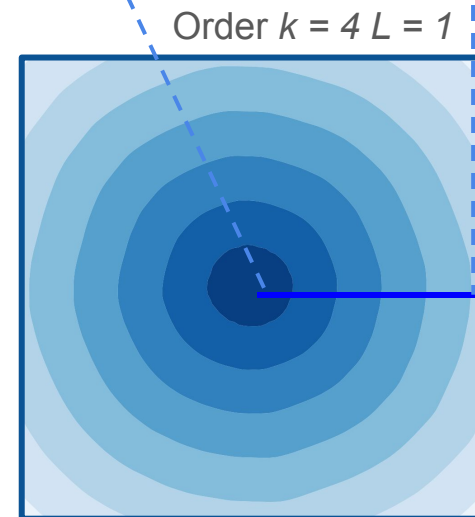
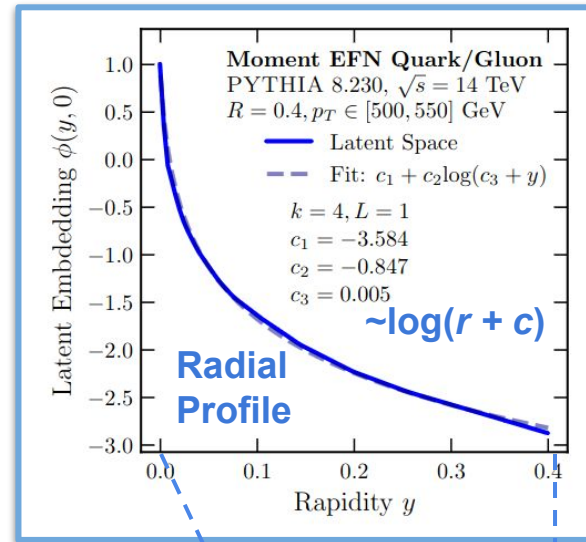
$$\Phi_{\mathcal{L}}(r) = c_1 + c_2 \log(c_3 + r)$$

This defines a **log angularity** observable, related to the $\square \rightarrow 0$ limit of ordinary angularities

We can even see non-perturbative physics!

$$c_3 \sim \frac{\Lambda_{\text{QCD}}}{p_T R} \sim 0.001$$

Just this alone is enough to get an IRC-safe quark-gluon classifier with an AUC ~ 0.83 !



The Moment of Truth

Learning
Uncertainties the
Frequentist Way

Can you Hear the
Shape of a Jet?

The Moments of
Clarity: Streamlining
Latent Spaces

Isolating Unisolated
Upsilon: Finding
Particles

Conclusion

We can extract physics from this!

$$\Phi_c(r) = c_1 + c_2 \log(c_3 + r)$$

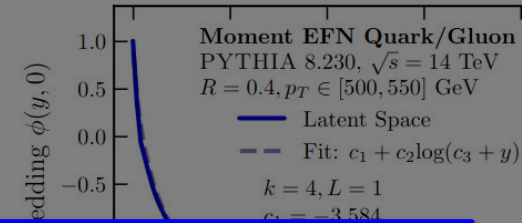
End of Part III

By choosing:

1. A human-readable function space (1 dimension)
2. A multiplication-based reparameterization to cram as much info into that one dimension as possible

We can construct a simple closed-form observable, from which we can even read off the IR divergence!

Just this alone is enough to get an IRC-safe quark-gluon classifier with an AUC ~ 0.83 !



Part IV: Re-Normalizing Flows

Taming Logs and Perturbation Theory in QCD



Basic Premise

Fact: Rikab isn't very good at hardcore QFT calculations.

But I want to make sensible predictions for the distributions of observables anyways!

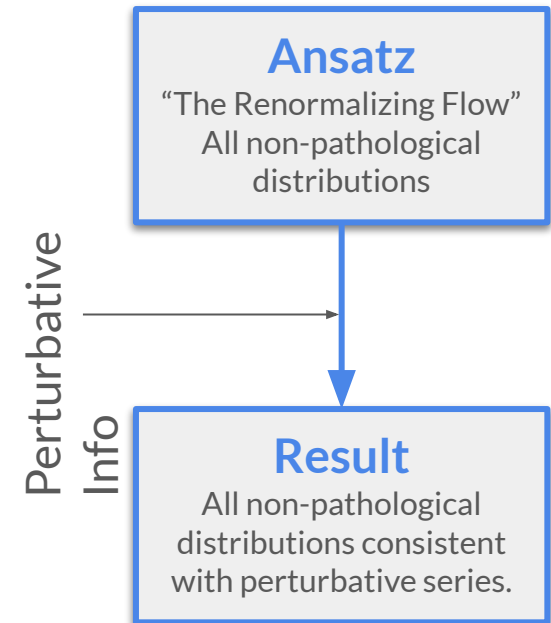
Instead, let me just write down the *most general* form for a probability distribution of an observable $p(x)$ in terms of easily parameterized functions *that are easy to Taylor-expand*.

This ansatz will be guaranteed to be positive, normalized, finite, and smooth* – **non-pathological!**

Then, I can just match to other people's fixed order calculations, and have a "free" resummed* calculation, with my ignorance about QCD built into higher-order functions!



Large logs? IR Divergences?
Resummation? SCET? Plus
functions? Sudakov Shoulders? Too
much for me!



*Piecewise

**Not literal logarithmic resummation, just “a” resummation

Perturbation Theory in QCD

Now that we've seen plenty of new observables x , let's try to actually calculate them!
We'll perturb in a small parameter α to calculate the distribution of x :

$$p(x|\alpha) = \underbrace{p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha)}_{\text{Fixed-order calculation}} + \underbrace{\mathcal{O}(\alpha^{M+1})}_{\text{Unknown}}$$

Example: Thrust [Singular parts Becher, Schwartz; [0803.0342](#)]

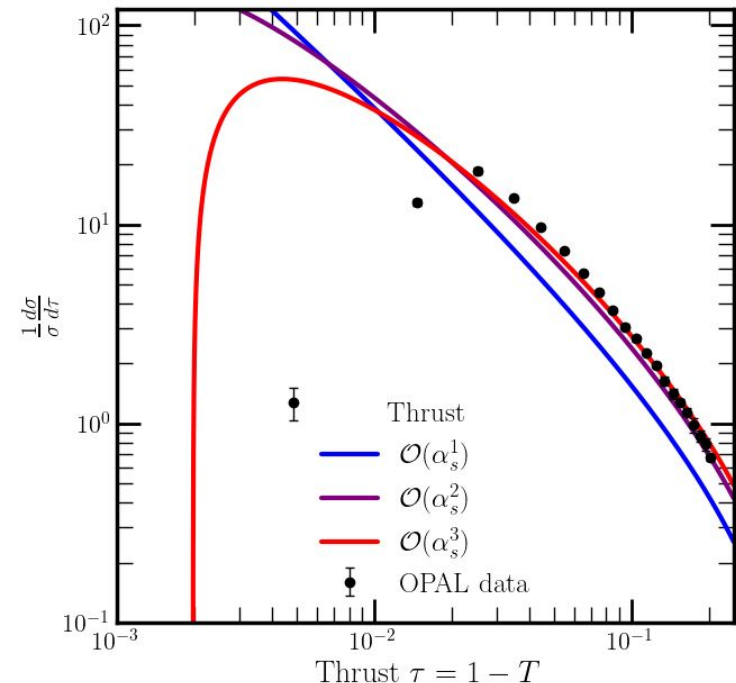
$$p_1 = \frac{1}{\tau} \{C_F [-4 \ln \tau - 3]\}$$

$$p_2 = \frac{1}{\tau} \left\{ C_F^2 \left[8 \ln^3 \tau + 18 \ln^2 \tau + (13 - 4\pi^2) \ln \tau + \frac{9}{4} - 2\pi^2 + 4\zeta_3 \right] \right. \\ \left. + C_F T_F n_f \left[-4 \ln^2 \tau + \frac{22}{9} \ln \tau + 5 \right] \right. \\ \left. + C_F C_A \left[11 \ln^2 \tau + \left(-\frac{169}{18} + \frac{2\pi^2}{3} \right) \ln \tau - \frac{57}{4} + 6\zeta_3 \right] \right\},$$

$$p_3 = \frac{1}{\tau} \left\{ C_F^3 \left[-8 \ln^5 \tau - 30 \ln^4 \tau + \ln^3 \tau \left(-44 + \frac{40\pi^2}{3} \right) + \ln^2 \tau (-88\zeta_3 + 24\pi^2 \right. \right. \\ \left. \left. - 27) + \ln \tau \left(-c_{2C_F}^S - 96\zeta_3 - \frac{53\pi^4}{90} + \frac{79\pi^2}{6} - \frac{17}{2} \right) + 16\pi^2\zeta_3 - 39\zeta_3 - 132\zeta_5 \right] \right. \\ \left. + C_F^2 T_F n_f \left[-4 \ln^4 \tau + \frac{22}{9} \ln^3 \tau + \frac{11}{3} \ln^2 \tau + \frac{11}{3} \ln \tau + \frac{11}{3} \right] \right. \\ \left. + C_F T_F^2 n_f^2 \left[-4 \ln^3 \tau + \frac{22}{9} \ln^2 \tau + \frac{11}{3} \ln \tau + \frac{11}{3} \right] \right\},$$

... [many more terms]

For comparison: **OPAL** Data



Perturbation Theory in QCD

Now that we've seen plenty of new observables x , let's try to actually calculate them!
We'll perturb in a small parameter α to calculate the distribution of x :

$$p(x|\alpha) = \underbrace{p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha)}_{\text{Fixed-order calculation}} + \underbrace{\mathcal{O}(\alpha^{M+1})}_{\text{Unknown}}$$

Example: Thrust [Singular parts Becher, Schwartz; [0803.0342](#)]

$$D_A(\tau) = \frac{1}{\tau} \{C_F [-4 \ln \tau - 3]\}$$

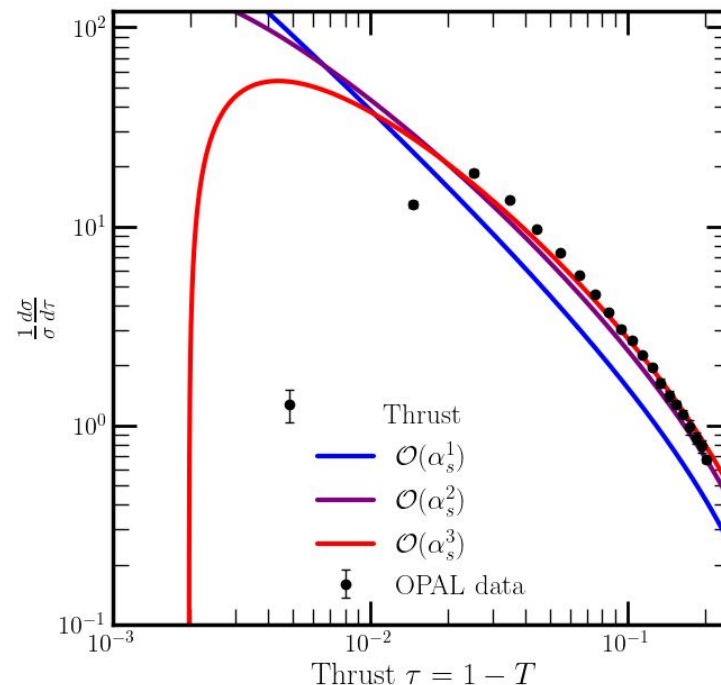
$D_B(\tau)$ This expansion has some pathologies “real life” (all-orders) doesn't have ...

1. Non-finite
2. Non-smooth* (I will never see a plus or delta function in binned data!)
3. Non-positive
4. Non-normalizable

$D_C(\tau)$

Large logs can spoil convergence! Lets fix it.

For comparison: **OPAL** Data



Our Solution: The Renormalizing Flow^{*}

Given a Fixed Order prediction for an observable x depending on α

$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

Fixed-order calculation – you give us this.

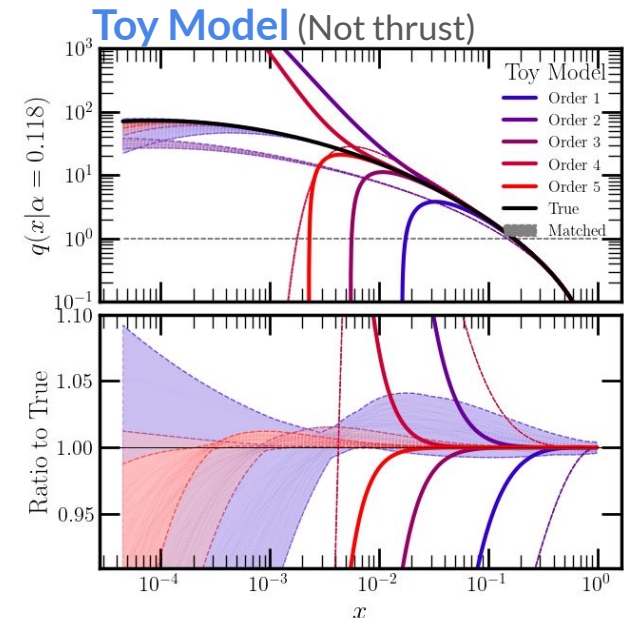
Unknown

We can **guess** the higher order coefficients by using the **Renormalizing Flow Ansatz**, which is parameterized by functions $g_m(t)$:

$$q(t|\alpha) = (g_* \alpha^{m_*}) e^{-\sum_m g_m(t) \alpha^m - \int_0^t dt' (g_* \alpha^{m_*})} \exp\left\{\sum_m g_m(t') \alpha^m\right\}$$

Which is *guaranteed* to fix all of the pathologies:

- | | | |
|---------------------|---|--------------------|
| 1. Non-finite | → | Finite! |
| 2. Non-smooth | → | Smooth! |
| 3. Non-positive | → | Positive! |
| 4. Non-normalizable | → | Normalized! |



The Renormalizing Flow: Details

1. **Probability Fact:** A positive, smooth, normalized, and finite PDF of a single variable t , where t ranges from 0 to ∞ , can *always* be written as:

$$q(t|\alpha) = f(t, \alpha) e^{-\int_0^t dt' f(t', \alpha)}$$

for *any* function f that is always non-negative and whose integral in t diverges.

2. **Analysis Fact:** All positive functions f that are analytic in α can be written as the *exponential* of a function g , given by*:

$$g(t, \alpha) = -\log\left(g_*(t)\alpha^{m^*}\right) + \sum_{m=0} g_m(t)\alpha^m$$

3. **Scaling:** An observable g is transformed into t :

$$t = \log\left(\frac{1}{x}\right) \quad \text{This is where Logs come from!}$$

These three ingredients together lead to our Ansatz!

Matching

$$q(t|\alpha) = (g_* \alpha^{m_*}) e^{-\sum_m g_m(t) \alpha^m - \int_0^t dt' (g_* \alpha^{m_*}) \exp\{\sum_m g_m(t') \alpha^m\}}$$

The **Renormalizing Flow** Ansatz is parameterized by functions $g_m(t)$

For a fixed order calculation up to order M :

For $m \leq M$, perturbative matching requires:

$$g(t, \alpha) = -\log(p_{m_*}(t) \alpha^{m_*}) + \sum_k \frac{1}{k} \left(\left[-\sum_{m>m_*}^M \frac{p_m(t)}{p_{m_*}(t)} \alpha^{m-m_*} \right]^k - \left[\sum_{m=m_*}^M \int_0^t dt' p_m(t') \alpha^m \right]^k \right) + \mathcal{O}(\alpha^{M+1})$$


 We divide out the lowest power of α – this power is m_*

This looks scary ... but this is trivial to implement with **autodiff** ~ **ML magic**!

For $m > M$, choose *anything you want*^{*} for $g_m(t)$: neural networks, random guessing, zero, ...

The choice of higher order $g_m(t)$ is perturbative uncertainty – *all* possible positive, smooth, finite, and normalized completions of the fixed order theory fit here!

We'll treat $g_m(t)$ as polynomials in t for simplicity – *logarithmic expansion in x* .

^{*}One condition – $g_m(t)$ must be non-positive for large t

Toy Example: Exponential

Consider the “platonic” distribution

$$p_{\text{Ideal}}(t|\alpha) = \alpha \exp(-\alpha t)$$

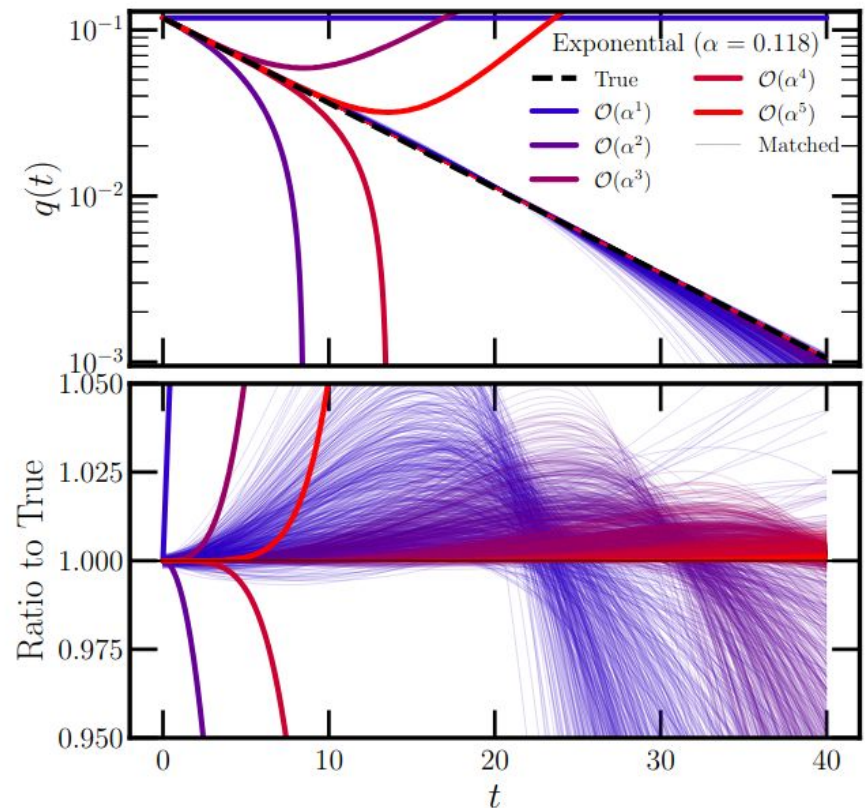
But we only have access to a finite fixed-order expansion

$$p_{\text{FO}}^{(M)}(t|\alpha) = \alpha \sum_m^{M-1} \frac{(-\alpha t)^m}{m!}$$

Let's apply the renormalizing flow!
We'll pick some prior for the higher order coefficients g_{mn} .

Solid: Fixed Order (Pathological)

Bands: Distribution of the Renormalizing Flow for different random choices of g_{mn}



Every curve in the bands is a *non-pathological* valid distribution that matches the FO!

Toy Example: Exponential

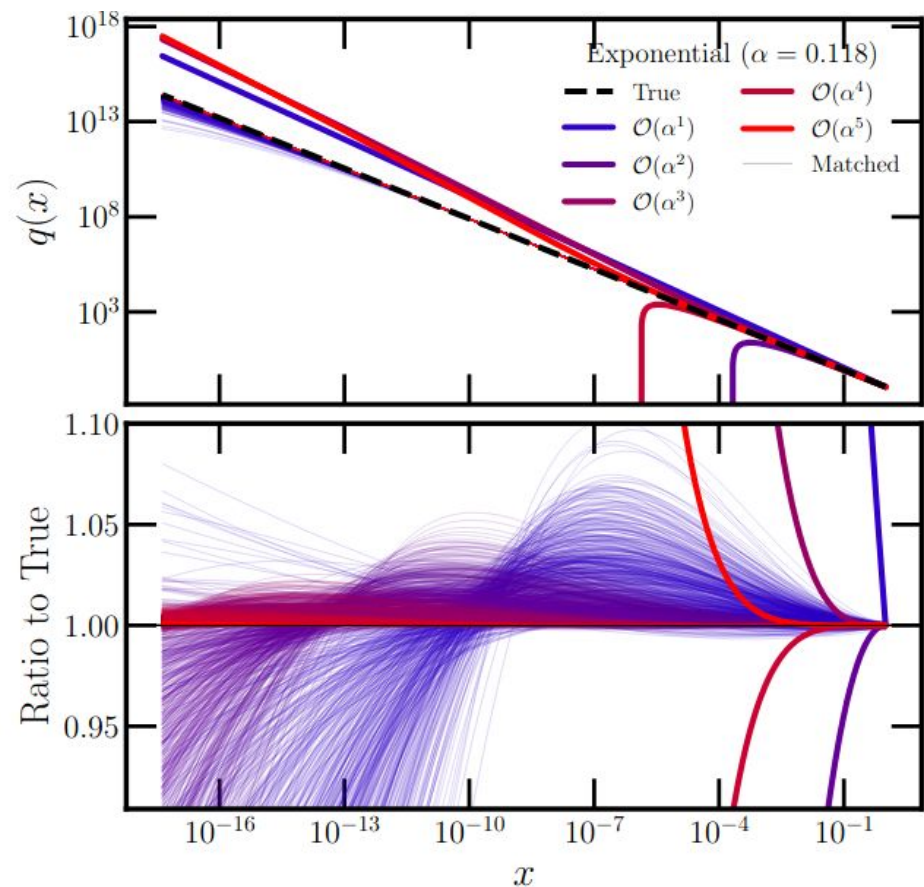
An exponential distribution in t space is a power-law distribution in x space with a slope of α .

The F.O. distributions are only good for large x , and quickly blow up or become negative for small x .

But the Renormalizing Flow *never* will!
It will always be well behaved.

Since the Ansatz is universal, the *true* answer corresponds to *some* choice of higher order g , but we don't know which — but no matter what we guess, our guess is always a valid distribution!

Solid: Fixed Order



Every curve in the bands is a *non-pathological* valid distribution that matches the FO!

Toy Example II: Angularity-Like

Consider the “platonic” distribution

$$p^{\text{LL}'}(\rho) = \frac{\alpha_s C_F}{\pi} \frac{1}{\rho} \left[\log \frac{1}{\rho} \right] \exp \left(-\frac{\alpha_s C_F}{2\pi} \left[\log^2 \frac{1}{\rho} \right] \right)$$

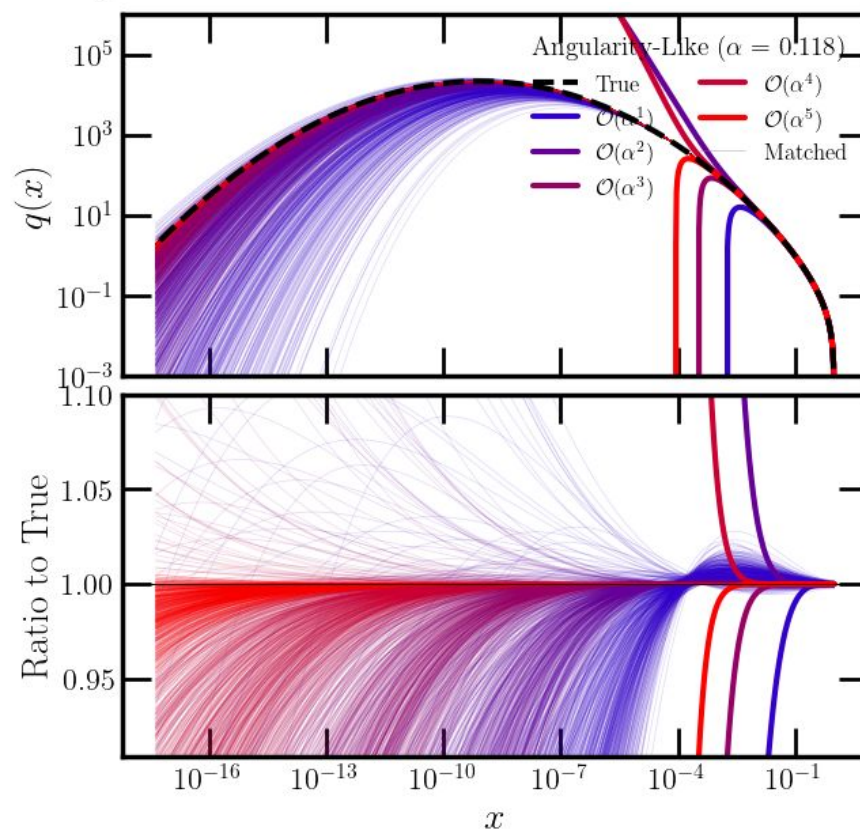
E.g. Let’s pretend that Leading-Log jet angularities are the way the world really works!

We can play the same game, only having access to fixed-order info.

We get *pretty good all-orders* predictions!

Solid: Fixed Order

Bands: Distribution of the Renormalizing Flow for different random choices of g_{mn}



Every curve in the bands is a *non-pathological* valid distribution that matches the FO!

Toy Example III: Angularity-Like + Running

Let's take the same fixed-order predictions from before, but now let α run! (Contrived Example, but bear with me!)

Letting α run with xQ effectively just changes higher order coefficients in a complicated way— but *autodiff magic* makes it easy, and we don't have to think about it!

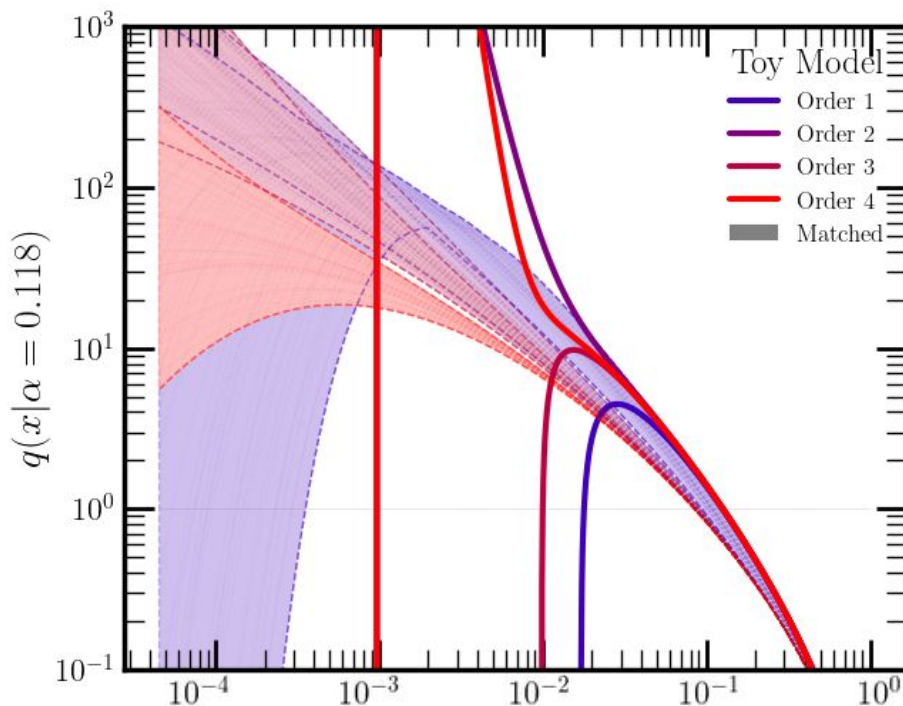
No longer an easy closed-form. We are flying blind!

Weird Fixed Order $\rightarrow$ Tamed Renormalizing Flows!

Solid: Fixed Order

Bands: Distribution of the Renormalizing Flow for different random choices of g_{mn}

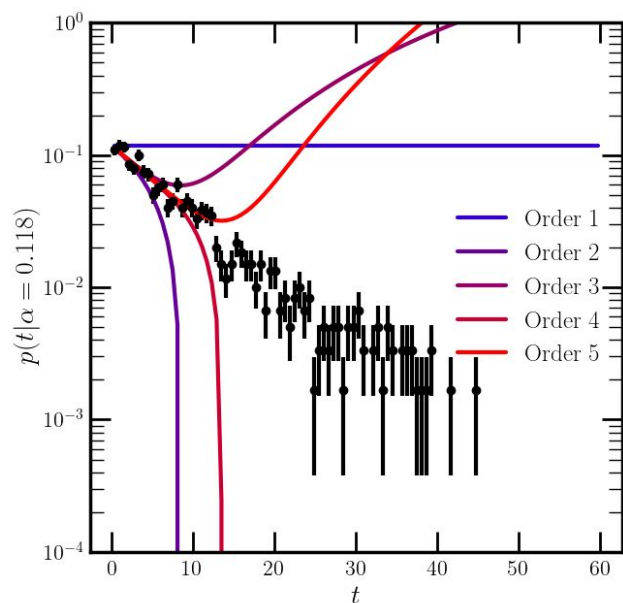
We still use $\alpha(m_Z)$ to define the conditional distribution $\frac{4\pi}{\alpha_s(\mu_0^2)} - \frac{4\pi}{\alpha_s(Q^2)} = \beta_0 \ln\left(\frac{\mu_0^2}{Q^2}\right)$



Every curve in the bands is a *non-pathological* valid distribution that matches the FO!

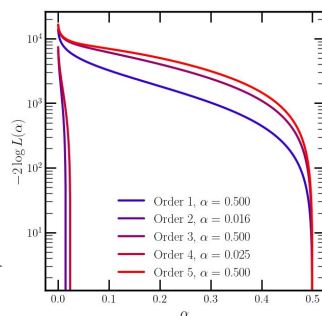
Inference: Fitting α From Data (Toy)

Since q are genuine, non-pathological PDFs, it's easy to *directly* do maximum-likelihood (or similar) inference!

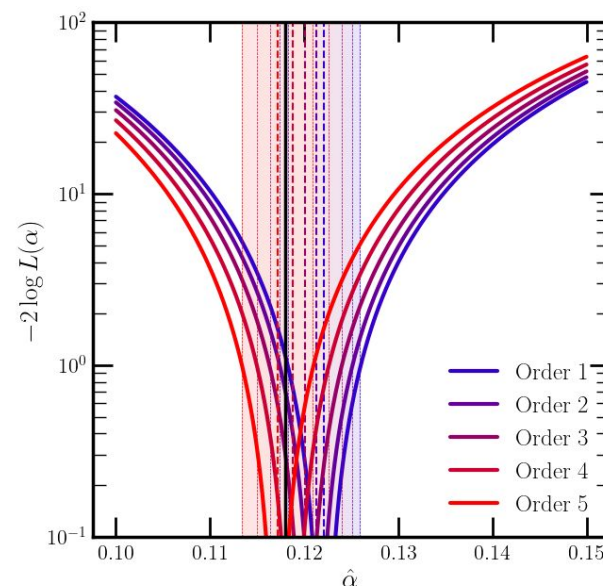


Direct F.O.
likelihoods are so
pathological that
MLE is nonsensical

Fit with $q(x|\alpha)$ directly
(Full likelihood)



Cutoffs, binning, and
other regulators
needed!



CLAIMER

What we are claiming

- **Claiming:** Given a distribution p , pathological or otherwise, we can find a regulated distribution q that is normalized, smooth, positive, and normalized that is perturbatively matched to p .
- **Claiming:** This is a universal approximator: any valid q can be reached with a choice of higher order g , including the “true” one.
- **Claiming:** For particular g 's, we can get meaningful q 's that look like theory calculations and tame large logs!

DISCLAIMER

What we are *not* claiming

- **Not Claiming:** The perturbative structure is *easy* to maintain numerically or that numeric artifacts are manageable.
 - But if you restrict to reasonable choices of g , like $g \sim N(0,1)$, then the results seem reasonable!
- **Not Claiming:** The envelopes give a meaningful uncertainty on higher orders.
 - The envelopes only tell you what *possible* higher order physics is consistent with positivity and normalizability — it *constrains* physics
- **Not Claiming:** This beats logarithmic resummation.
 - This is an all-orders summation. A genuine logarithmic resummation is always better — but for our method, you don't need to know *any* physics!

Key Point: This is *not* a shortcut to doing QCD, just a new way to parameterize things to guarantee nice properties. There is no free lunch, you still have to do physics!

What we are still thinking about...

1. Sometimes, we don't have a simple a fixed order expression in closed form. We are currently investigating if it is possible to interface with fixed-order matrix element calculators, potentially in an autodifferentiable way!
2. We have yet to apply this framework to a non-toy physics observable. Hopefully, coming soon!
3. The higher order g 's envelope the true distribution. Can we motivate a prior for them and derive meaningful statistical uncertainties? Can we treat them as nuisance parameters for inference?
4. We only explored 1D distributions in this talk. Is it possible to do more?

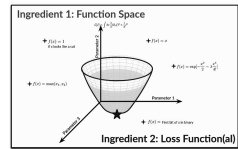
Part V: Conclusion

Even if you missed some details from each of the four vignettes I went over, I hope you took away:

- ML is not a black box, it's just numerical functional analysis.
- We can **completely** control what the ML does through the function space and loss functional – so physically meaningful results can be **guaranteed**.
- We can use this control to solve collider physics problems involving subtle correlations, high dimensional structures, or complicated simulations *that we could not have solved (as optimally) before*.

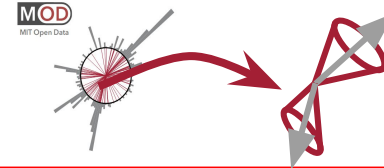
Part 0

Introduction



Part I

Learning **Uncertainties** the **Frequentist** Way

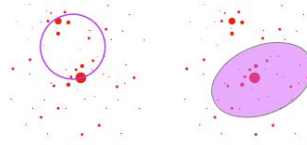


[RG, Nachman, Thaler, [2205.05084](#)]

[RG, Nachman, Thaler, [2205.05084](#)]

Part II

Can you Hear the **Shape** of a Jet?

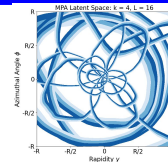


[Ba, Dogra, RG, Tasissa, Thaler, [2302.12266](#)]

[RG, Larkoski, Thaler, [2410.05379](#)]

Part III

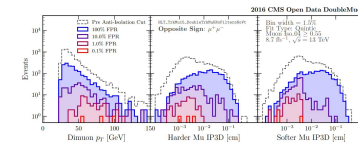
Moment of Clarity: Streamlining Latent Spaces



[RG, Osathapan, Thaler, [2403.08854](#)]

Part IV

Isolating Unisolated Upsilon: Finding Particles



[RG, Mastandrea, Nachman, Thaler, [2502.14036](#)]

Part V
[You are Here!]

Conclusion

Backup

Learning MLC

How do we calculate f ?

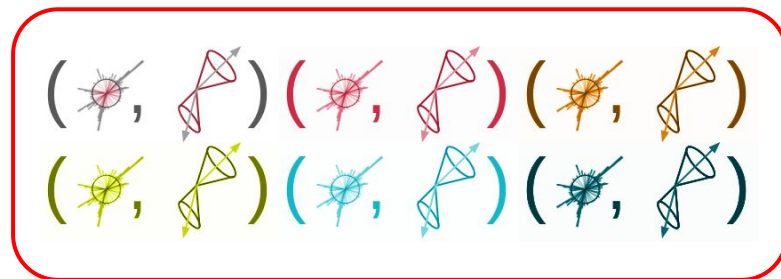
$$\begin{aligned} f_{\text{MLC}}(x) &= \operatorname{argmax}_z p_{\text{train}}(x|z) \\ &= \operatorname{argmax}_z \log \underbrace{\frac{p_{\text{train}}(x, z)}{p_{\text{train}}(x)p_{\text{train}}(z)}}_{T(x, z)} \end{aligned}$$

The function T is the likelihood ratio between $p(x, z)$ and $p(x)p(z)$.

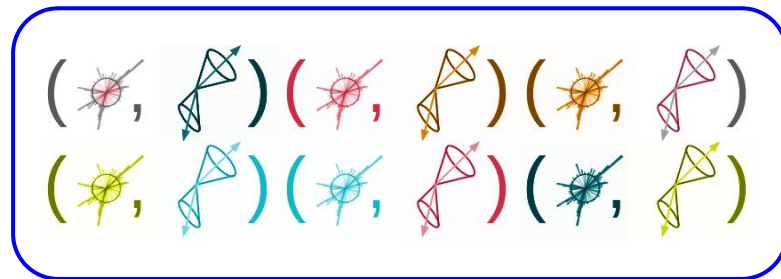
↓
Neyman–Pearson

T is the **optimal classifier** between (x, z) pairs and shuffled (x, z) pairs!

Class P



Class Q



Classify between **P** and **Q**!

Aside: Mutual information

A measure for non-linear interdependence is the **Mutual Information**:

$$\begin{aligned} I(X; Z) &= \int dx dz p(x, z) \log \frac{p(x, z)}{p(x) p(z)} \\ &= \mathbb{E}_{\text{train}} T(X, Z) \end{aligned}$$

Answers the question: How much information, in terms of bits, do you learn about Z when you measure X (or vice versa)?

When doing calibration this way, we get a measure of the **correlation** between X and Z , *for free*.

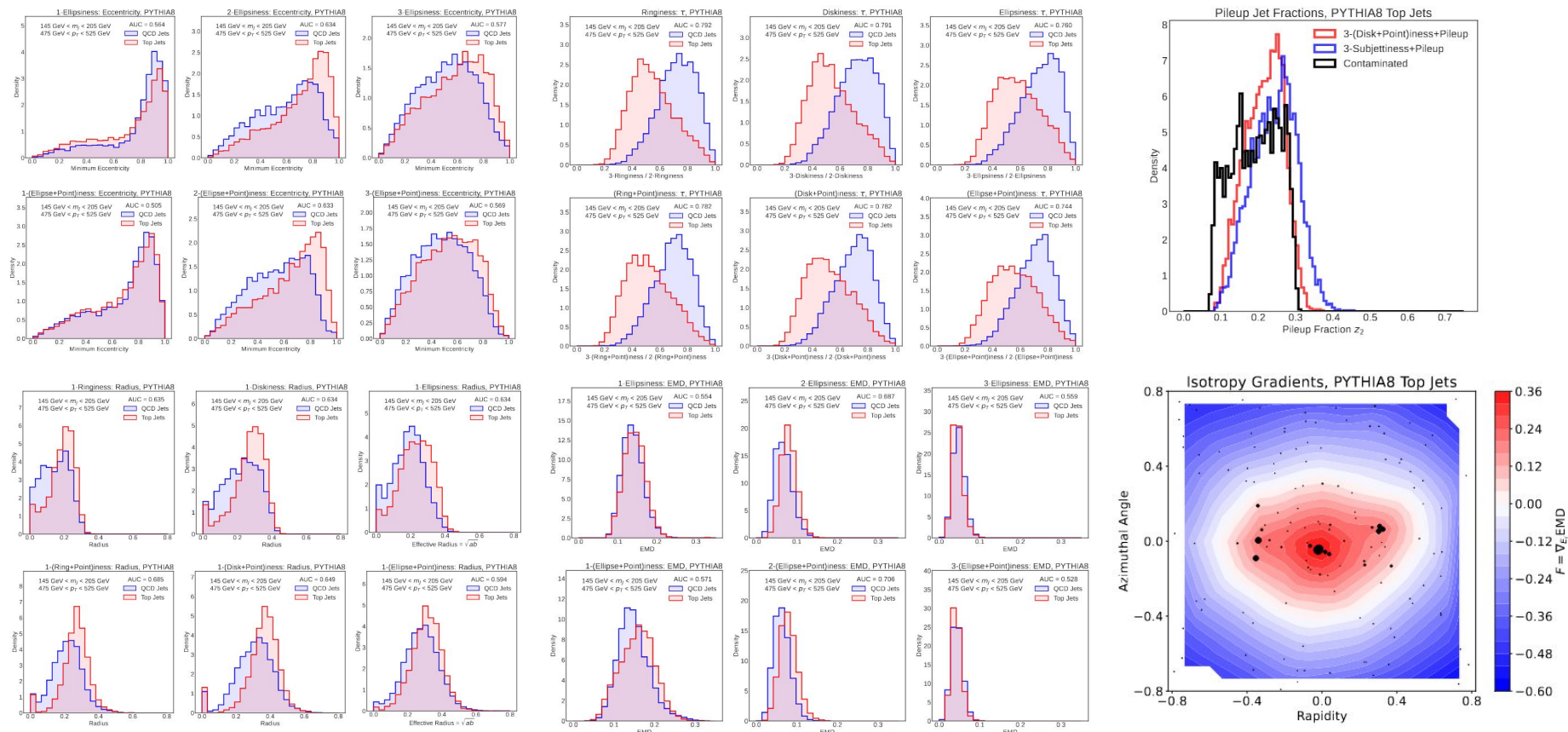
The general strategy for *all* of these problems will be:

1. Identify what properties we want our solution function f to have: *e.g. prior-independent, nice Gaussian limit, contains particular statistical information, IRC-safety, certain geometrical properties, easy function composition, easily plottable limits, physical constraints like smoothness or positivity ...*
2. Design a space of parameterized functions F that manifestly satisfy the properties you want. *e.g. If I want Gaussian functions in z , $\log(T(x,z)) = A(x) + (B(x) - z)C^{-1}(x)(B(x) - z)$ parametrizes all of them!*
3. In conjunction, design a “Loss Functional” $\mathcal{L}[f]$ over F whose minimum satisfies the properties you want. *e.g the Wasserstein Metric guarantees IRC-safety! The DV-loss is prior-free! We know how to do Euler-Lagrange, so just check!*
4. Perform the minimization! *This is where the actual ML comes in – but these are just incidental numeric details!*

If you don't remember the details of each physics example I show in this talk, I want you to remember this overall theme:

Key Takeaway: This is *all you need* for learning physics from machines!

New IRC-Safe Observables



... **Lots** of extractable information!

Automatic Grooming with Shapes

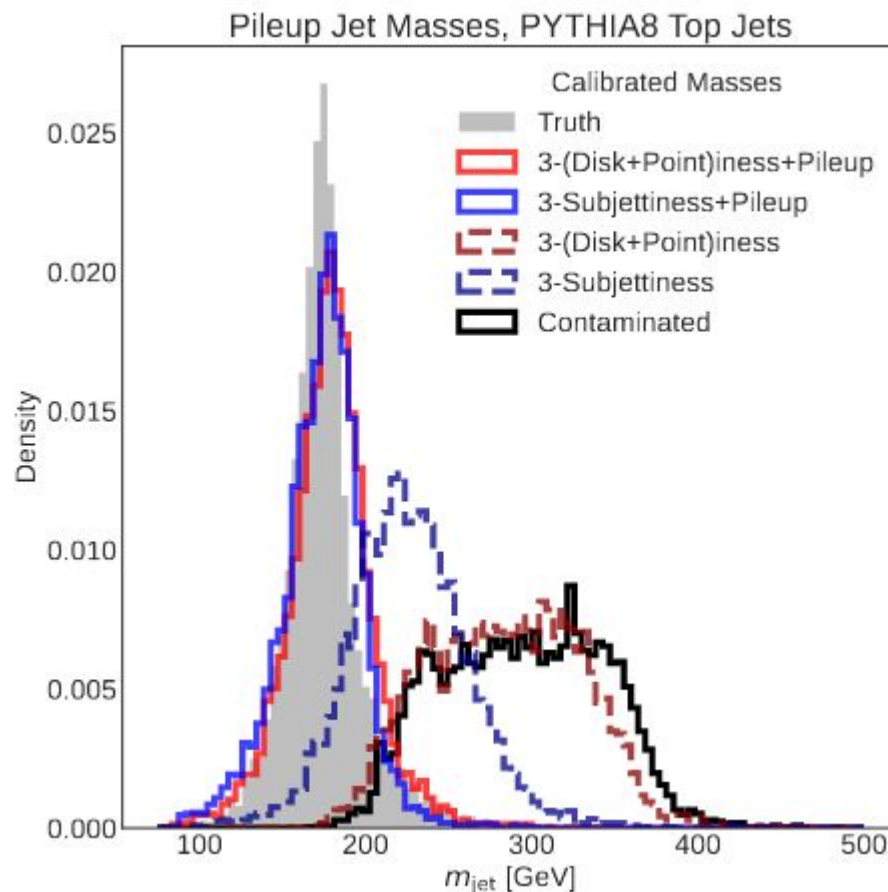
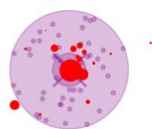
Use shapes to approximate events and extract masses – model pileup with a uniform background with floating weight!

No external hyperparameters, unlike softdrop.
Only need to assume pileup is uniform!

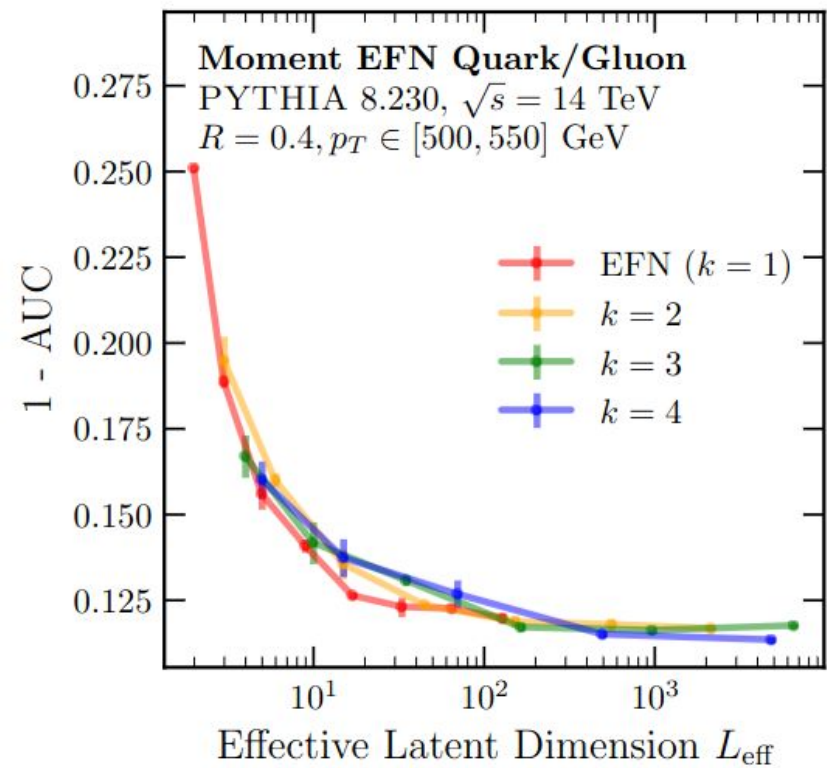
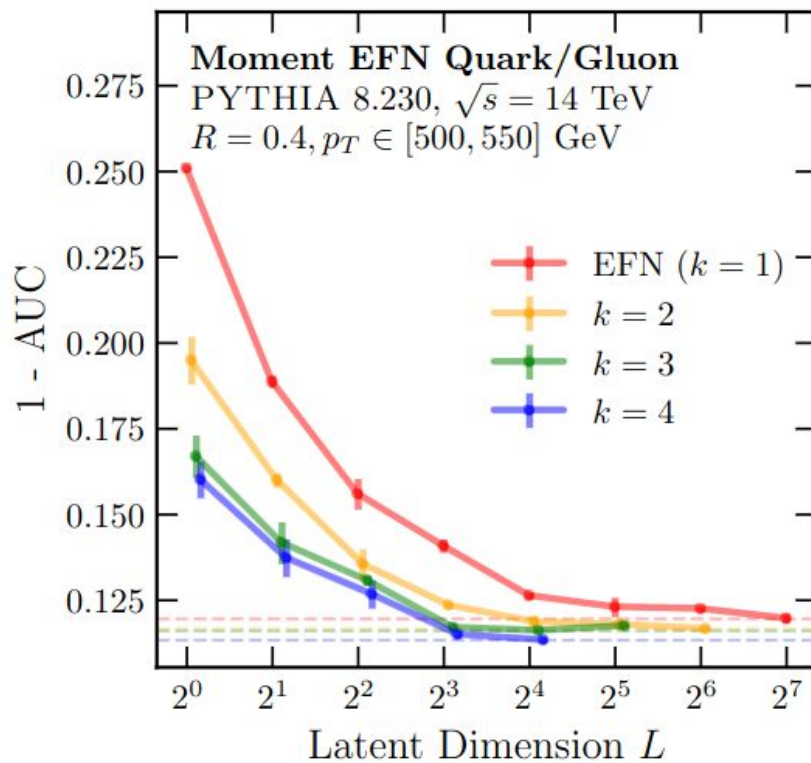
Contaminate top jets with 5-30% extra energy spread uniformly in an 0.8×0.8 plane

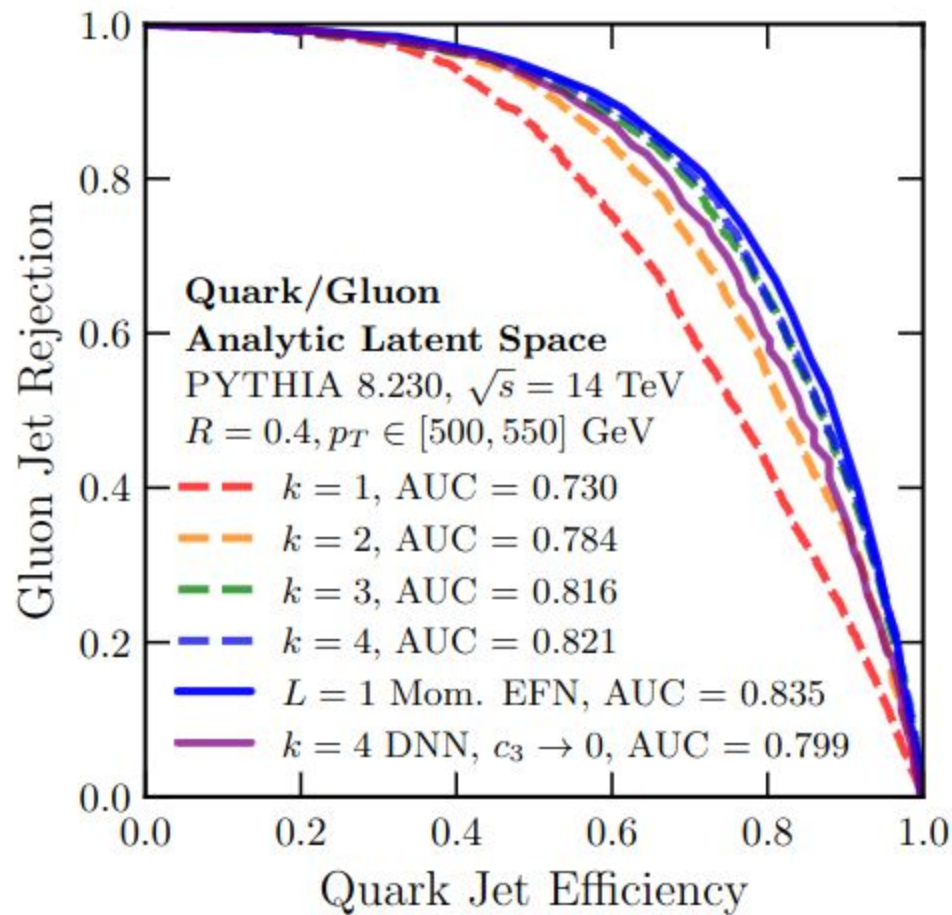
Consider 4 shapes:

- 3-Subjettiness
- 3-Subjettiness + Pileup
- 3-(Disk+Point)iness ←
- 3-(Disk+Point)iness + Pileup

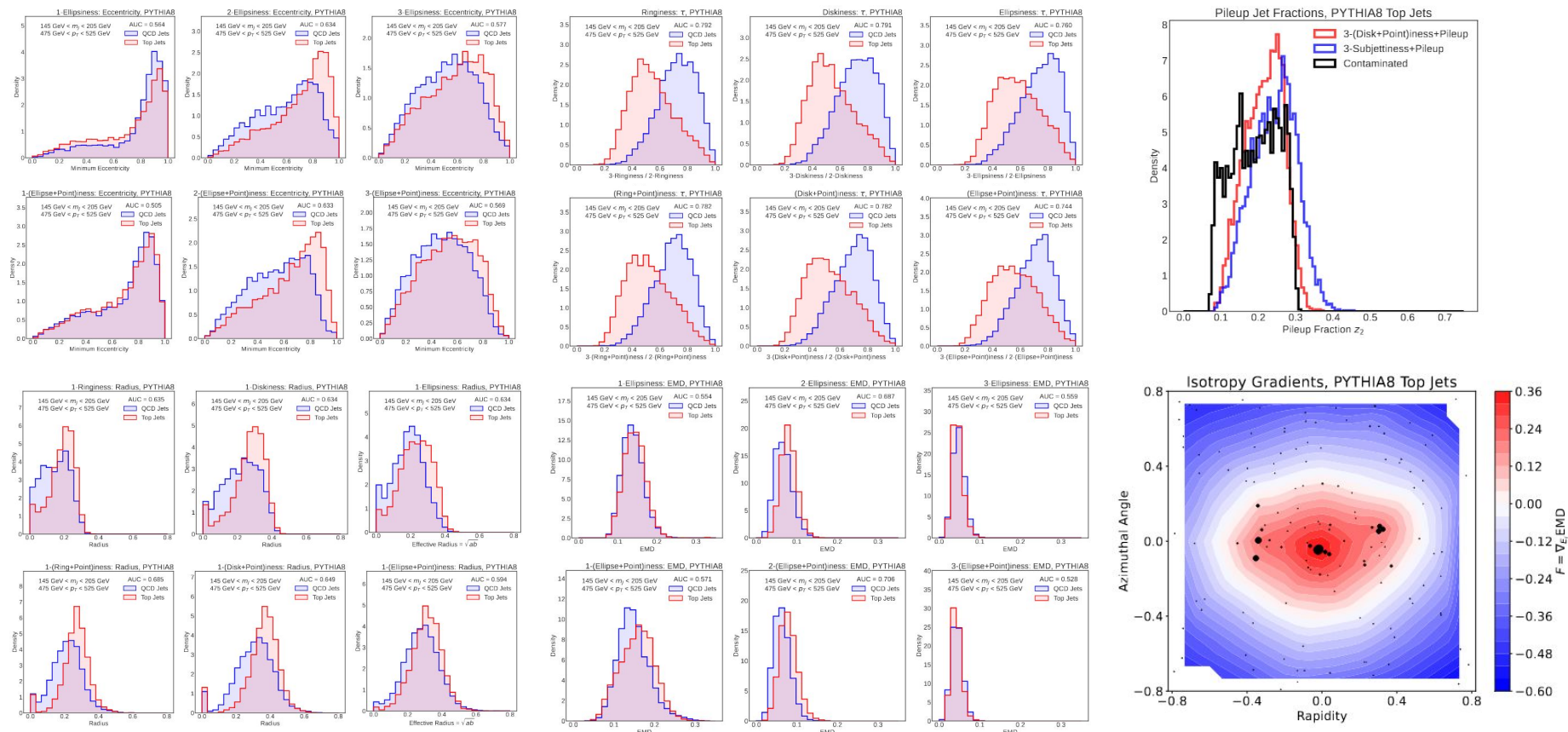


Can also consider ellipses instead of disks – only marginally better performance





New IRC-Safe Observables



... **Lots** of extractable information!

Automatic Grooming with Shapes

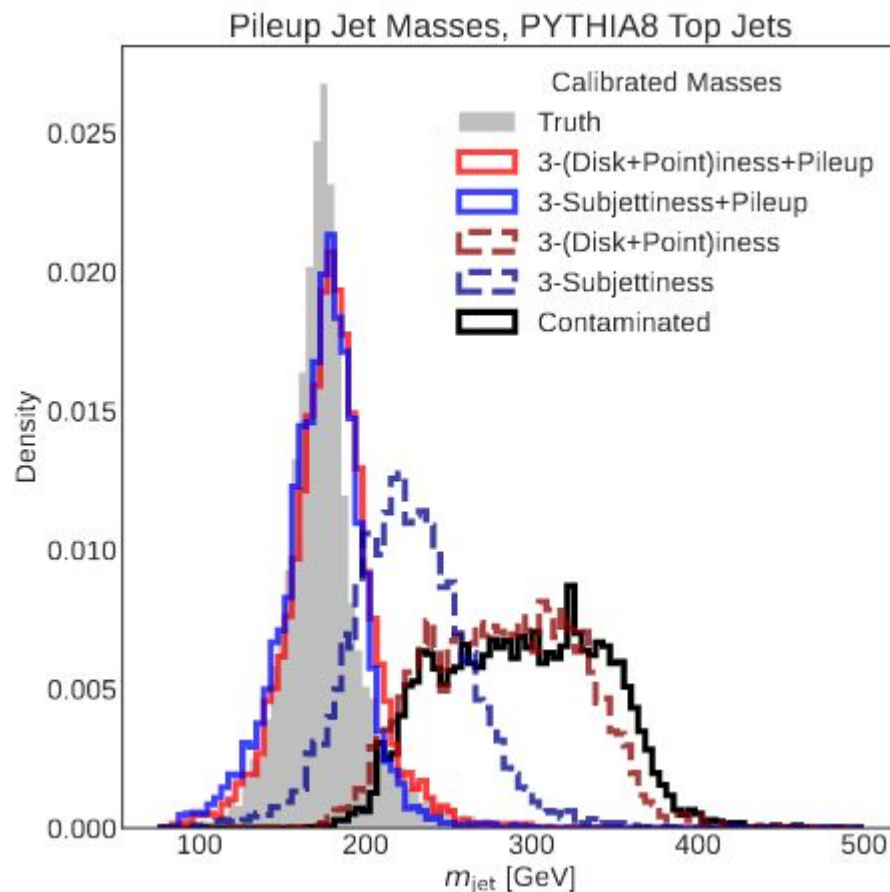
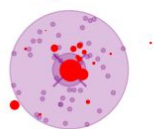
Use shapes to approximate events and extract masses – model pileup with a uniform background with floating weight!

No external hyperparameters, unlike softdrop.
Only need to assume pileup is uniform!

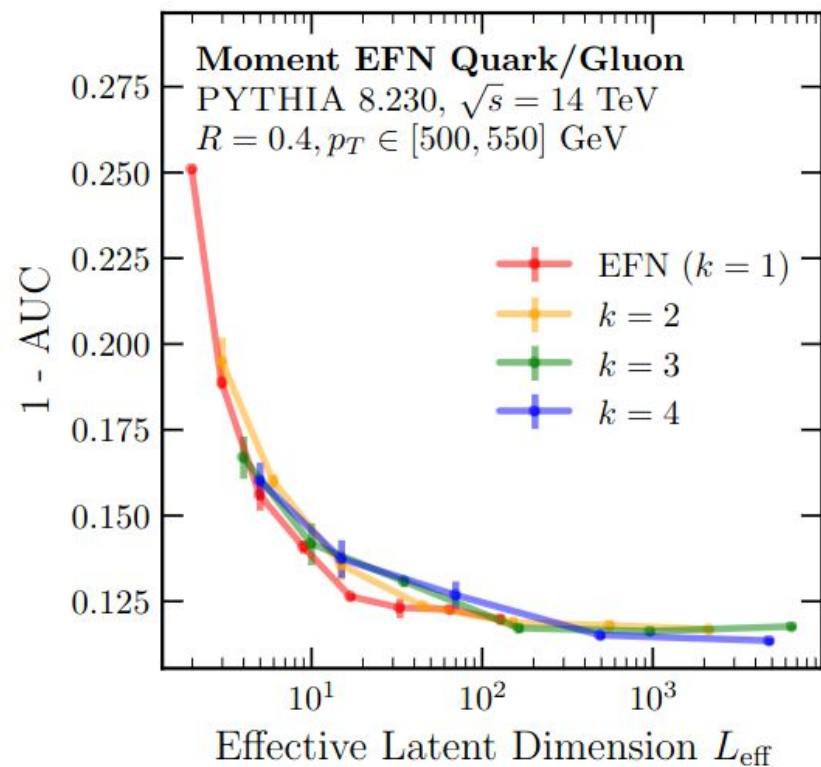
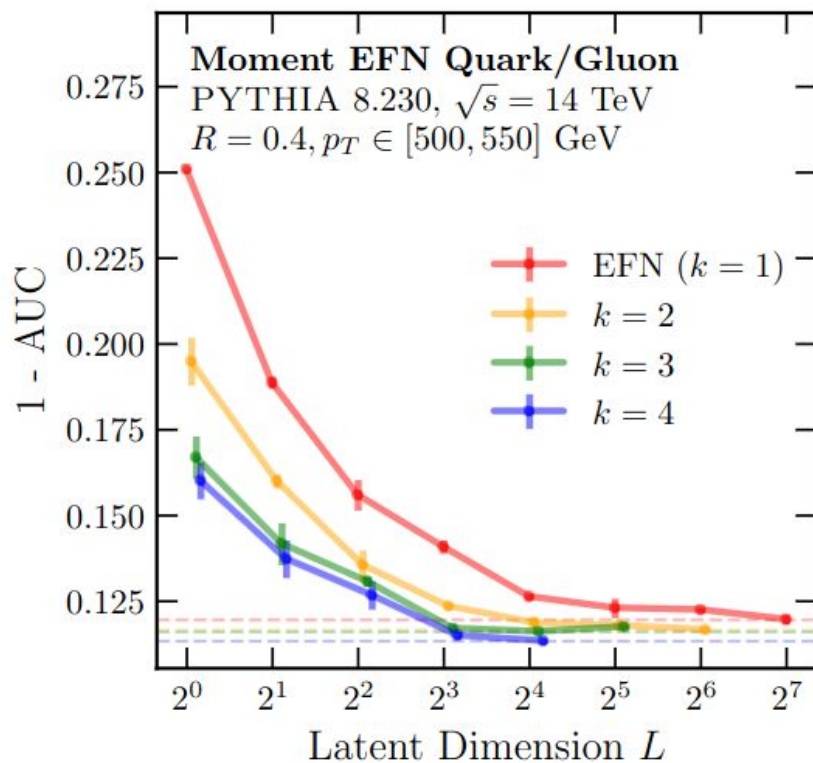
Contaminate top jets with 5-30% extra energy spread uniformly in an 0.8×0.8 plane

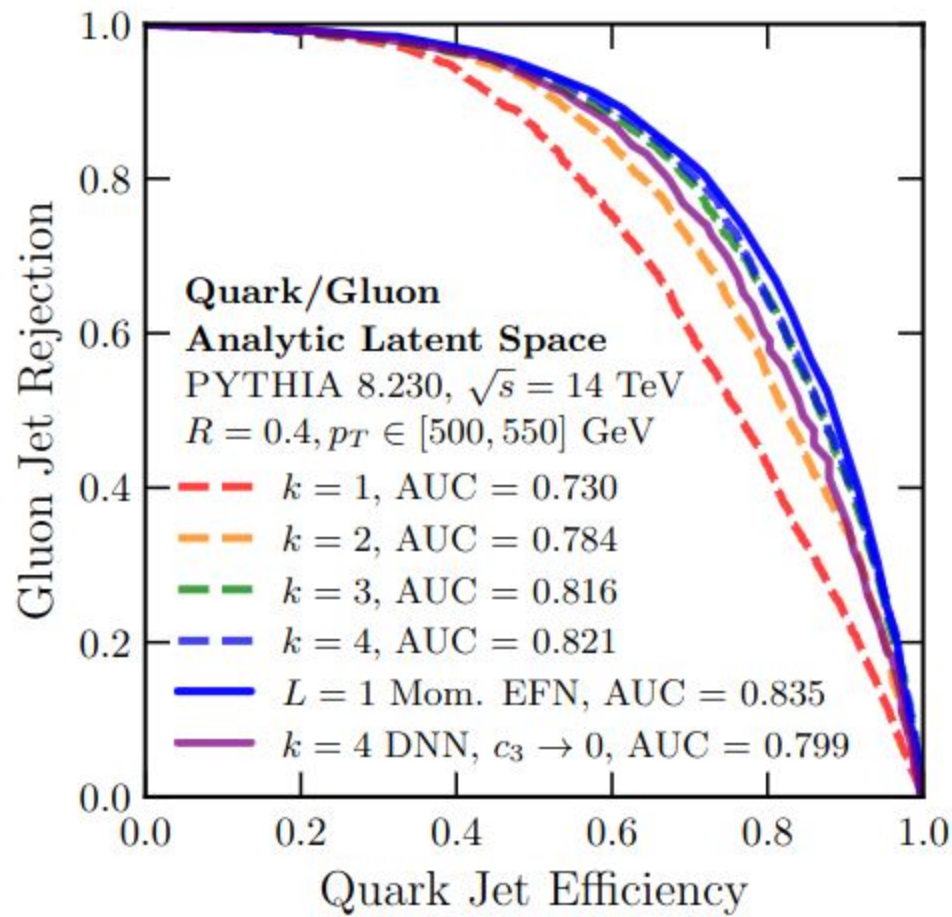
Consider 4 shapes:

- 3-Subjettiness
- 3-Subjettiness + Pileup
- 3-(Disk+Point)iness ←
- 3-(Disk+Point)iness + Pileup



Can also consider ellipses instead of disks – only marginally better performance





The Spectral EMD

Similar to the ordinary EMD, the **Spectral EMD (SEMD)**¹ is an IRC-saf metric between events or jets, computed on their **spectral representations**:

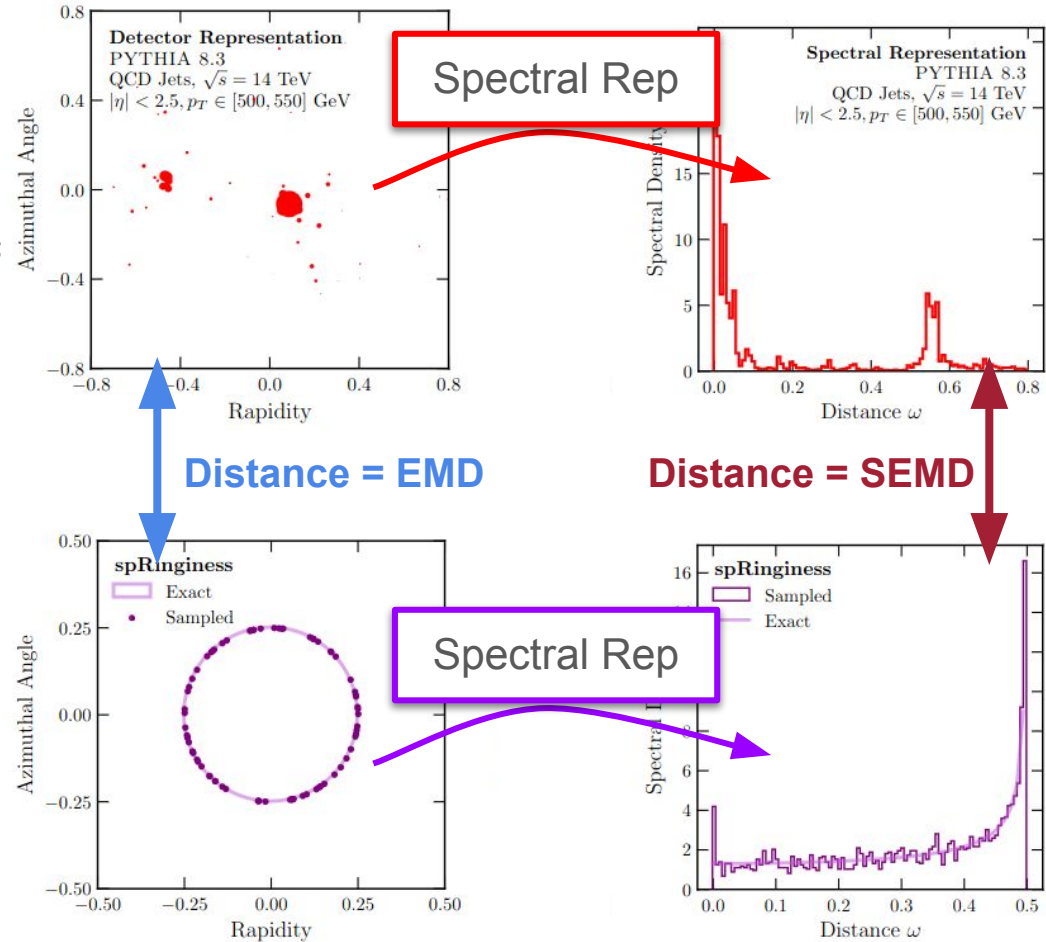
$$s(\omega) \equiv \sum_{i,j \in \mathcal{E}} E_i E_j \delta(\omega - \omega_{ij})$$

= list of energy-weighted pairwise distances

Then the p -SEMD is given by:

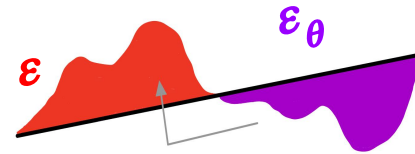
$$\text{SEMD}_p(s_A, s_B) \equiv \int_0^{E_{\text{tot}}^2} dE^2 \left| S_A^{-1}(E^2) - S_B^{-1}(E^2) \right|^p$$

↑
Inverse of integral of $s(\omega)$



The SEMD automatically respects all isometries of the metric ω . In this case, the SEMD is invariant under rotations or translations of either event or jet

Technical Details ...



For $p = 2$, possible problem on the sp

For events A, B , the p **spectral EMD** is defined as (1D OT!):

EMD = Work done to move "dirt" optimally

$$\text{SEMD}_{\beta,p}(s_A, s_B) \equiv \int_0^{E_{\text{tot}}^2} dE^2 |S_A^{-1}(E^2) - S_B^{-1}(E^2)|^p$$

$$\begin{aligned} \text{SEMD}_{\beta,p=2}(s_A, s_B) = & \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 \\ & - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l (\min [S_A(\omega_n^+), S_B(\omega_l^+)] - \max [S_A(\omega_n^-), S_B(\omega_l^-)]) \\ & \times \Theta(S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta(S_B(\omega_l^+) - S_A(\omega_n^-)), \end{aligned}$$

S = **cumulative spectral function**

$\pm$ indicates whether or not to include ω in the sum

The trick: Sum over pairs n of particles within each event.

Looks like $O(N^4)$, but with clever sorting & indexing in 1D, reduces to **$O(N^2)$** !

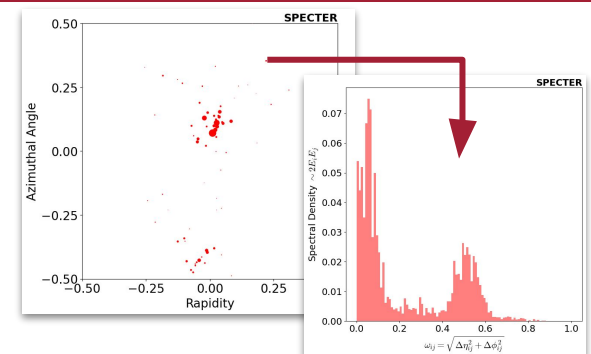
The **spectral density function**

$$s(\omega) = \sum_{i=1}^N \sum_{j=1}^N E_i E_j \delta(\omega - \omega(\hat{n}_i, \hat{n}_j))$$

Pairwise Distances

Reduces events to 1D, while preserving all* information about the event, up to translations and rotations.

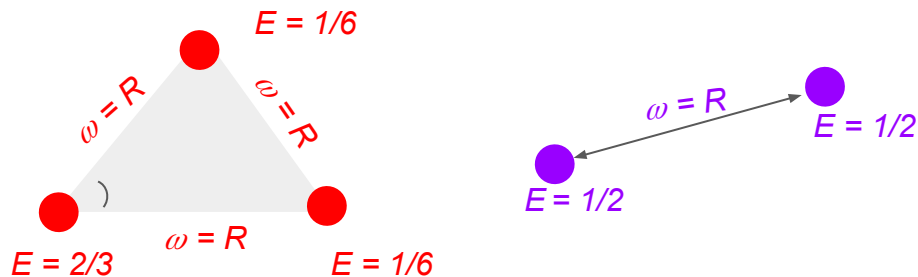
*up to measure 0, but important degeneracies, ask me about this later!



EMDs, ...

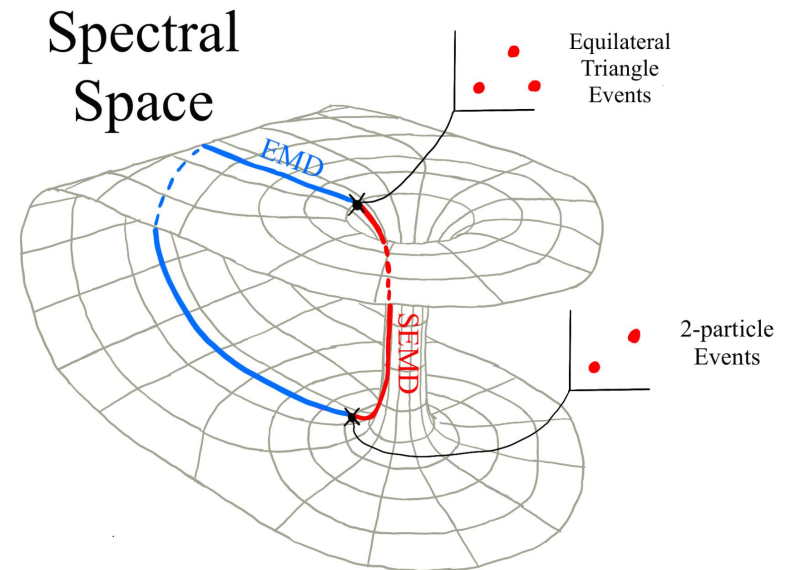
The EMD vs. The SEMD

The SEMD is *topologically different* from the ordinary EMD!



This three particle event looks very different from this two particle event, and thus they have a large EMD.

But their SEMD is zero! The spectral function only cares about pairwise distances and degenerate configurations can occur.

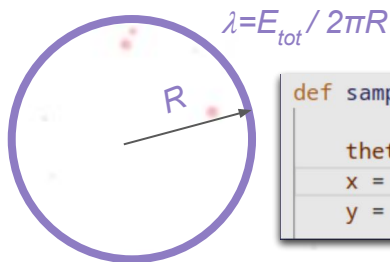


The space of events gets "pinched" at degenerate configurations when looking at only their spectral representations

This can come into play when events have 3 hard prongs, e.g. top jets!

Full Example: How “ring-like” are jets?

Step 1: Define the shape with parameters

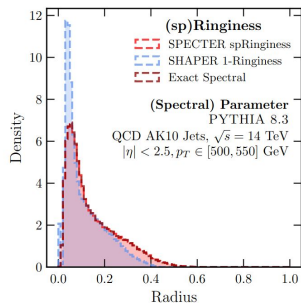
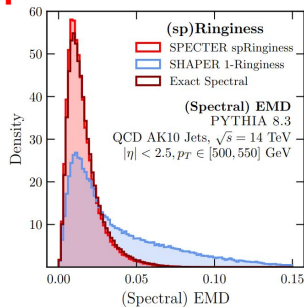


```
def sample_circle(params, N, seed):
    thetas = jax.random.uniform(seed, shape=(N,))
    x = params["Radius"] * jnp.cos(thetas)
    y = params["Radius"] * jnp.sin(thetas)
```

Unlike ordinary EMD, not necessary to specify center / orientation!

Pictured: 100k Jets, PYTHIA 8 QCD Jets

Step 5: Plots!

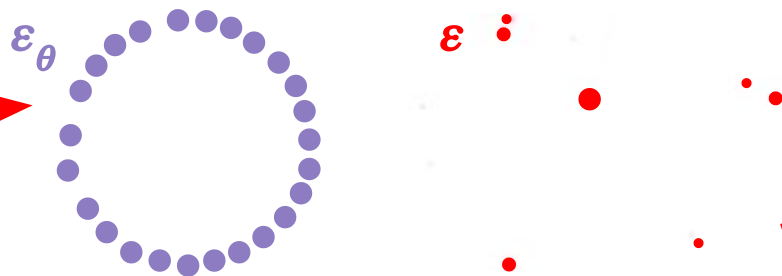


SPECTER

Our code framework
for these calculations



Step 2: Sample from Parameterized Shapes

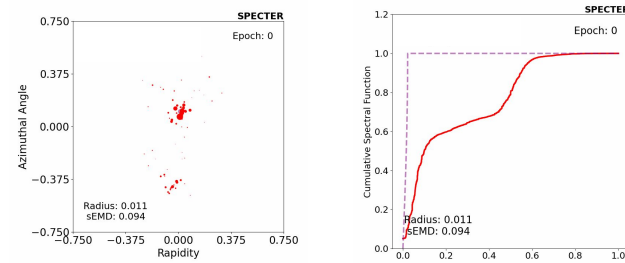


Step 3: Calculate the spectral metric between events and shapes

$$\text{SEMD}_{\beta, p=2}(s_A, s_B) = \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l (\min[S_A(\omega_n^+), S_B(\omega_l^+)] - \max[S_A(\omega_n^-), S_B(\omega_l^-)]) \times \Theta(S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta(S_B(\omega_l^+) - S_A(\omega_n^-))$$

Key difference from previous work: We use the SEMD, *not* the EMD!

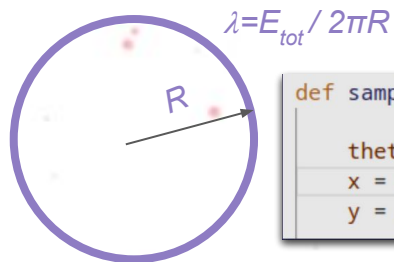
Step 4: Minimize w.r.t. parameters using grads



Pictured: Animation of optimizing for the radius R

Full Example: How “ring-like” are jets?

Step 1: Define the shape with parameters



Unlike ordinary EMD, not necessary to s

```
def samp
    theta
    x = p
    y = p
```

The spectral EMD, and its optimization, are often partially or **completely solvable** in closed form!

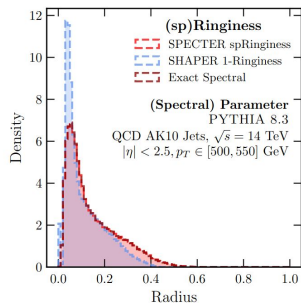
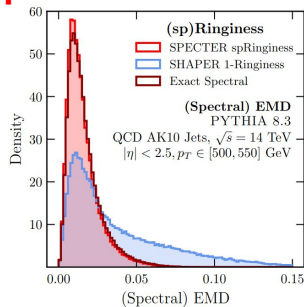
$$R_{\text{opt}} = \frac{2}{\pi} \sum_{\substack{n \in \mathcal{E}^2 \\ \omega_n < \omega_{n+1}}} \omega_n \left[\sin \left(\frac{\pi}{2E_{\text{tot}}^2} \sum_{\substack{n \leq m \in \mathcal{E}^2 \\ \omega_m < \omega_{m+1}}} (2EE)_m \right) - \sin \left(\frac{\pi}{2E_{\text{tot}}^2} \sum_{\substack{n < m \in \mathcal{E}^2 \\ \omega_m < \omega_{m+1}}} (2EE)_m \right) \right]$$

$$\text{SEMD}_{\beta, p=2}(s, s_{\text{jet ring}}) = \sum_{i < j \in \mathcal{E}} 2E_i E_j \omega_{ij}^2 - 2E_{\text{tot}}^2 R_{\text{opt}}^2$$

For many shapes, we can completely short circuit having to perform expensive optimization over an optimal transport problem entirely!

Pictured: 100k Jets, PYTHIA 8 QCD Jets

Step 5: Plots!



SPECTER

Our code framework for these calculations



Step 2: Sample from Parameterized Shapes

Alternatively...

the spectral metric between

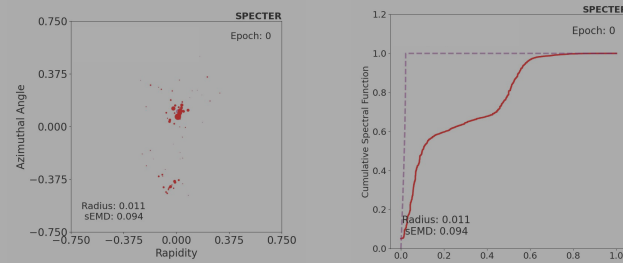
$$2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2$$

$$(\omega_n^+, S_B(\omega_l^+)) - \max[S_A(\omega_n^-), S_B(\omega_l^-)]$$

$$- S_B(\omega_l^-)) \Theta(S_B(\omega_l^+) - S_A(\omega_n^-)) ,$$

Key difference from previous work: We use the SEMD, *not* the EMD!

Step 4: Minimize w.r.t. parameters using grads



Pictured: Animation of optimizing for the radius R

Hearing Jets (sp)Ring

Runtimes (NVIDIA A100 GPU):

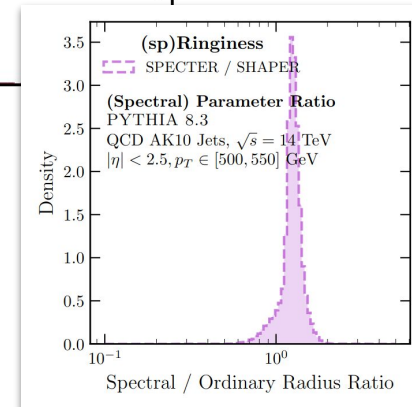
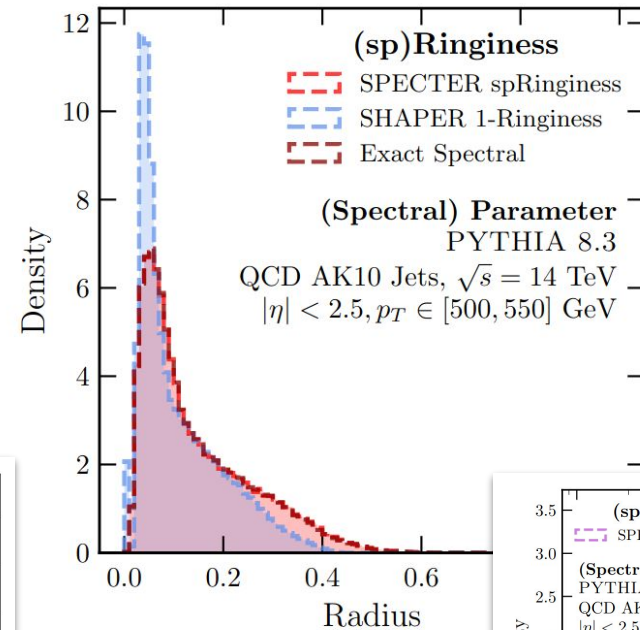
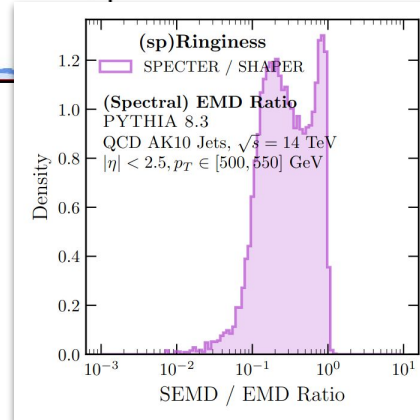
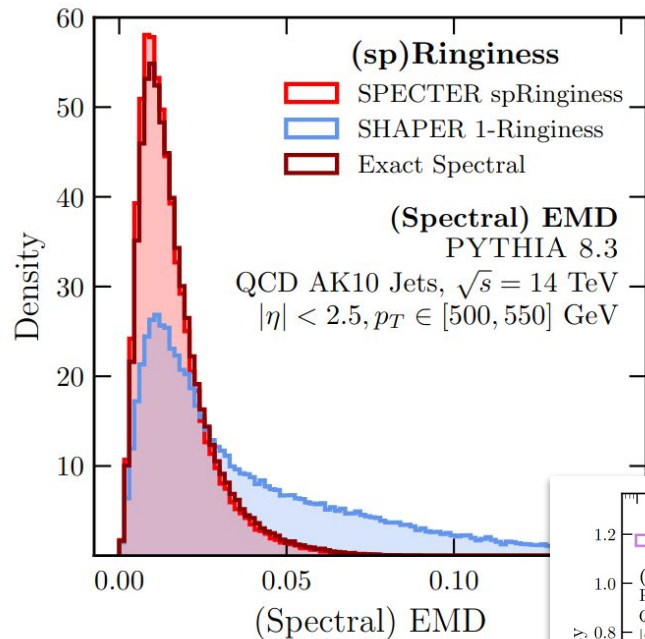
SHAPER (EMD): ~ 36 hours / 100k events

Generalized SPECTER: ~55 seconds / 100k events

Closed Form SPECTER: ~ < 0.1 seconds / 100k events

The SEMD and EMD are qualitatively different, but give similar radii!

They probe the same event length scale



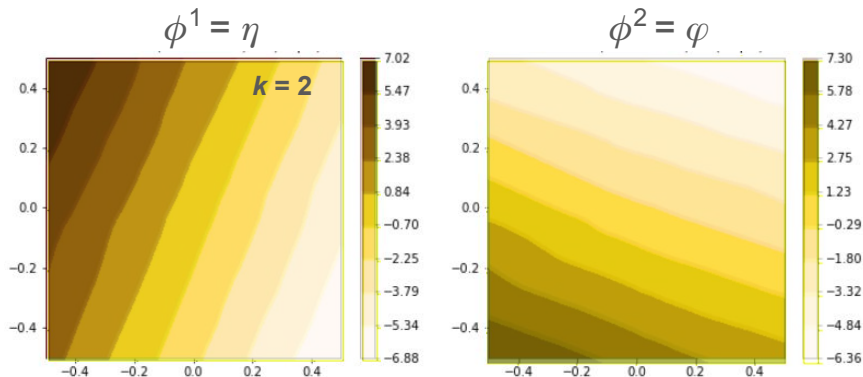
This is actually a much simpler version of the attention mechanism. It turns out **you don't even need attention** – ask me later!

Multiplication is all You Need

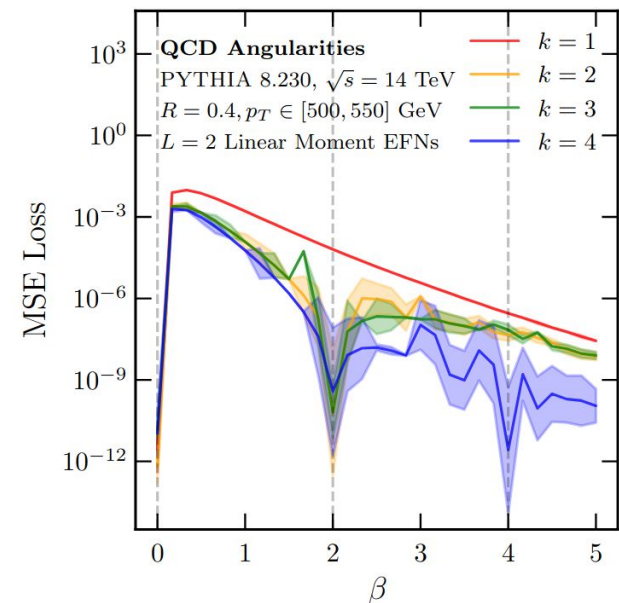
In the moment language, even integer^{*} β jet angularities $\Leftrightarrow k = \beta^{\text{th}}$ moments!

$$\begin{aligned}\lambda^{(\beta)}(\mathcal{P}) &= \sum_i z_i (\eta_i^2 + \phi_i^2)^{\beta/2} \\ &= \langle \eta^\beta \rangle_{\mathcal{P}} + \langle \phi^\beta \rangle_{\mathcal{P}} + \text{Cross Moments}\end{aligned}$$

Learns the simplest latent representations!

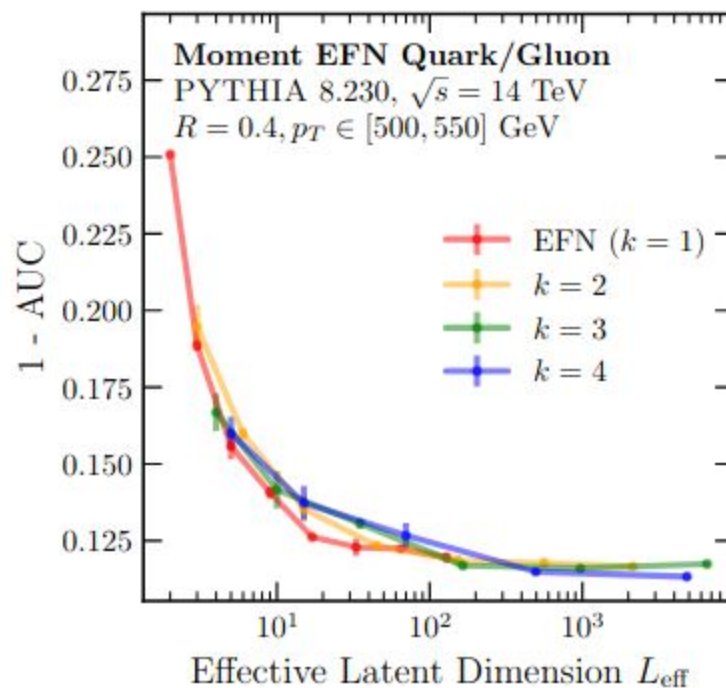
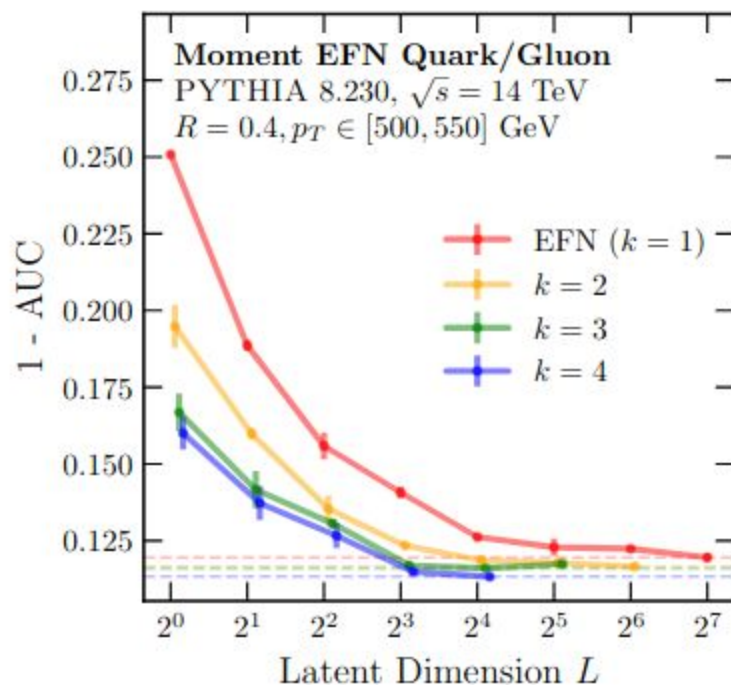


Encourage learning a diagonal basis in (η, φ) using L1 Regularization on F

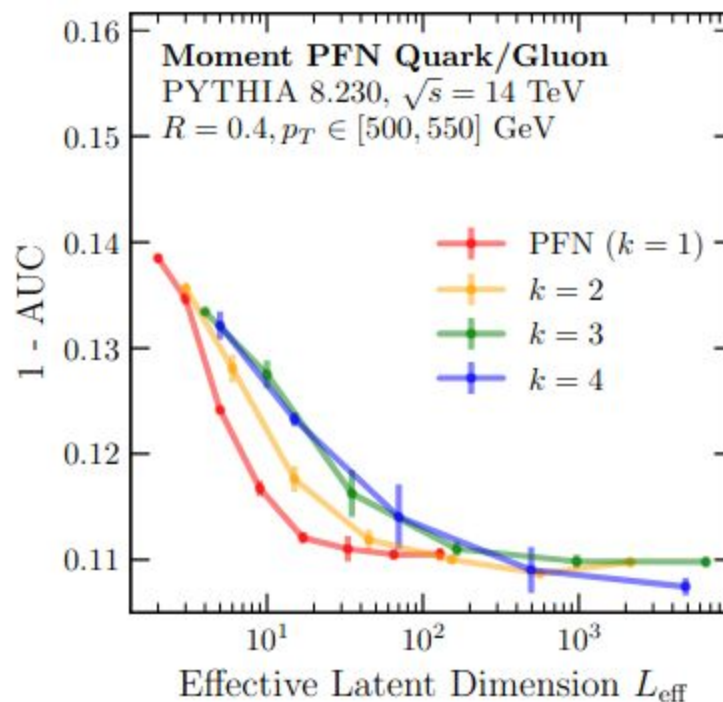
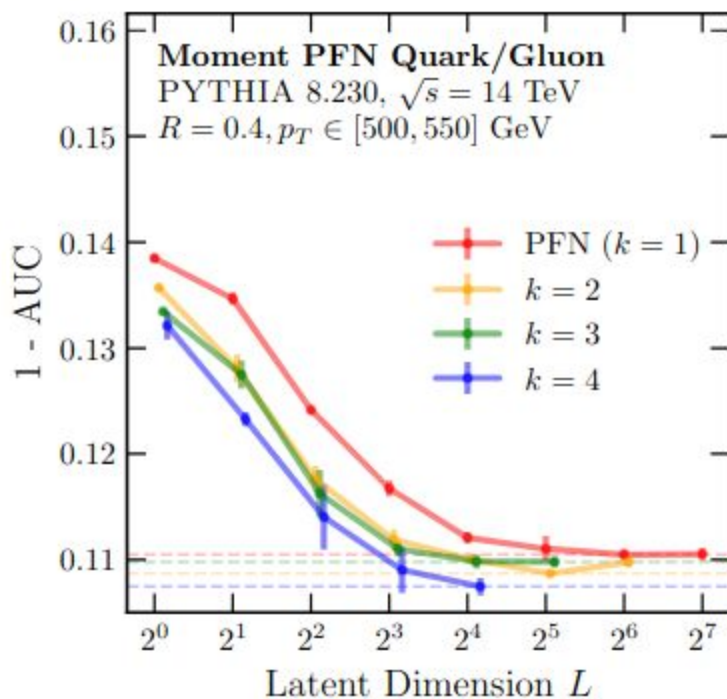


The lesson: Basic observables are naturally represented with multiplication, so build it in!

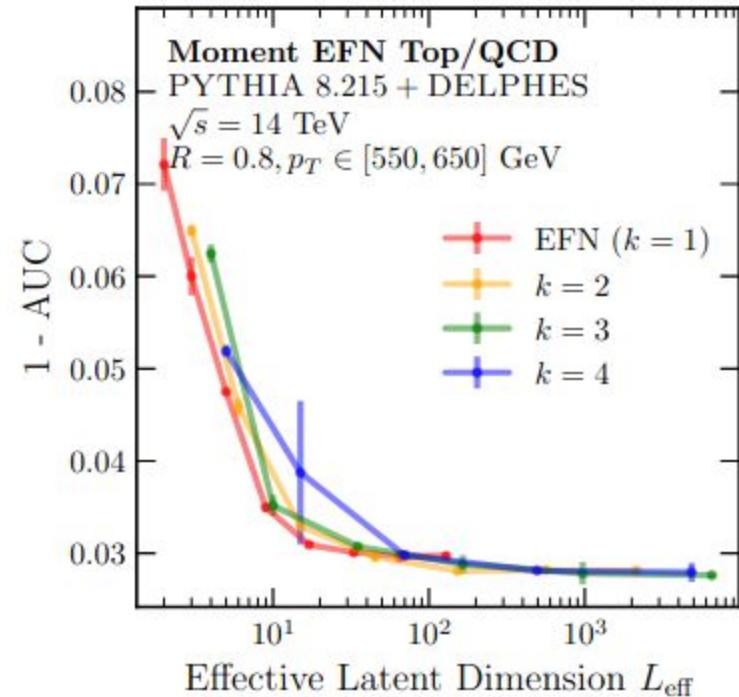
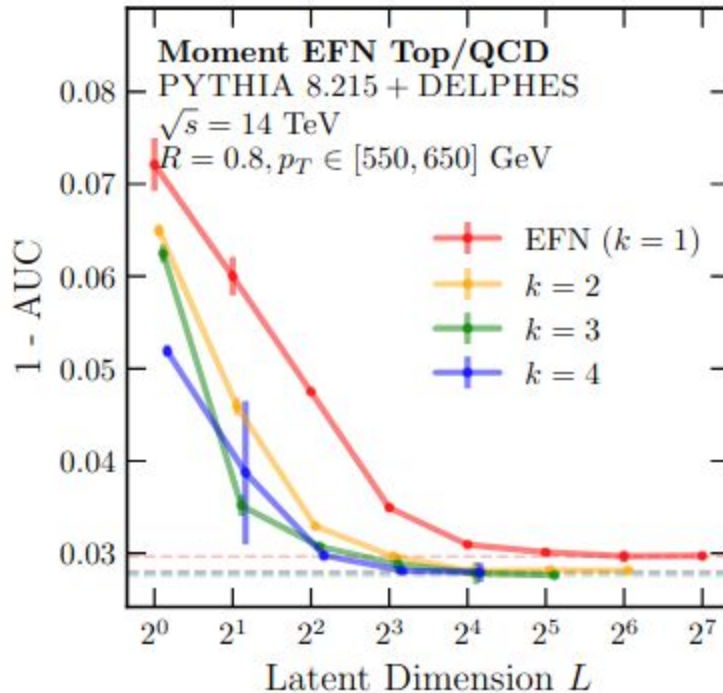
Quark/Gluon EFN



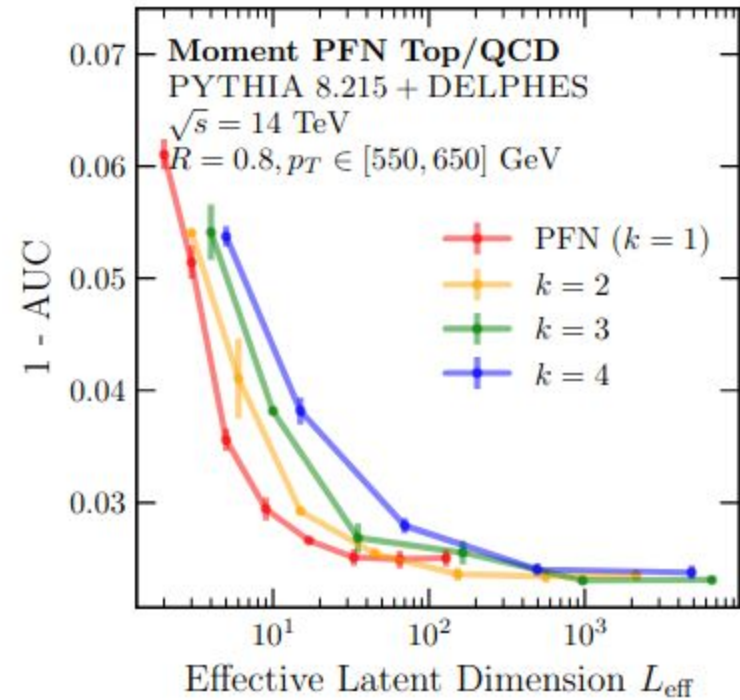
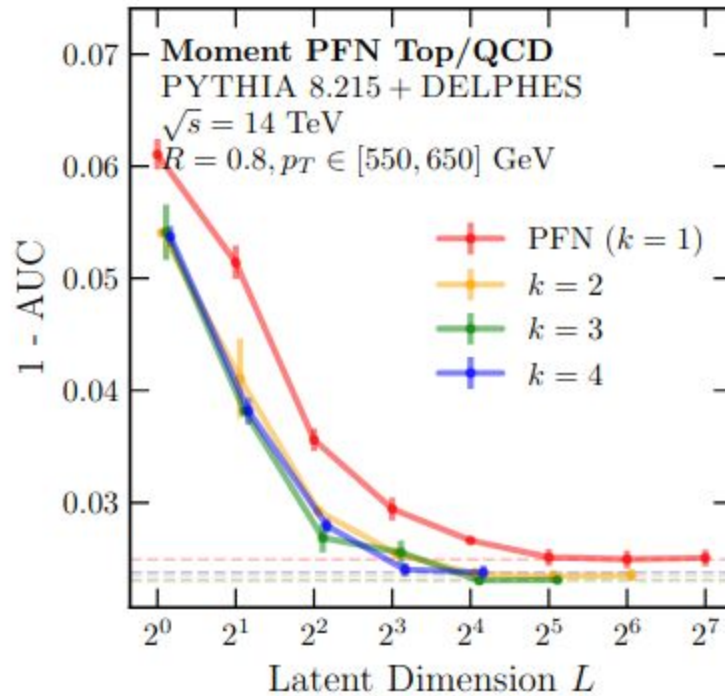
Quark/Gluon PFN



Top/QCD EFN



Top/QCD PFN



Same-sign dimuons

